Notes: Multiscale Analysis of sEEG Functional Connectivity via the Laplacian Renormalization Group

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Contents

1 Introduction

The mesoscopic organization of the human brain is neither fixed nor monolithic: functional interactions among neuronal populations span multiple spatial and temporal scales, and they reconfigure as cognitive demands change [1, 3]. A central open question in network neuroscience is whether—and at which hierarchical scales—this organization *persists* through a cognitive episode or *resets* to a baseline configuration once the task ends [5, 7]. This question can be cast explicitly as one of ergodicity: given a well-defined pre-task resting state, does the FC community structure return to that state after the task, or does the system retain a lasting, task-induced imprint? Answering it requires a framework that can (i) resolve hierarchical structure at arbitrary resolution without assuming a topological null model, (ii) operate on frequency-specific networks, and (iii) provide a principled notion of distance between multiscale architectures that is valid across all levels of community granularity.

These notes are the direct methodological update of a companion document, hereafter $\text{N}\textsc{otes}_{\text{MSC}}$, which developed the full Laplacian Renormalization Group (LRG) analysis pipeline for intracranial sEEG recordings of epileptic patients performing a transitive inference task [9, 10]. The scientific question, experimental design, patient cohort, LRG framework, inter-phase comparison methodology, and statistical testing protocol are all *unchanged*. The single revision concerns the FC estimation step: the original Magnitude-Squared Coherence (MSC) estimator is here replaced by the imaginary part of coherency, $\text{Im}[\text{Coh}_{ij}(f)]$ [8], hereafter denoted Imaginary Coherence (ImCoh) or |ImCoh| when the band-averaged magnitude is taken. This is a standard methodological requirement when working with common-reference intracranial recordings, where zero-phase-lag coupling artifacts arise from the shared recording reference and from volume conduction—both of which produce a purely real cross-spectral density and therefore vanish exactly in the imaginary part [2, 8]. Adopting ImCoh is the principled and physically motivated response to this known limitation. All LRG outputs, metric families, unanimity maps, and scalar results are correspondingly updated, and a direct MSC-vs-ImCoh comparison is provided wherever it is informative.

The experimental design comprises four recording phases—Pre-task resting state (rsPre), learning phase (taskLearn), testing phase (taskTest), and Post-task resting state (rsPost)—enabling within-subject comparisons between baseline rest, active learning, generalization over novel stimulus pairs, and post-task rest. For each phase, FC is estimated via ImCoh, which isolates phase-lagged coupling between channel pairs and naturally decomposes into canonical Electroencephalography (EEG) frequency bands (δ, θ, α, β, low-γ, high-γ) without requiring band-pass filtering. The result is, for every patient, a collection of weighted networks indexed by the triplet (phase, band, patient), each of which is passed through the LRG pipeline described in $\text{N}\textsc{otes}_{\text{MSC}}$.

The analysis exploits three simultaneous axes of multiscale variation:

- **Topology** — the hierarchical community structure resolved by the LRG framework at arbitrary diffusion scales τ;
- **Frequency** — six canonical EEG bands, each capturing a distinct dynamical scale of electrophysiological coupling;
- **Space** — the three-dimensional Montreal Neurological Institute (MNI) positions of the implanted sEEG contacts, onto which multiscale communities are re-embedded for anatomical interpretation.

For each band, we compare the multiscale community structure across the four recording phases by evaluating distances between LRG-derived ultrametric matrices at every hierarchical scale $k = 2, \ldots, \lfloor N/2 \rfloor$. The resulting scale-resolved Variation of Information profiles, combined with a strict cross-patient unanimity criterion (cf. $\text{N}\textsc{otes}_{\text{MSC}}$, §5), form the evidence base for the central question: does a given frequency band exhibit persistent, resetting, or null behaviour across the cognitive episode, and at which community granularities does this behaviour live? All numerical results reported in **??** were independently verified by the computational pipeline before any text was written.

The remainder of these notes is organized as follows.

Section § 2 documents the ImCoh-based FC estimation pipeline and the direct comparison with MSC outputs.

Section § 3 provides a self-contained recap of the LRG pipeline and comparison framework, with key figures regenerated under ImCoh; full derivations and metric-family analysis are in $\text{N}\textsc{otes}_{\text{MSC}}$.

Section § 4 presents the substrate phase-distance audit on the |ImCoh| matrices directly, before any diffusion or community detection, establishing a directionally consistent task-shaped pattern in four of six bands that survives drift controls. Section § 5 carries the same rest–task–rest geometry through the LRG pipeline and identifies the band-resolved structure of the trace at the diffusion layer: a controlled cohort claim at α, β, and low-γ via the per-pair correlation on $\mathcal{D}(\tau)$ (within-probe BH-$q = 0.027$ at $m = 6$), with a per-band geometric decomposition (β multifacet, low-γ heights-only, α per-pair only) and an anatomical headline at low-γ ctx-lh-fusiform that survives Bonferroni correction across the (band, region) cell space. The substrate-and-LRG layered reading, with the controlled per-pair claim of § 5.3 as the load-bearing cohort statement and the per-patient module taxonomy of § 5.6 as the structural illustration, is the central contribution of the present notes.

2 Functional Connectivity Estimation

2.1 Frequency bands as dynamical scales

Electrophysiological interactions are structured across characteristic frequency ranges corresponding to distinct dynamical timescales. We reconstruct one FC network per canonical EEG band and compare their multiscale organization across recording phases. Let a frequency band be an interval $\mathcal{B} = [f_{\min}, f_{\max}]$ with $0 \le f_{\min} < f_{\max} \le f_{\mathrm{Ny}}$, where $f_{\mathrm{Ny}} = f_s/2$ is the Nyquist frequency. The six canonical bands used throughout are reported in 1. These cutoffs are fixed once and reused for all patients and phases. All subsequent steps

Band	**Label**	**Range (Hz)**
Delta	`delta`	0.53–4 Hz
Theta	`theta`	4–8 Hz
Alpha	`alpha`	8–13 Hz
Beta	`beta`	13–30 Hz
Low gamma	`low_gamma`	30–80 Hz
High gamma	`high_gamma`	80–300 Hz

Table 1: Canonical frequency bands used for FC estimation.

are performed independently for each band $\mathcal{B}^{(b)}$, yielding one weighted adjacency matrix per (band, phase, patient) triplet.

2.2 From magnitude-squared coherence to imaginary coherence

2.2.1 Coherence and its limitations on common-reference sEEG.

For a multichannel sEEG recording $\mathbf{X} \in \mathbb{R}^{N \times T}$, spectra are estimated with Welch's averaged periodogram method [11] using a `hann` window with `nperseg` $= 4096$, frequency resolution $\Delta f = 0.5$ Hz, and 50 % overlap. The cross-spectral density between channels i and j is $S_{ij}(f)$, with auto-spectra $S_{ii}(f)$ and $S_{jj}(f)$. The magnitude-squared coherence is

$$\mathrm{MSC}_{ij}(f) = \frac{|S_{ij}(f)|^2}{S_{ii}(f) S_{jj}(f)} \in [0, 1], \tag{1}$$

and the band-averaged FC weight is $W_{ij}(\mathcal{B}) = |\mathcal{K}_{\mathcal{B}}|^{-1} \sum_{k \in \mathcal{K}_{\mathcal{B}}} \mathrm{MSC}_{ij}(f_k)$, where $\mathcal{K}_{\mathcal{B}} = \{k : f_k \in \mathcal{B}\}$. The MSC captures all phase-consistent coupling, however, including components arising from the recording setup rather than neural interactions. In the present sEEG dataset, each electrode shaft uses a dedicated reference contact located on that same shaft, so all contacts belonging to a given shaft share a common reference signal. This injects an identical electrical artifact into every same-shaft channel simultaneously, producing zero-phase-lag correlations within each shaft that are a property of the recording montage rather than of neural activity. Volume conduction further compounds this: electrical fields from a neural source propagate effectively instantaneously through brain tissue, generating additional zero-lag coupling between physically adjacent contacts. Both mechanisms produce a purely real cross-spectral density for the artifactual component—since their phase lag is zero, $\mathrm{Im}(S_{ij}(f)) = 0$ for those components—and consequently inflate

MSC_{ij} systematically for same-shaft pairs. As shown in Fig. 1, the same-shaft to cross-shaft mean weight ratio under MSC ranges from 1.5× to 8.9× across patients and bands (cohort mean 3.2×). This bias is not a flat offset at the edge-weight level but grows with the number of LRG communities n, reaching $\sim 6\times$ on average at $n = 30$ (Fig. 2), meaning it contaminates the coarse mesoscopic community structure most severely—precisely the scales most relevant to the scientific question.

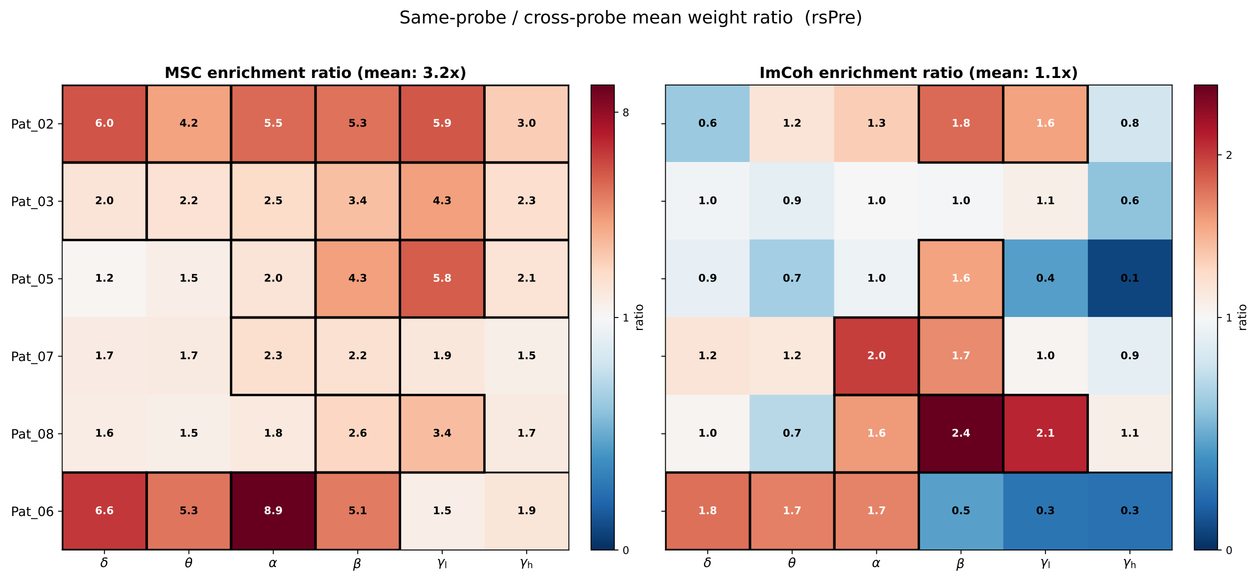


Figure 1: Same-shaft to cross-shaft mean weight ratio for MSC (left) and |ImCoh| (right), computed at the edge-weight level for all six patients and six frequency bands (rsPre phase). Cells with thick black borders exceed $2\times$ (left panel) or $1.5\times$ (right panel). Under MSC the ratio ranges from $1.14\times$ to $7.51\times$ (cohort mean $2.59\times$); under $|\mathrm{ImCoh}|$ every $(\mathrm{patient}, \mathrm{band})$ cell falls below $1.5\times$ (cohort mean $0.99\times$). The two colorbars use independent scales to reflect the qualitative difference in bias magnitude.

2.2.2 Imaginary coherence as an artifact-immune FC estimator.

Since the artifact contributions to $S_{ij}(f)$ are purely real, taking the imaginary part removes them exactly. The imaginary coherence [8] is defined as

$$\mathrm{ImCoh}_{ij}(f) \equiv \mathrm{Im}[\mathrm{Coh}_{ij}(f)] = \frac{\mathrm{Im}(S_{ij}(f))}{\sqrt{S_{ii}(f)S_{jj}(f)}} \in [-1, 1], \tag{2}$$

and its magnitude $|\mathrm{ImCoh}_{ij}(f)| \in [0, 1]$ measures the strength of phase-lagged coupling between channels i and j at frequency f, free from any zero-lag component [2, 4, 8]. This immunity is mathematically exact: because volume conduction and common-reference coupling are instantaneous, their contribution to $S_{ij}(f)$ is real-valued at every frequency, and the imaginary part is identically zero for those contributions regardless of the signal amplitude, electrode geometry, or frequency band. Any nonzero $|\mathrm{ImCoh}_{ij}(f)|$ therefore certifies the presence of a genuine time-lagged neural interaction between the two channels.

For the purposes of constructing a LRG-compatible adjacency matrix we use the band-averaged magnitude

$$W_{ij}(\mathcal{B}) = \frac{1}{|\mathcal{K}_{\mathcal{B}}|} \sum_{k \in \mathcal{K}_{\mathcal{B}}} |\mathrm{ImCoh}_{ij}(f_k)| \in [0, 1], \tag{3}$$

hereafter denoted $|\mathrm{ImCoh}|$ for brevity. The resulting matrix is symmetric, non-negative, and has zero diagonal, satisfying the non-negativity requirement of the combinatorial Laplacian $L = D - W$ without any rectification step. Taking the magnitude before band-averaging discards the sign of the imaginary part, which encodes directionality of the phase lag but is not needed for the undirected community-detection framework pursued here; the convention follows Ewald et al. [4] and Bastos and Schoffelen [2]. The LRG framework requires only that edge weights be non-negative and that they reflect genuine coupling strength: both conditions are satisfied by |ImCoh| and violated by MSC in the presence of same-shaft artifacts. |ImCoh| is computed from the same Welch cross-spectral estimate as MSC with a single change in the final derivation step. No surrogate-based sparsification is applied: circular-shift surrogates are inappropriate for |ImCoh| because a time shift creates an artificial phase lag that |ImCoh| registers as genuine coupling, systematically inflating the surrogate null distribution [2]. Dense |ImCoh| matrices are passed directly to the LRG pipeline.

2.2.3 Quantification of bias removal.

The bias reduction achieved by switching from MSC to |ImCoh| is shown in Fig. 1 and Fig. 2. The cohort mean same-shaft enrichment ratio drops from 2.59× under MSC to 0.99× under |ImCoh|, the latter indistinguishable from no enrichment, and no (patient, band) cell under |ImCoh| exceeds 1.40× (Pat08, β). The scale-resolved analysis in Fig. 2 confirms that |ImCoh| enrichment remains close to 1× across $n = 3$–30 communities, while MSC enrichment rises steeply. Residual |ImCoh| enrichment at fine scales ($n \geq 15$) for some patients reflects genuine lagged short-range coupling between adjacent contacts, which is physically interpretable and does not constitute artifact. We note that Pat06 exhibits elevated spectral noise above approximately 30 Hz visible in Fig. 3; this artifact is present under both estimators but reduced under |ImCoh|, and is consistent with a known recording quality issue. Pat06 is retained in the cohort across all phases and bands; d_F values for Pat06 in the γ_l and γ_h bands are flagged as cohort outliers in § 4.6.

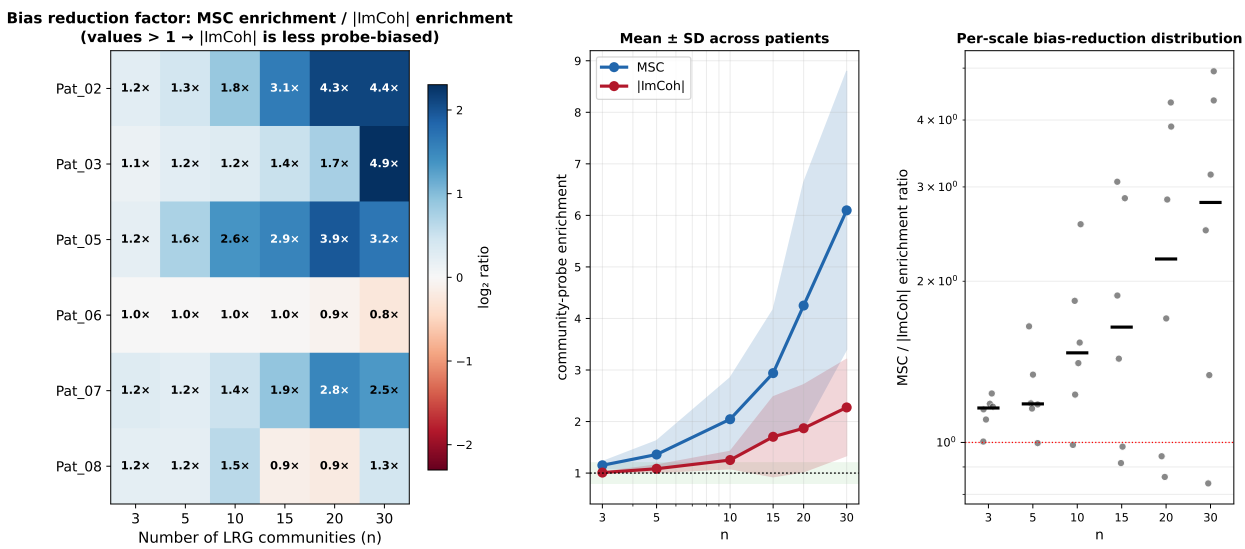


Figure 2: Community-scale dependence of same-shaft enrichment under MSC (blue) and |ImCoh| (red) for the β band (rsPre phase). *Left:* $\log_2$ bias-reduction factor (MSC/|ImCoh| enrichment ratio) per patient and scale n; values above zero indicate |ImCoh| is less biased. *Centre:* mean $\pm$ SD across patients; the MSC curve rises from $\sim 1\times$ at $n = 3$ to $\sim 6\times$ at $n = 30$, while |ImCoh| remains below $2.5\times$ throughout. *Right:* per-scale distribution of the MSC/|ImCoh| ratio; the median rises from $1.69\times$ at $n = 10$ to $2.80\times$ at $n = 30$. The monotone growth of the bias with n means that MSC-based LRG communities are most severely contaminated at the coarse scales most relevant to the scientific question.

2.3 Adjacency matrices, weight distributions, and network characterization

The |ImCoh| band-averaged magnitude defined in Eq. (3) yields a symmetric, non-negative, zero-diagonal matrix $W(\mathcal{B})$ for each (patient, phase, band) triplet, interpretable as a weighted undirected network. We characterize these networks at the edge, node, and spectral levels before entering the LRG pipeline, comparing throughout with the corresponding MSC matrices to document the structural consequences of the estimator switch.

2.3.1 Spectral profiles.

Figure 3 shows the cross-spectral profiles across all channel pairs for all six patients under MSC (top) and |ImCoh| (bottom). Under MSC the median coherence is broadly distributed across frequency with inter-patient variability spanning roughly two orders of magnitude; Pat02 and Pat08 exhibit a pronounced theta–alpha peak, Pat03 shows comparatively flat and elevated coherence across the full spectrum, and Pat05 decays sharply above the alpha band. Under |ImCoh| the profiles are qualitatively different: the median is lower and flatter, inter-patient variability is reduced, and the frequency-specific inflation visible in the MSC profiles is suppressed. Pat06 displays severe spectral noise above approximately 30 Hz in both panels; this recording artifact is reduced but not eliminated under |ImCoh| and is treated as documented in Section § 2.2.

2.3.2 Edge weight distributions: same-shaft vs cross-shaft.

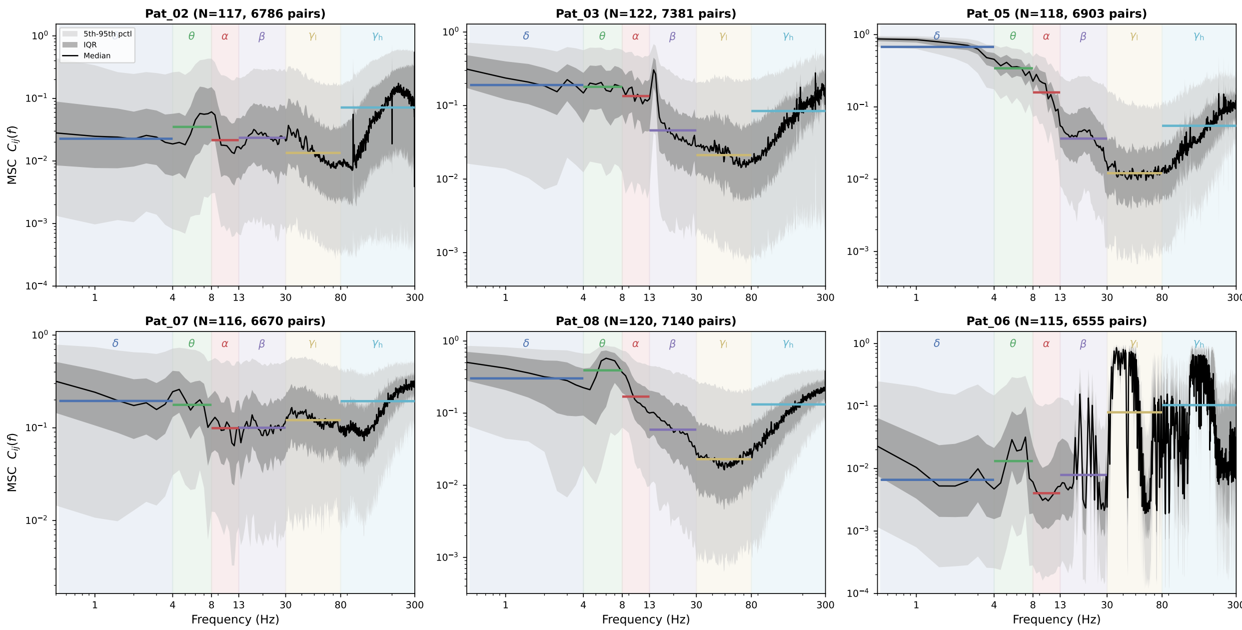
Figure 4 shows kernel density estimates of edge weights for same-shaft pairs (dashed) and cross-shaft pairs (solid) across all six patients for the β band. Under MSC every patient's same-shaft distribution sits to the right of the corresponding cross-shaft distribution by at least half a decade on the log scale. Under |ImCoh| the two distributions overlap for every patient; in several cases (e.g. Pat05, Pat06) the same-shaft distribution lies slightly below the cross-shaft one, indicating that within-shaft lagged coupling is weaker than cross-shaft coupling once zero-lag artifact is removed. This reversal is physically interpretable: contacts on the same shaft sample largely overlapping neural populations, leaving little scope for genuine lagged interactions, whereas cross-shaft pairs can capture long-range phase-lagged coupling between anatomically distinct regions.

2.3.3 Adjacency matrix structure across phases and bands.

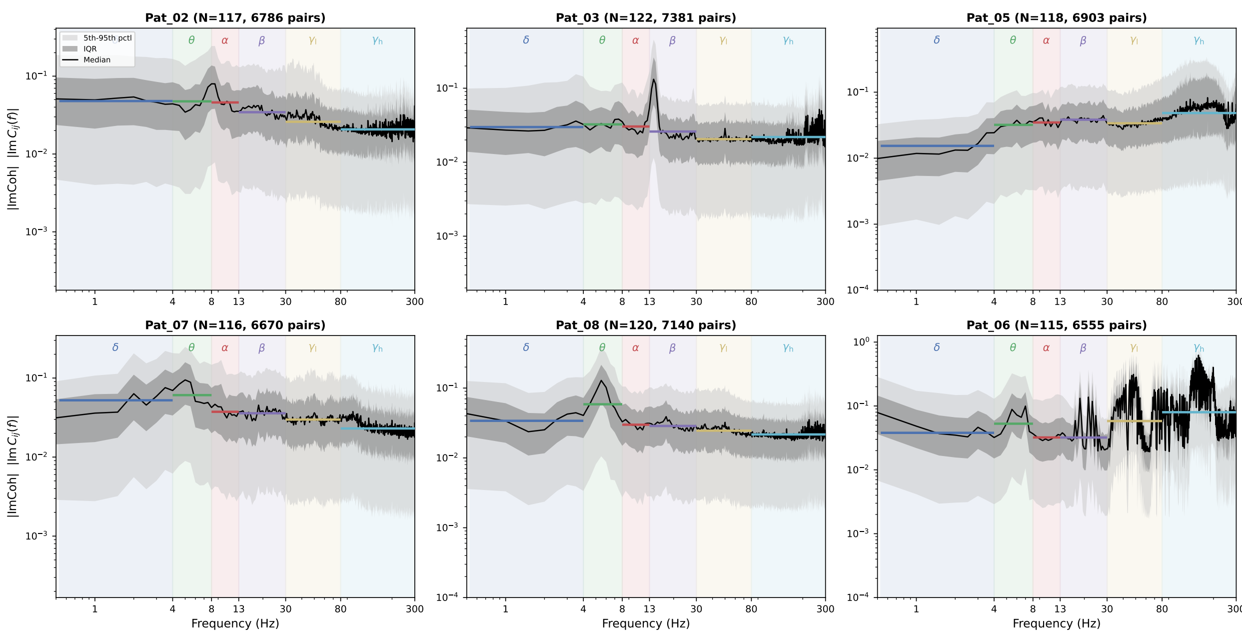
Figure 5 and Fig. 6 display the full $W(\mathcal{B})$ matrices for Pat02 and Pat05 respectively, arranged as phases (rows) by frequency bands (columns), for MSC (top) and |ImCoh| (bottom). Same-shaft blocks are outlined in yellow. Under MSC bright same-shaft blocks dominate the diagonal in most bands and phases, suppressing off-diagonal cross-shaft structure visually. Under |ImCoh| the same-shaft blocks are no longer elevated and cross-shaft coupling becomes visible: block-diagonal patterns, phase-dependent reorganization, and band-specific connectivity profiles emerge clearly. Comparing the two patients confirms that while the bias is systematic under MSC, the functional structure revealed by |ImCoh| is patient-specific, reflecting the distinct cortical regions sampled by each implant. A qualitative observation that will be formalized in Section **??** is already visible here: in the |ImCoh| panels the β band shows marked phase-to-phase variation in cross-shaft block structure, while the δ band remains comparatively stable across phases. Fig. 7 extends the comparison to all six patients for the β band and rsPre phase, with MSC in the top row and |ImCoh| in the bottom row. The same-shaft inflation is visible in every patient's MSC panel, confirming a cohort-level systematic effect rather than an idiosyncrasy of any single implant. The |ImCoh| row reveals the between-patient heterogeneity in functional connectivity that was partly masked by the artifact.

2.3.4 Network-level consequences: topology and Laplacian spectrum.

The adjacency matrix changes propagate to the network objects that feed the LRG pipeline. Fig. 8 shows force-directed network embeddings for three patients under MSC (top row) and |ImCoh| (bottom row), with the same layout algorithm and parameters applied to both estimators so that the two rows are directly comparable. Same-shaft edges are rendered in the shaft's color; cross-shaft edges are black. Under MSC the layout is dominated by same-shaft cliques: colored same-shaft edge bundles are thick, tightly wound, and spatially prominent, while cross-shaft black edges fade into a dense background that carries little visible structure. The network effectively organizes around electrode shaft identity rather than functional topology. Under |ImCoh| the picture reverses: same-shaft colored edges become sparse and thin, while cross-shaft black edges now carry the dominant weight, and the layout opens into a more distributed structure where no single shaft cluster monopolizes the embedding. The contrast is consistent across all three patients despite their different implant geometries, confirming that the same-shaft dominance under MSC is a recording artifact rather than a neural signal, and that |ImCoh| recovers a network whose strongest connections span anatomically distinct regions. Fig. 9 shows the Laplacian eigenvalue spectra for Pat05 under MSC and |ImCoh| across all six bands and four phases. The absolute eigenvalue scale differs substantially: MSC reaches $\lambda_{\max} \sim 20$–80 depending on band, while |ImCoh| reaches $\lambda_{\max} \sim 2$–18, reflecting the lower overall



(a) MSC cross-spectral distribution.



(b) $|\mathrm{ImCoh}|$ cross-spectral distribution.

Figure 3: Cross-spectral profiles for all six patients (rsPre phase). Each panel shows the median (black), interquartile range (dark grey), and 5th–95th percentile band (light grey) across all $\binom{N}{2}$ channel pairs on a log–log scale. Colored horizontal segments indicate the band-averaged value per canonical band. Top: MSC, values in $[10^{-4}, 10^{0}]$. Bottom: $|\mathrm{ImCoh}|$, values in $[10^{-4}, 10^{-1}]$. The $|\mathrm{ImCoh}|$ profiles are lower and flatter, with reduced inter-patient spread. Pat03 is recorded at $1024\,\mathrm{Hz}$ (others at $2048\,\mathrm{Hz}$) and shows elevated MSC across the full spectrum. Pat06 shows high-frequency spectral noise above $30\,\mathrm{Hz}$ in both panels.

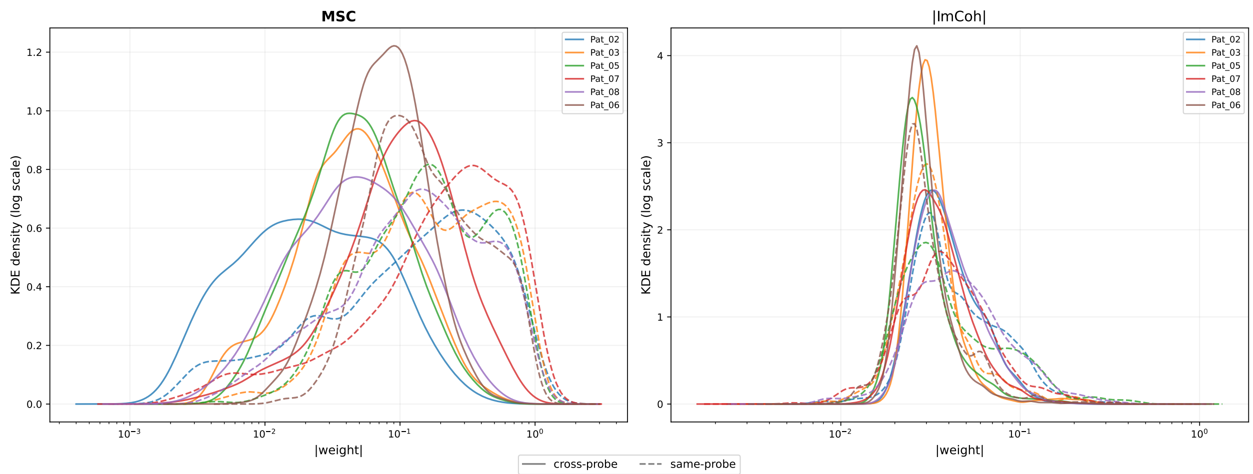


Figure 4: Kernel density estimates of β-band edge weights for same-shaft (dashed) and cross-shaft (solid) pairs, rsPre phase, all six patients overlaid by color. Left: MSC. Right: |ImCoh|. Under MSC same-shaft distributions are consistently shifted to higher weights in every patient. Under |ImCoh| the two distributions overlap, and in Pat05 and Pat06 the same-shaft distribution lies below the cross-shaft one, reflecting physical suppression of within-shaft zero-lag components.

weight scale. More consequentially for the LRG, the spectral gap λ_2 is sharper and more phase-stable under $|\mathrm{ImCoh}|$ in the β and θ bands, implying a more clearly defined resolution window $[\tau', \tau^*] = [1/\lambda_{\max}, 1/\lambda_{\mathrm{gap}}]$. The companion Fig. 10 confirms this: the $|\mathrm{ImCoh}|$ susceptibility peak $\tilde{C}(\tau)$ is sharper, of higher amplitude ($\tilde{C}^* \approx 2.5$ vs. ≈ 0.4 under MSC), and shifted approximately one decade to larger τ. These spectral differences imply that the two estimators yield qualitatively different LRG hierarchies, not merely rescaled versions of each other.

Section summary. The $|\mathrm{ImCoh}|$ substrate replaces the MSC substrate of $\textsc{Notes}_{\mathrm{MSC}}$ across every patient, phase, and band. The same-shaft enrichment of MSC (cohort mean 2.59×, rising to $\sim 6\times$ at $n = 30$ communities) is suppressed exactly to $\sim 1\times$ under $|\mathrm{ImCoh}|$, and the network-level consequences—force-directed layouts, edge-weight distributions, and Laplacian spectra—confirm that the artifact-corrected substrate is structurally distinct rather than rescaled. All subsequent analyses operate on the $|\mathrm{ImCoh}|$ matrices defined in Eq. (3).

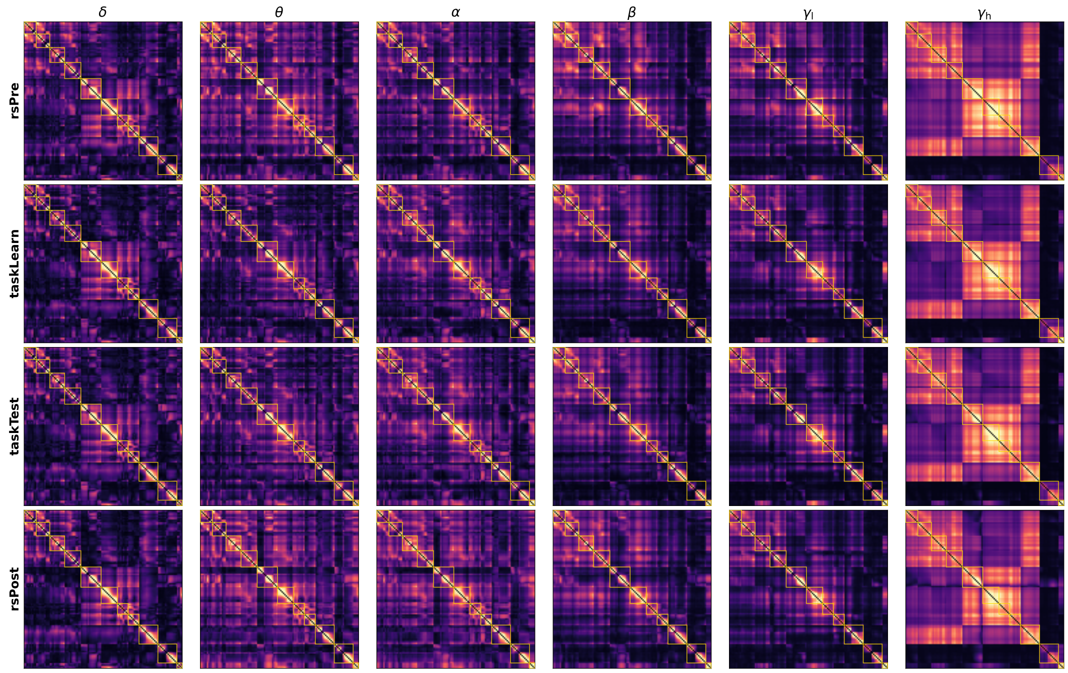
3 LRG Pipeline, Community Detection, and Comparison Framework

This section serves as a companion to $\textsc{Notes}_{\mathrm{MSC}}$ § 3–§ 5, which contain the full derivations of the LRG framework, the multiscale community detection pipeline, and the inter-phase comparison methodology. No derivations are repeated here. What this section adds is: (i) a minimal self-contained recap of the key objects needed to read the results; (ii) a direct MSC vs $|\mathrm{ImCoh}|$ comparison of LRG outputs showing that the structural change is fundamental; and (iii) a revisited metric concordance analysis under $|\mathrm{ImCoh}|$.

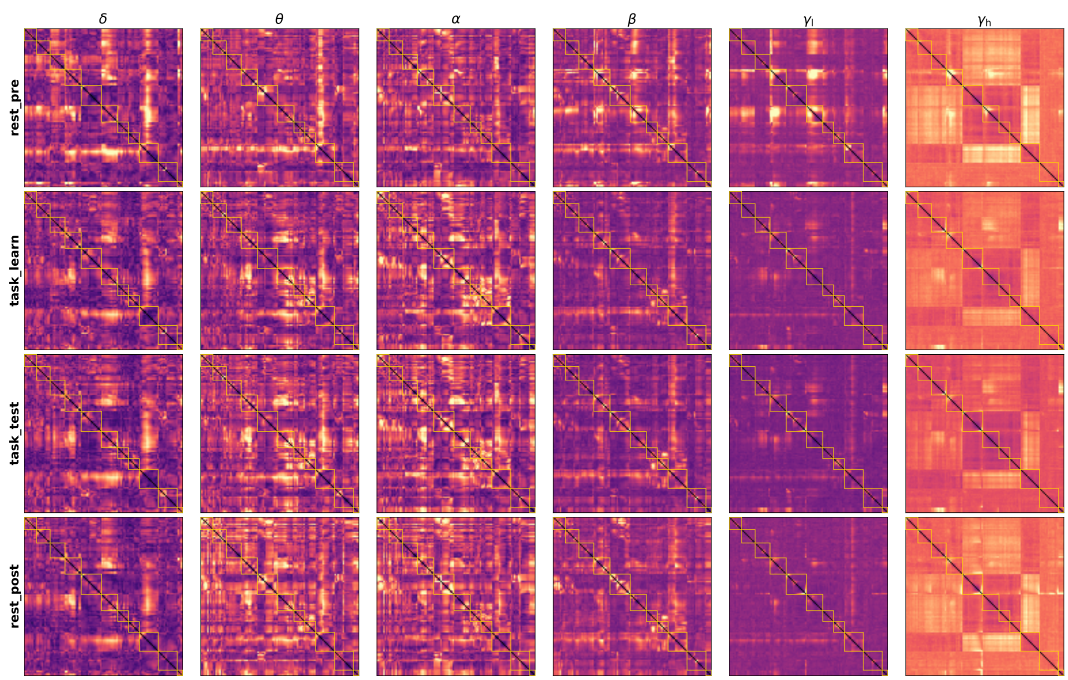
3.1 Key objects: a self-contained recap

For each (patient, phase, band) triplet, the $|\mathrm{ImCoh}|$ adjacency matrix $W(\mathcal{B})$ constructed in § 2 defines a weighted undirected graph. The combinatorial Laplacian $L = D - W(\mathcal{B})$ governs a continuous-time diffusion process on the network, parameterized by a diffusion time $\tau > 0$ that acts as a resolution parameter. The Laplacian density matrix

$$\hat{\rho}(\tau) = \frac{e^{-\tau L}}{\mathrm{Tr}[e^{-\tau L}]} \tag{4}$$

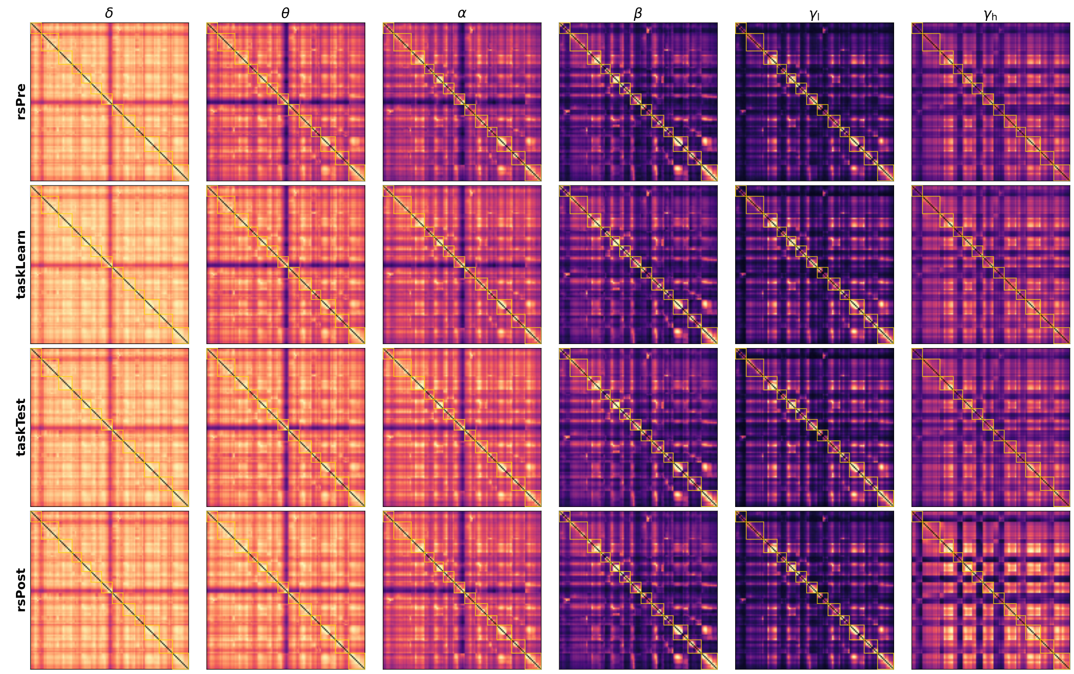


(a) MSC adjacency matrices, Pat02.

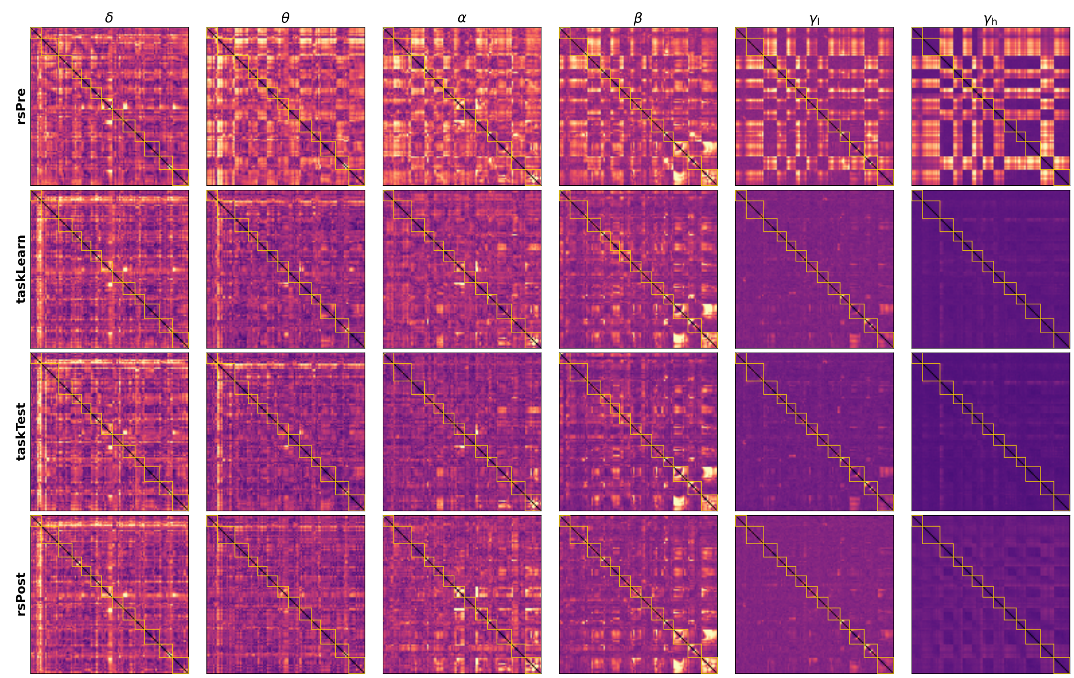


(b) $|\mathrm{ImCoh}|$ adjacency matrices, Pat02.

Figure 5: Band-resolved FC matrices $W(\mathcal{B})$ for Pat02 ($N = 117$ channels), phases (rows) by frequency bands (columns). Yellow boxes outline same-shaft diagonal blocks. Top: MSC, colorscale $[0, 0.6]$. Bottom: $|\mathrm{ImCoh}|$, colorscale $[0, \max]$ per band. Under MSC same-shaft blocks dominate and suppress off-diagonal structure. Under $|\mathrm{ImCoh}|$ same-shaft blocks are no longer elevated and cross-shaft functional structure becomes visible, including band-specific and phase-dependent patterns.



(a) MSC adjacency matrices, Pat05.



(b) $|\mathrm{ImCoh}|$ adjacency matrices, Pat05.

Figure 6: Same as Fig. 5 for Pat05 ($N = 118$ channels). The implant geometry differs substantially from Pat02 and the off-diagonal coupling structure revealed by $|\mathrm{ImCoh}|$ is correspondingly different, confirming that the underlying functional FC architecture is patient-specific. The same-shaft inflation under MSC is present in every band and phase, confirming its origin in the recording montage rather than neural activity.

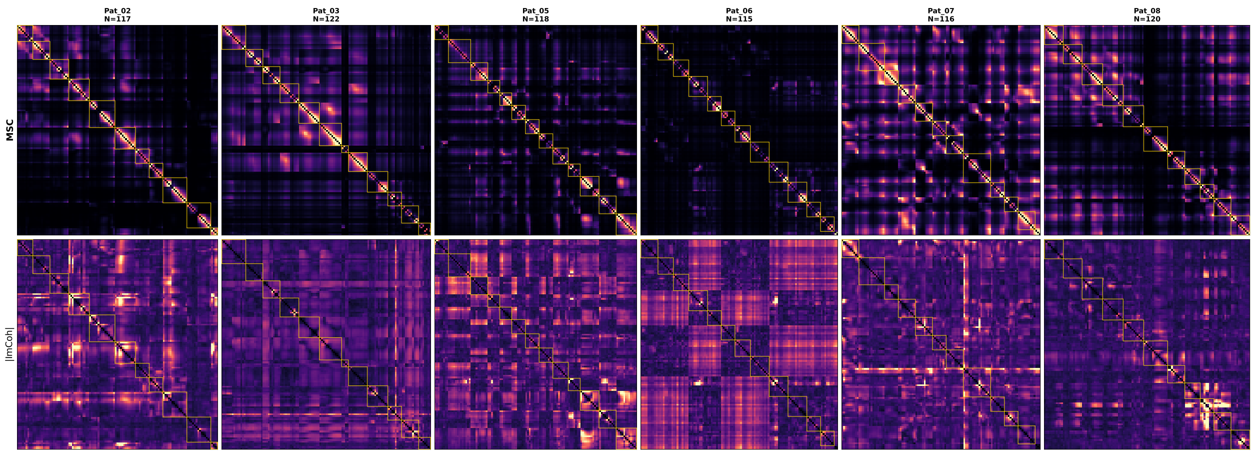


Figure 7: Band-resolved FC matrices for all six patients (β band, rsPre). Top row: MSC; bottom row: $|\mathrm{ImCoh}|$. Each matrix is scaled to its own maximum; patient channel counts N are indicated above each column. Yellow boxes outline same-shaft blocks. Same-shaft inflation is present in every MSC panel and absent in every $|\mathrm{ImCoh}|$ panel.

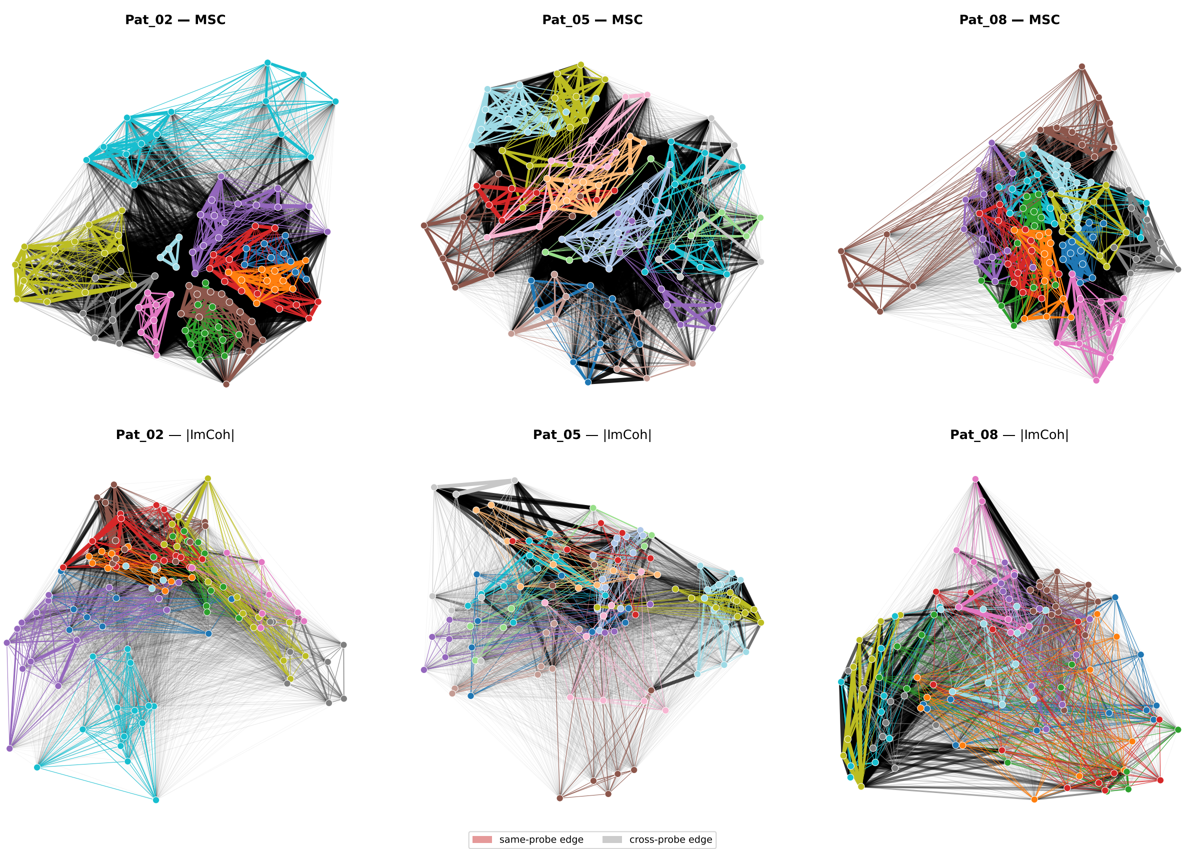


Figure 8: Force-directed network embeddings of the β-band FC network for Pat02, Pat05, and Pat08 (rsPre). The same layout algorithm and parameters are applied to both estimators. Top row: MSC; bottom row: $|\mathrm{ImCoh}|$. Same-shaft edges are colored by shaft identity; cross-shaft edges are black. Under MSC, thick colored same-shaft bundles dominate the layout and cross-shaft structure is suppressed. Under $|\mathrm{ImCoh}|$, same-shaft edges are sparse and the strongest connections cross shaft boundaries, reflecting genuine lagged cross-regional coupling.

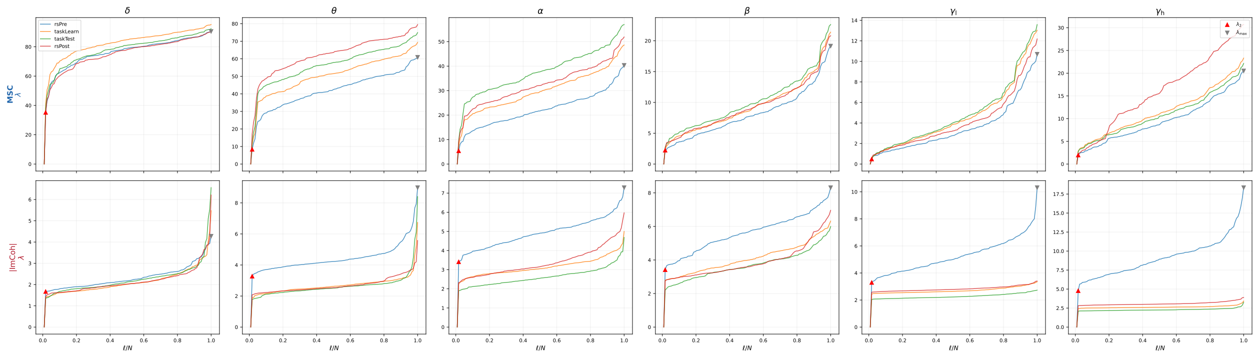


Figure 9: Sorted Laplacian eigenvalues λ_ℓ for Pat05 across all six bands (columns) and four phases (colors), for MSC (top row) and $|\mathrm{ImCoh}|$ (bottom row). Red triangles mark λ_2 (spectral gap); grey triangles mark $\lambda_{\max}$. The absolute scale is substantially lower under $|\mathrm{ImCoh}|$, and the phase-to-phase spread of the spectrum is reduced in the β and θ bands, indicating a more stable and interpretable LRG resolution window.

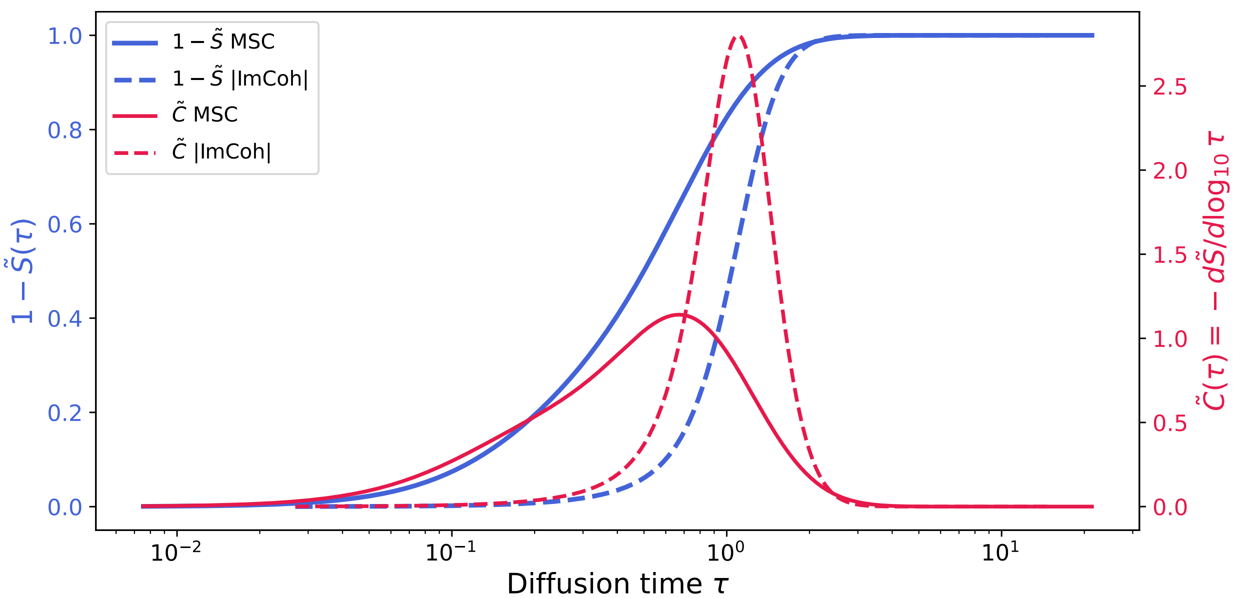


Figure 10: Normalized entropy $1-\tilde{S}(\tau)$ (blue, left axis) and entropic susceptibility $\tilde{C}(\tau)$ (red, right axis) for Pat08, α band, taskLearn, comparing MSC (solid) and $|\mathrm{ImCoh}|$ (dashed). The $|\mathrm{ImCoh}|$ susceptibility peak is sharper, higher ($\tilde{C}^* \approx 2.5$ vs. ≈ 0.4 under MSC), and shifted approximately one decade to the right in τ, reflecting the different weight distributions of the two estimators and the correspondingly different LRG resolution window.

encodes internode communicability at scale τ (cf. $\text{NOTES}_{\text{MSC}}$ Eq. (4)). Inverting its off-diagonal entries yields the communication distance

$$\mathcal{D}_{ij}(\tau) = \frac{1-\delta_{ij}}{K_{ij}(\tau)}, \qquad K(\tau) = e^{-\tau L}, \tag{5}$$

from which a dendrogram $\mathcal{T}(\tau)$ is obtained by average-linkage clustering ($\text{NOTES}_{\text{MSC}}$ § 4.2); the cophenetic distance induced on $\mathcal{T}(\tau)$ is ultrametric by construction, while $\mathcal{D}(\tau)$ itself is not. The analysis is anchored at $\tau' = 1/\lambda_{\max}$, the finest scale resolved by the propagator, where $\lambda_{\max}$ is the largest eigenvalue of L. Coarser scales $\tau \gg \tau'$ wash out the heterogeneity of $W(\mathcal{B})$ that carries the trace at the substrate level, as developed in § 5.1. The partition stability index

$$\Psi(n;\tau) = \mathcal{N}\left[\log_{10}\Delta_n(\tau) - \log_{10}\Delta_{n+1}(\tau)\right] \tag{6}$$

quantifies how much logarithmic room a partition into n communities has before the next merge event, with $\mathcal{N}$ a normalization constant (Notes$_{\text{MSC}}$ Eq. (8)). Inter-phase comparison uses the Variation of Information evaluated across all dendrogram cut levels $k = 2, \ldots, \lfloor N/2 \rfloor$, combined with a strict cross-patient unanimity criterion; these are defined in full in Notes$_{\text{MSC}}$ § 5.4.

3.2 MSC vs |ImCoh|: fundamentally different LRG structures

3.2.1 Dendrogram structure.

Figure 11 shows the normalized average-linkage dendrograms for three (patient, band) combinations under MSC (top row, log Δ axis) and |ImCoh| (bottom row, linear Δ axis). Leaves are colored by electrode shaft identity, and the horizontal dashed line marks the selected community cut.

Under MSC the branch structure is vertically stratified: long stretches of the height axis contain no merges, followed by bursts of merges clustering at distinct heights, producing a well-defined preferred scale at $n^* \approx 23$–24 communities. Inspecting the leaf coloring reveals that this preferred scale corresponds to the scale at which contacts from the same electrode shaft co-merge: the coarse-level bifurcations of the MSC dendrogram group contacts by probe geometry rather than by functional similarity. This is consistent with the enrichment results of § 2.2: at $n = 20$ the mean same-shaft enrichment under MSC reaches 5.5× (Fig. 12a). The previously reported dominant mesoscale of MSC-based LRG analyses was therefore an artifact of electrode shaft geometry, not a property of brain functional organization.

Under |ImCoh| the picture changes qualitatively. Merges are distributed quasi-uniformly along the height axis: branch lengths are short and no preferred scale dominates. This is the signature of a genuinely multi-scale system with comparable structure at many cut levels, rather than a single privileged mesoscale. The dendrogram does not collapse into a few large probe-following clusters but instead resolves a larger number of smaller communities ($n^* \approx 20$–22) whose leaf coloring crosses shaft boundaries. A methodological consequence follows directly from this flat landscape: the Ψ index, which was designed under the assumption that a meaningful stability peak exists, loses its standard interpretation under |ImCoh|. When applied without a cap on n, it selects $n^* \approx N$ (the degenerate partition into singletons), which is the algorithmic signature of a flat stability curve. The cuts shown in Fig. 11 are therefore conventional cuts chosen for visual comparability with MSC, not Ψ-optimal cuts in the strict sense. Any partition-level analysis under |ImCoh| should be reported as a function of n, not at a single selected n^*. This is precisely what the scale-resolved VI(k) framework of Notes$_{\text{MSC}}$ § 5.4 provides, and it is the reason that framework is adopted throughout the results rather than Ψ-based single-partition comparisons. The MSC dendrogram's apparent preferred scale at $n^* \approx 23$–24 is itself the algorithmic signature of the same-shaft bias: the spectral gap that drives the Ψ peak under MSC is generated by probe geometry, not by neural organization. Once the bias is removed by switching to |ImCoh|, the gap closes, and as we develop in § 5.1, Ψ no longer selects an interior scale anywhere across the cohort.

3.2.2 Brain-space community structure.

Figure 12a and Fig. 12b show the LRG community partitions projected onto the glass-brain (axial view) for Pat02, Pat03, and Pat05 at $n = 5$ and $n = 20$ communities, under MSC and |ImCoh| respectively. Under MSC at $n = 5$ the communities already show a pronounced same-shaft preference (mean enrichment 1.40×), and at $n = 20$ the enrichment reaches 4.21×: communities trace electrode trajectories spatially, with contacts on the same shaft grouped together regardless of their brain location. Under |ImCoh|, the same-shaft enrichment drops to 1.05× at $n = 5$ and 1.75× at $n = 20$. The residual |ImCoh| enrichment at $n = 20$ reflects genuine lagged short-range coupling between adjacent contacts on the same shaft, not artifact: since |ImCoh| is exactly zero for any zero-lag component, any nonzero value is physically attributable to true neural interactions. The community structure under |ImCoh| is correspondingly more anatomically distributed, with contacts from different shafts and different brain regions grouped together, and the pattern differs between patients in a way that reflects each implant's coverage geometry. Meaningful multi-community structure under |ImCoh| emerges only at $n = 20$ and above: at $n = 10$ the partition collapses to a single dominant community plus a few singletons, which is itself diagnostic of the flat hierarchical landscape described above.

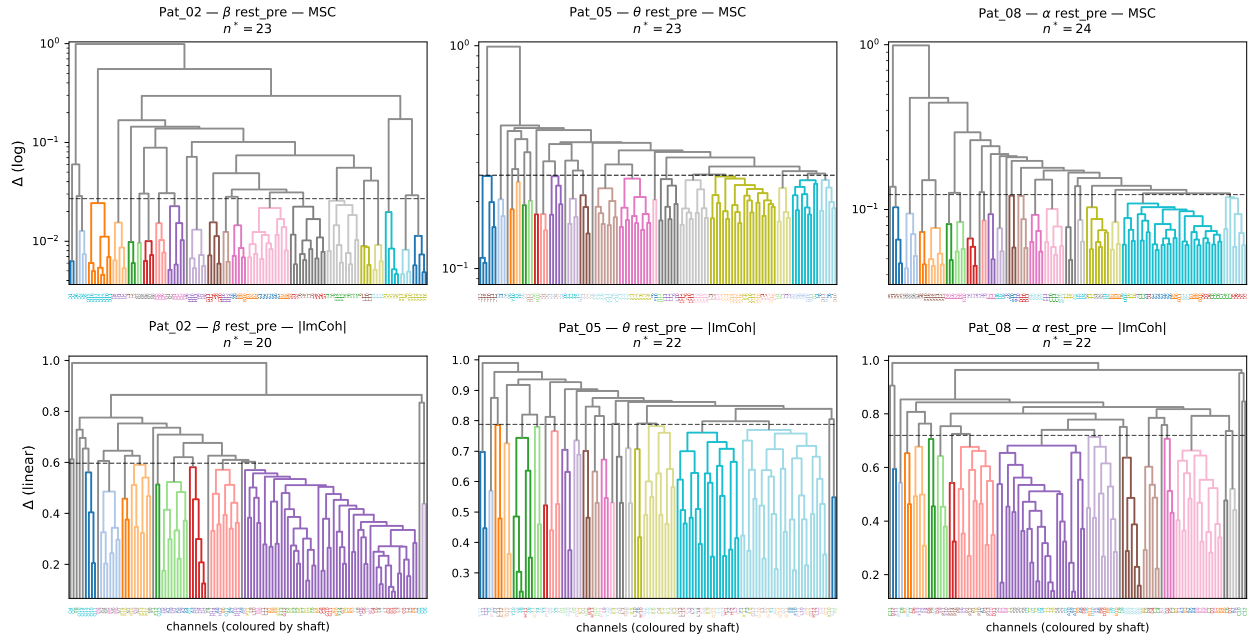
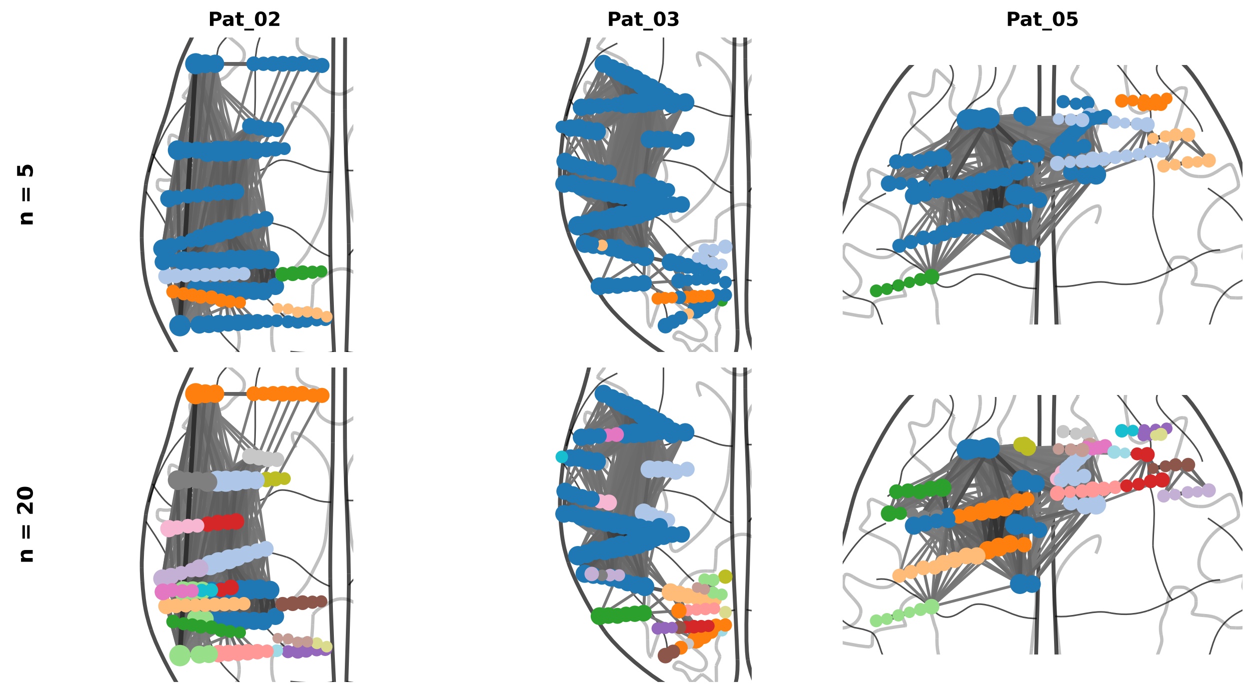
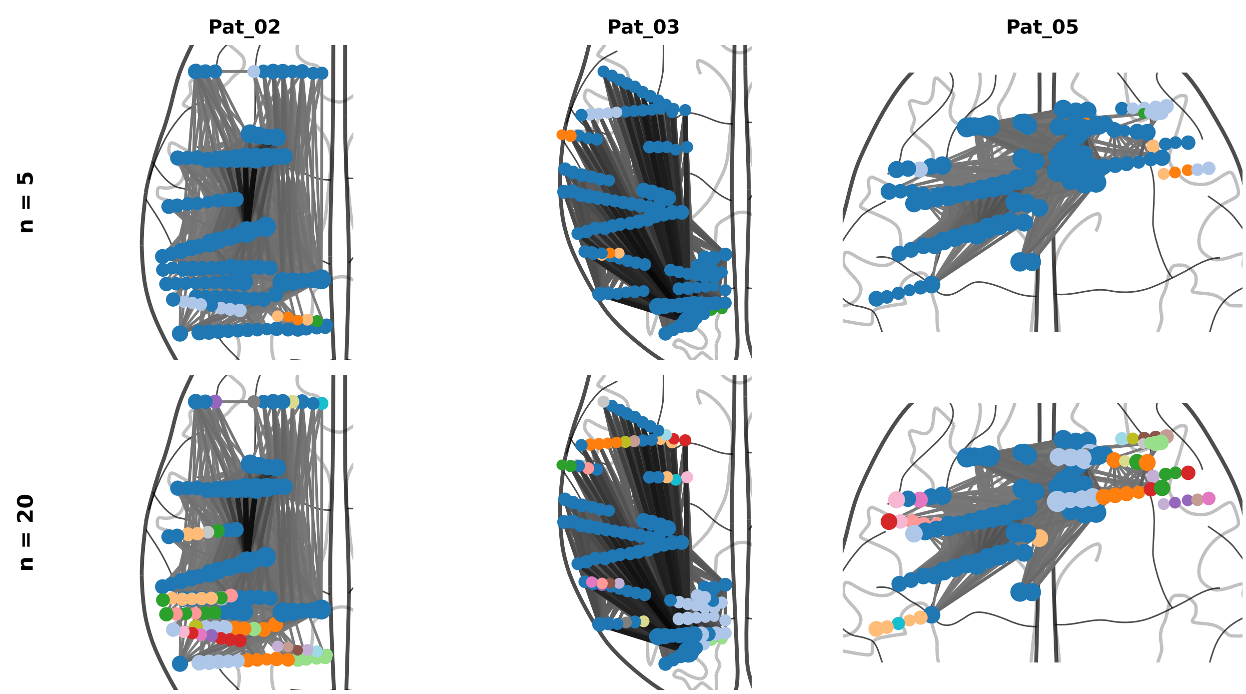


Figure 11: Normalized average-linkage dendrograms for three (patient, band) combinations: (Pat02, β), (Pat05, θ), (Pat08, α), all rsPre. Top row: MSC, Δ axis on a log scale; bottom row: $|\mathrm{ImCoh}|$, Δ axis on a linear scale. Leaves are colored by electrode shaft identity. Dashed horizontal lines mark the selected community cut. Under MSC, coarse branches follow shaft membership and the stability index yields a clean $n^* \approx 23$–24. Under $|\mathrm{ImCoh}|$, merges are distributed uniformly and community cuts cross shaft boundaries; the n^* values shown (20–22) are conventional cuts chosen for visual comparability, not Ψ-optimal cuts (see text).

Section summary. The $|\mathrm{ImCoh}|$-based LRG pipeline is structurally different from the MSC-based pipeline of $\textsc{Notes}_{\mathrm{MSC}}$, not a rescaling. The MSC dendrogram's apparent preferred scale $n^* \approx 23$–24 is the algorithmic signature of the same-shaft bias and dissolves once $|\mathrm{ImCoh}|$ is adopted. The resulting flat hierarchical landscape forces a methodological adaptation, developed in § 5.1, in which the load-bearing object becomes the communication distance $\mathcal{D}(\tau)$ rather than any partition derived from cuts of the dendrogram.



(a) MSC LRG community partitions projected onto the glass-brain (axial view) for Pat02, Pat03, and Pat05 (β band, rsPre). Top row: $n = 5$ communities; bottom row: $n = 20$. Node color encodes community assignment; node size encodes strength; arcs show the top $10\,\%$ strongest edges. Same-shaft enrichment at $n = 5$: mean $1.40\times$; at $n = 20$: mean $4.21\times$. Communities follow electrode shaft trajectories rather than anatomical organization.



(b) $|\mathrm{ImCoh}|$ LRG community partitions, same layout as Fig. 12a. Same-shaft enrichment at $n = 5$: mean $1.05\times$; at $n = 20$: mean $1.75\times$. The same-shaft bias is suppressed by approximately $2.4\times$ at $n = 20$ relative to MSC ($2.1\times$ averaged across all reported scales). Communities are anatomically distributed and patient-specific. Multi-community structure becomes informative only at $n \geq 20$, consistent with the flat hierarchical landscape of the $|\mathrm{ImCoh}|$ dendrogram.

4 Raw functional connectivity phase-distance audit

4.1 Why look at raw FC first

The LRG machinery developed in § 3 can unfold communication-flow structure that is invisible to direct inspection of the contact matrices: by progressively filtering Laplacian eigenmodes, the diffusion propagator $K(\tau)$ reveals patterns of internode communicability whose mesoscopic organization is not readable from the entries of $W(\mathcal{B})$ alone. This is precisely the analytical leverage we expect to exploit in the search for band-dependent task traces. Before invoking it, however, we ask a more basic question at the substrate level: do the |ImCoh| matrices themselves carry a band-dependent task trace across the four recording phases? This is the first step of the trace investigation. Establishing what is already visible at the edge level fixes a baseline against which any subsequent LRG-level result can be read as either confirming, sharpening, or surfacing structure beyond what the raw matrices show.

The audit presented in this section is deliberately conservative. We work directly on the upper-triangular entries of $W(\mathcal{B})$, without diffusion, without dendrograms, and without partition cuts, comparing phases pairwise with a small set of distances chosen so that their disagreements are diagnostic rather than redundant. The substrate audit and the subsequent LRG step are complementary: a band-dependent trace may live at the edge level, at the LRG level, or both, and the two views check each other.

The remainder of this section is organized as follows. § 4.2 fixes the substrate and the three distances used to compare phases, together with the trace scalar that summarizes the phase triangle and the within-baseline null that sets the noise floor. § 4.3 reports the four-phase geometry, which constitutes the strongest cohort-level positive control. § 4.4 gives the per-band trace verdict. § 4.5 presents two cross-checks that strengthen the trace reading against the alternative of monotone session drift. § 4.6 lists the deferred controls and frames what the audit does and does not establish.

4.2 Substrate, distances, and trace scalar

The substrate of the audit is the |ImCoh| adjacency matrix $W(\mathcal{B})$ constructed in § 2, taken without any further processing. For two phases ϕ, ϕ' at a fixed band, we extract the upper-triangular edge-weight vectors $a^{(\phi)} = \mathrm{vec}_{\triangle}\big[W^{(\phi)}(\mathcal{B})\big]$ and $a^{(\phi')} = \mathrm{vec}_{\triangle}\big[W^{(\phi')}(\mathcal{B})\big]$ and compare them through three distances chosen so that their disagreements are diagnostic.

The first is the Spearman rank distance,

$$d_S = 1 - \rho_S\big(a^{(\phi)}, a^{(\phi')}\big), \qquad\qquad \rho_S(x, y) = \rho_P\big(\mathrm{rk}(x), \mathrm{rk}(y)\big), \tag{7}$$

where $\mathrm{rk}(\cdot)$ returns the rank vector of its argument and ρ_P is the Pearson correlation coefficient defined below. Because d_S depends only on the ordering of edges, it is invariant under any monotone rescaling of the weights and shielded from uniform amplitude shifts that arise from arousal or impedance fluctuations. It is the load-bearing axis of the audit.

The second is the Pearson distance,

$$d_P = 1 - \rho_P\big(a^{(\phi)}, a^{(\phi')}\big), \qquad\qquad \rho_P(x, y) = \frac{\sum_e (x_e - \bar{x})(y_e - \bar{y})}{\sqrt{\sum_e (x_e - \bar{x})^2}\sqrt{\sum_e (y_e - \bar{y})^2}}, \tag{8}$$

where $\bar{x}$ and $\bar{y}$ denote the means of the two edge-weight vectors and the sums run over all upper-triangular edges. d_P is magnitude-weighted: it gives more leverage to high-coherence edges than d_S does. It serves as a convergence check on d_S rather than an independent axis, since the two are correlated by construction.

The third is the normalized Frobenius distance,

$$d_F = \frac{\big\|a^{(\phi)} - a^{(\phi')}\big\|_2}{\sqrt{\big\|a^{(\phi)}\big\|_2 \cdot \big\|a^{(\phi')}\big\|_2}}, \tag{9}$$

which is amplitude-only and most directly responsive to the kind of monotone session drift a structural trace must be distinguished from.

The four-phase rest–task–rest geometry is summarized through a single scalar per (patient, band, distance),

$$T_d = d(\text{taskTest}, \text{rsPost}) - d(\text{rsPre}, \text{taskTest}), \tag{10}$$

defined identically for each of the three distances. A negative T_d indicates that the post-task resting state sits closer to the task state than the pre-task resting state does, which is the directional signature of a *trace*: the cognitive episode left a structural mark that the network carries into the post-task rest. We emphasize that $T_d < 0$ is a directional pattern, not yet a hypothesis test: drift, baseline anomalies, and genuine task-induced reorganization can all produce it, and the work of disambiguating them is taken up in § 4.3 and § 4.5.

To establish a noise floor against which the cross-phase distances can be read, we build a within-rsPre split-half null per (patient, band) by dividing the baseline recording into 50 non-overlapping segment pairs, each pair contributing one within-baseline distance. The resulting empirical distribution defines a robust z-score for any observed cross-phase distance. The role of this null is screening only, rejecting the hypothesis that an observed distance is indistinguishable from within-baseline sampling jitter; it does not control for monotone session drift, which would inflate every cross-phase distance jointly with the time gap separating the phases, and the drift-specific controls are deferred to § 4.6. We do not use the z-scores from this null as a measure of how strong the trace is: the null is very narrow, so even small cross-phase changes produce very large z-scores. We therefore report distance ratios and absolute distance values throughout, and use the z-score only to check that the observed distance is larger than what the baseline alone produces.

We note here a methodological point that recurs across the figures of this section: Pat03 was recorded at 1024 Hz against 2048 Hz for the rest of the cohort. Rank-based and ratio-based quantities are dimensionless with respect to the sampling rate, so Pat03 is included in every cohort summary, but it is plotted with a distinct marker (orange triangle) to flag the acquisition asymmetry to the reader.

4.3 Four-phase geometry

Before reading the trace scalar, we examine the full four-phase distance geometry. The four phases yield six pairwise distances per (patient, band, distance class), of which the most informative are the within-task pair $d(\text{taskLearn}, \text{taskTest})$, the within-rest pair $d(\text{rsPre}, \text{rsPost})$, and the trace pair $d(\text{taskTest}, \text{rsPost})$. The relative ordering of these three is the strongest cohort-level positive control of the audit, and it does not require a hypothesis test to interpret.

Figure 12 shows the per-pair cohort distributions on the rank distance d_S, with each band sorted independently from smallest to largest cohort median to make the ordering immediately readable. The cohort pattern is consistent across all six bands: the within-task pair sits at the smallest cohort-median distance, the within-rest pair sits at or near the largest, and the trace pair occupies the position immediately above the within-task pair, closer to within-task than to within-rest. This sequence—within-task tightest, trace pair next, within-rest widest—is the geometric statement of the trace: rsPost sits in the rest–task continuum displaced toward the task state, not toward the pre-task baseline. The fact that the two task phases share a common architecture distinguishing them from any rest-containing pair, while the two resting states are not interchangeable, is itself a non-trivial structural statement at the substrate level. The same ordering serves as a positive control against the alternative that the apparent trace is an artifact of monotone session drift. Pure drift would inflate every cross-phase distance proportionally with the time gap separating the phases, and would in particular inflate the within-task and within-rest pairs equally, since both span comparable temporal extents within their respective phase blocks. The empirical asymmetry, with within-task systematically tighter than within-rest by a factor of approximately two on the cohort median in every band, is incompatible with a drift-only account: the two task phases hold together more tightly than the two rest phases do, which requires a structural rather than temporal explanation.

Figure 13 represents the same four-phase geometry as chord diagrams per band and per distance class, with phase positions on the baseline determined by a one-dimensional spring layout whose stiffnesses are inversely proportional to the squared distances. The redundancy-free representation confirms that the within-task tightest, within-rest widest ordering holds for d_P as well, and that d_F shows the same qualitative pattern with the expected drift-driven inflation of the long-gap pairs.

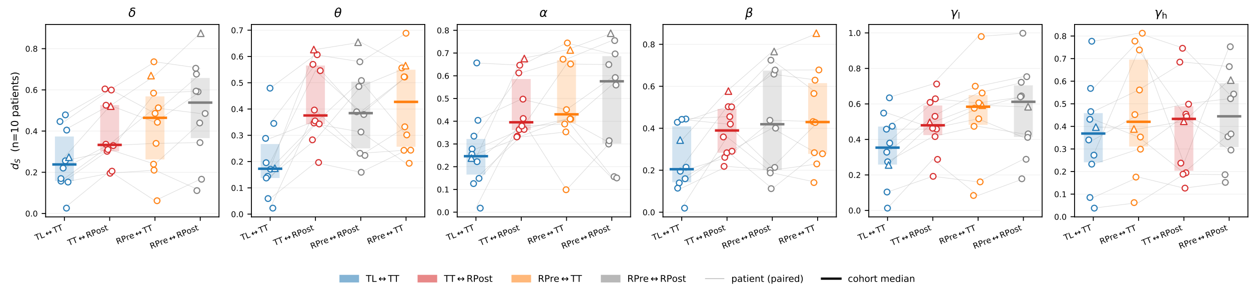


Figure 12: Four-phase geometry on the rank distance d_S, per band. Each panel shows the four reference phase pairs sorted from smallest to largest cohort median; per pair, the translucent box reports the interquartile range, the bold line the cohort median, and jittered dots the per-patient values, with thin gray lines linking each patient's four points across pair positions. Pair colors are fixed across bands: blue = within-task (taskLearn↔taskTest), red = trace pair (taskTest↔rsPost), orange = pre-to-task (rsPre↔taskTest), gray = within-rest (rsPre↔rsPost). In every band the within-task pair is the leftmost (tightest) and the within-rest pair sits at the right, with the trace pair typically in second position.

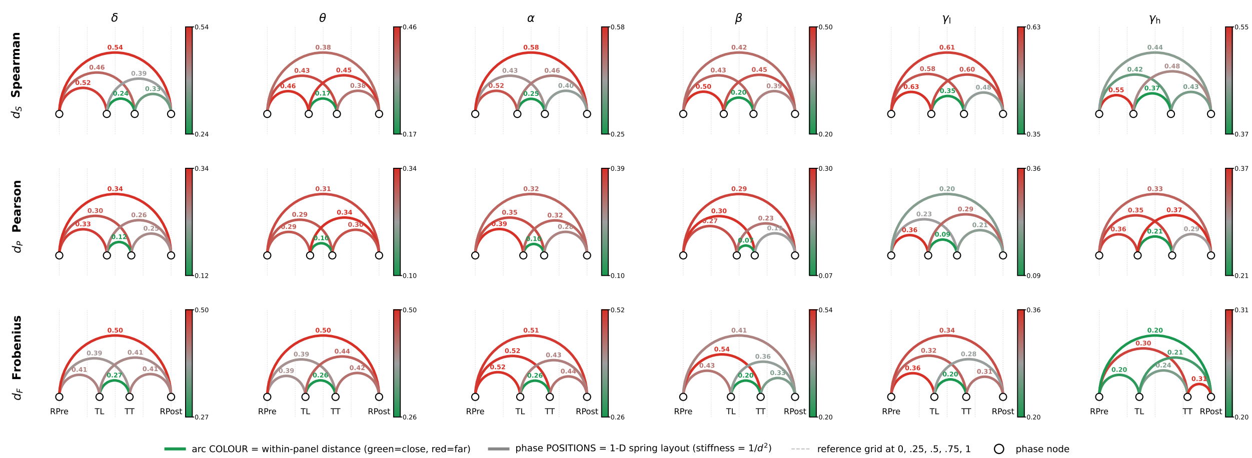


Figure 13: Chord-diagram representation of the four-phase distance geometry, per band (columns) and per distance class (rows: d_S, d_P, d_F). The four phase nodes sit on a baseline in temporal order; their horizontal positions are determined by a one-dimensional spring layout with stiffness $w_{ij} = 1/d_{ij}^2$, so closer pairs pull harder and the equilibrium positions visually encode the geometry. The six unique pair distances are drawn as semicircular arcs above the baseline; arc color reports the within-panel cohort-median distance on a green-to-red scale, with per-panel colorbars for local interpretation. Cohort-median values are annotated at each arc apex.

4.4 Per-band trace verdict

We now read the trace scalar T_d per band and per patient. The headline question is directional: in how many patients, and in which bands, does the post-task resting state sit closer to the task state than the pre-task resting state does? A negative T_d is the directional signature of a trace, with the geometric ordering of § 4.3 ruling out the pure-drift alternative as a cohort-level explanation.

Figure 14 shows the cohort layout on d_S directly, with one point per patient per band in the plane of $(d_S(\text{rsPre}, \text{taskTest}), d_S(\text{taskTest}, \text{rsPost}))$. Points below the identity line are exactly those with $T_d^{(d_S)} < 0$ and define the trace zone of the figure. In four of the six bands—δ, α, β, and low-γ—between six and eight of the ten patients fall in the trace zone, with α the strongest single cell at 8/10. The θ band shows only three patients in the trace zone, while high-γ is borderline at 5/10 with a marginally positive cohort-median $T_d^{(d_S)}$. The cohort means and one-sigma covariance ellipses, plotted on the same axes, place the cohort centroid below the identity line in the four trace bands and on or above it in θ and high-γ. Fig. 15 reports

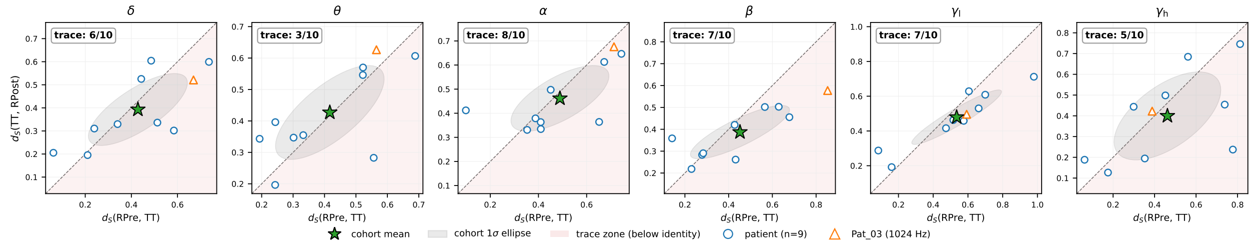


Figure 14: Trace scatter on the rank distance d_S, per band. Each panel shows one point per patient ($n = 10$; Pat03 marked as orange triangle as a sampling-rate methodological outlier) in the plane of $d_S(\text{rsPre}, \text{taskTest})$ versus $d_S(\text{taskTest}, \text{rsPost})$. Points below the dashed identity line fall in the red-shaded trace zone, where the post-task rest is closer to the task state than the pre-task rest is. The green star marks the cohort mean and the gray ellipse the cohort 1σ covariance ellipse. The trace count $n_{\text{trace}}/10$ is annotated per panel.

the per-band T_d distributions across all three distances on independent vertical axes, since the per-band variability in d_F reaches an order of magnitude larger than in d_S and d_P for some patients. Each per-patient point is colored green when $T_d < 0$ (trace direction) and red otherwise, with the cohort median annotated and the trace count badged at the top of each column. The cross-distance pattern is informative: for δ, α, β, and low-γ, the trace count on d_S is reproduced or strengthened on d_P and d_F, with β showing seven patients negative on d_S, seven on d_P, and eight on d_F. For θ, the trace count rises from three on d_S to six on d_F, the gap between rank-only and amplitude-only distances suggesting that θ carries an amplitude-driven cross-phase shift but no rank-level reorganization. The high-γ row shows extreme per-patient $T_d^{(d_F)}$ values, dominated by Pat06 and Pat05, that have to be read in light of the substrate's reduced edge-weight dynamic range in this band, as discussed in § 2.3. The per-band verdict, summarized by the trace counts on d_S and

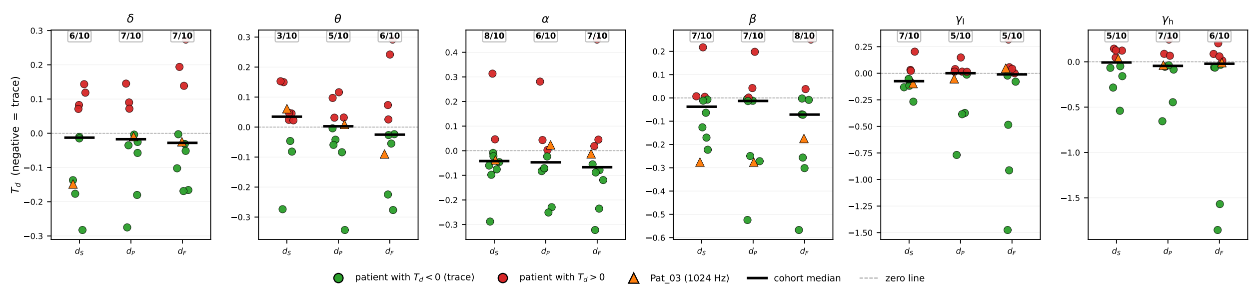


Figure 15: Per-band T_d distributions for all three distances, with independent vertical axes per band to accommodate the order-of-magnitude differences in d_F variability. Each dot is one patient, green for $T_d < 0$ (trace direction) and red otherwise; Pat03 is marked as orange triangle. Black bars indicate cohort medians; the trace count $n_{\text{trace}}/10$ is annotated at the top of each column.

the cohort-median $T_d^{(d_S)}$, is the following. The α band yields the strongest single-cell trace count at eight patients out of ten, with a cohort-median $T_d^{(d_S)}$ of -0.042 that is internally consistent across the cohort distribution. The β band reaches seven patients on d_S with cohort-median -0.038, and is the convergence band on which all three distances agree, as developed in § 4.5. The low-γ band shows seven patients in the trace zone with the largest cohort-median magnitude on d_S of any band (-0.074). The δ band is directionally consistent at six patients out of ten but the cohort-median $T_d^{(d_S)}$ is small in magnitude (-0.013) relative to the per-patient spread, indicating a directional pattern that is dominated by between-patient heterogeneity rather than by a tight common shift. The θ band fails to reach the directional threshold on the rank distance and is read as drift-only. The high-γ band is borderline. The cohort is heterogeneous in implant geometry, with each patient sampling a different set of brain regions, so the cross-patient counts above are conservative: any band that produces consistent directional behavior across implants reflects a regularity that survives the geometric heterogeneity, while bands at or near 5/10 carry an indication worth investigating further rather than a clean null. We frame this collectively as a directionally consistent task-shaped pattern in four of six bands, not yet a corrected-significance result; the formal significance discussion is deferred to § 4.6.

Band	**Indication**	$n^{(d_S)}_{\text{trace}}$	$\widetilde{T}^{(d_S)}_d$	$n^{(d_P)}_{\text{trace}}$	$n^{(d_F)}_{\text{trace}}$
δ	trace	6/10	−0.013	7/10	7/10
θ	drift-only	3/10	+0.035	5/10	6/10
α	trace	8/10	−0.042	6/10	7/10
β	trace	7/10	−0.038	7/10	8/10
γ_{l}	trace	7/10	−0.074	5/10	5/10
γ_{h}	borderline	5/10	−0.007	7/10	6/10

Table 2: Per-band trace summary on the raw |ImCoh| substrate. n_{trace} is the number of patients (out of ten) with negative trace scalar T_d for each of the three distances; $\widetilde{T}^{(d_S)}_d$ is the cohort median of the per-patient trace scalar T_d on the rank distance. The chance level under no signal is 5/10, but with a cohort heterogeneous in implant geometry, even counts in the 5–6/10 range carry an indication that motivates further investigation rather than indicating the absence of structure. The *indication* column applies a deliberately coarse threshold rule on the rank distance only—"trace" for $n^{(d_S)}_{\text{trace}} \geq 6$, "drift-only" for $n^{(d_S)}_{\text{trace}} < 5$, "borderline" otherwise—intended as a reading aid rather than a hypothesis test, and we frame the corresponding bands as task-shaped indications rather than as established results.

4.5 Convergence and structural-vs-drift cross-check

The verdict of § 4.4 is directional but not yet diagnostic. Two cross-checks sharpen the reading: a within-distance-triplet convergence between d_S and d_P, which tests whether rank-only and magnitude-weighted views agree on the trace, and a cross-distance contrast between d_S and d_F, which tests whether the trace can be explained by pure amplitude drift.

Figure 16 shows the cohort layout on the trace scalar in the $(T^{(d_S)}_d, T^{(d_P)}_d)$ plane per band, with the lower-left quadrant identifying patients in the trace direction on both axes. The β band is the convergence cell, with seven patients in the trace zone on d_S and seven on d_P, the largest joint count of any band. The α and low-γ bands also show concurrent trace direction on the two axes, while θ sits near the origin on both. The cohort Spearman correlation between $T^{(d_S)}_d$ and $T^{(d_P)}_d$ is high in every band, in the range 0.77–0.95, confirming that the two distances are not orthogonal axes but co-confirming views: the value of d_P is in agreeing or disagreeing with d_S, not in spanning an independent dimension. Fig. 17 addresses the drift

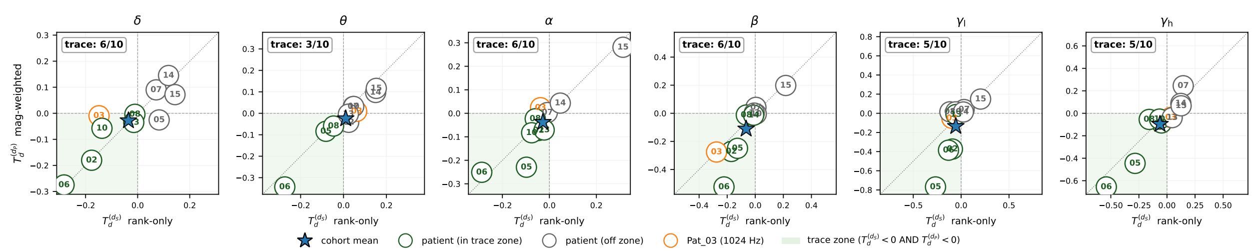


Figure 16: Convergence between rank-only $T^{(d_S)}_d$ and magnitude-weighted $T^{(d_P)}_d$, per band. Each numbered circle is one patient (last two digits of the identifier inside); circles with green edges fall in the green-shaded lower-left quadrant where both axes are in the trace direction, gray-edged otherwise; Pat03 carries an orange edge. The blue star marks the cohort mean and the dotted line is the identity. Trace counts are annotated per panel.

alternative directly. Pure monotone session drift would shift the absolute amplitudes of $W(\mathcal{B})$ coherently across phases, inflating d_F systematically with the time gap, while leaving the rank ordering of edges—and therefore d_S—unaffected. Under a drift-only account, the $(T^{(d_S)}_d, T^{(d_F)}_d)$ cohort cloud would therefore be uncorrelated, a circular blob centered at $T^{(d_S)}_d \approx 0$. The empirical pattern is the opposite. The cohort cloud is elongated along the identity line in every band, with per-patient Spearman correlations between the two axes in the range 0.78–0.92 and sign-agreement counts of seven to nine of ten patients. The trace moves coherently on rank and amplitude axes, which is incompatible with pure amplitude drift as the sole driver of the trace direction. The two cross-checks combined narrow the interpretive space for the trace pattern

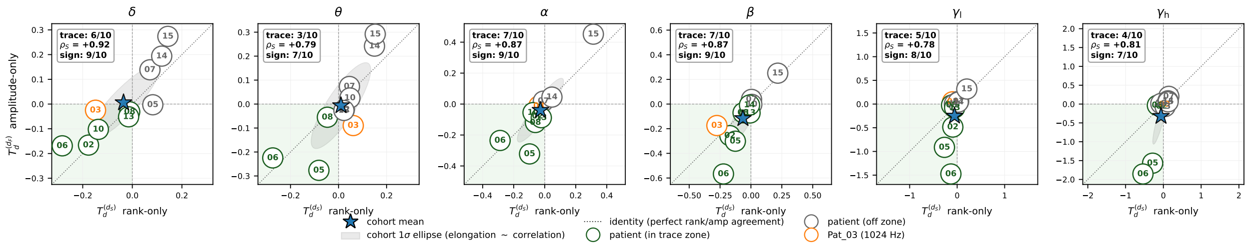


Figure 17: Structural-shift versus amplitude-drift contrast. Each numbered circle is one patient in the plane of rank-only $T_d^{(d_S)}$ and amplitude-only $T_d^{(d_F)}$ per band; green-edged circles fall in the trace zone (lower-left, both axes negative). The blue star marks the cohort mean and the gray ellipse the cohort 1σ covariance ellipse, whose elongation along the dotted identity line directly visualizes the rank-amplitude correlation. The per-band Spearman ρ_S between the two axes and the sign-agreement count are annotated per panel. Under a pure drift account the ellipse would be circular and centered at zero on the rank axis; the empirical elongation is incompatible with that account in every band.

of § 4.4: the rank-magnitude convergence shows the trace is not specific to a single distance class, and the rank-amplitude correlation rules out pure amplitude drift as a cohort-level explanation. What remains is taken up in § 4.6.

4.6 Caveats and deferred controls

The audit of § 4.3–§ 4.5 establishes a directionally consistent task-shaped pattern in four of six bands, with the geometric ordering and the rank-amplitude correlation jointly ruling out the pure-drift alternative as a cohort-level explanation. We now state explicitly what the audit does and does not establish, and which controls remain deferred.

Multiple-comparison-corrected significance is not reached. Per-cell one-sided Wilcoxon signed-rank tests on the cohort T_d vector, with alternative $T_d < 0$, yield a smallest uncorrected p-value of ≈ 0.04 for the $\beta \times d_F$ cell, and no cell survives Benjamini–Hochberg correction across the eighteen (band, distance) combinations: the smallest BH q-value is ≈ 0.29. The headline of this section is therefore framed throughout as a directional-consistency claim, not as a hypothesis test at $\alpha = 0.05$.

Amplitude-only distances are fragile to single patients. The cohort $T_d^{(d_F)}$ values are dominated by Pat06 in low-γ and high-γ, where its amplitude-only triangle asymmetries are an order of magnitude above the rest of the cohort, and the formal Wilcoxon significance on the d_F axis collapses when Pat06 is removed. The rank-distance trace counts on d_S are unaffected by the same removal, with six to eight of ten patients in the trace zone in the four trace bands either way. This is a methodological argument against reading direct metric-space distances such as the Frobenius distance as load-bearing on small heterogeneous cohorts: a single patient with an outlier amplitude scale can move the cohort scalar by an order of magnitude, while rank-based distances absorb it.

Drift controls. The within-rsPre split-half null of § 4.2 only checks that observed cross-phase distances are larger than the noise inside the baseline recording. It says nothing about the more dangerous alternatives: the four phases are recorded back-to-back in a single session, so a slow drift in signal quality could inflate every cross-phase distance for non-task reasons, and any peculiarity of the post-task rest could make it look different from the pre-task rest in ways that have nothing to do with the task. Three additional controls address these directly.

Control 1 — is there drift inside each rest phase? We split each rest phase into four consecutive chunks, compute all within-rest distances, and fit a linear model of distance against temporal gap. A large slope and a high R^2 would mean within-rest distances grow with time gap, which is the hallmark of monotone drift. The cohort-median R^2 on d_S sits at or below 0.10 in twelve of eighteen (band, distance) cells; the largest single value is 0.20 for α. The drift-only prediction also overshoots the actually observed cross-phase distances by factors of 1.4–2.1× on the rank distance (cohort medians on d_S: $\widetilde{d}_S(rsPre, rsPost)$ sits at 48–71% of the drift-model prediction), so even where a small drift slope exists, it does not account for the inter-phase distances we report.

Control 2 — is post-rest as stable as pre-rest? We mirror the within-rsPre split-half null on rsPost and compare the two null widths. If rsPost were unstable or had a transient post-task anomaly, its within-phase distances would scatter much more widely. The ratio of the two null widths, measured by their median absolute deviation, lies between 1.1 and 1.3 in every trace band; no cell crosses the 2× threshold. The two rest phases are equally stationary at the relevant scale.

Control 3 — is the late-task to early-post distance shorter than the late-pre to early-post distance? The cleanest test of task specificity is to compare two distances that span the same temporal gap and use the same amount of data per side. We take the late half of rsPre and the early half of rsPost (a long-gap baseline-to-baseline pair with no task between the halves), and compare it to the late half of taskTest and the early half of rsPost (the corresponding task-to-rest pair). If the network's post-rest configuration was reshaped by the task, the second distance should be shorter than the first; if it was not, the two should be similar. The directional pass rate on d_S is seven out of ten patients in α, β, and low-γ; six out of ten in δ; and at the binomial five-out-of-ten null in θ and high-γ. The two null bands recover the same "no signal" verdict that the audit had already given them, while the four trace bands either pass the stricter task-specificity bar (α, β, low-γ) or sit one short (δ).

Band	drift R^2 (med)	rsPost/rsPre MAD	xb sym $n/10$	**post-controls indication**
δ	0.05	1.09	6/10	soft trace
θ	0.02	1.31	5/10 (null)	drift-only
α	0.20	1.20	7/10	trace (1 caveat: drift $R^2 = 0.20$)
β	0.09	1.24	7/10	trace, controls pass cleanly
γ_l	0.06	1.22	7/10	trace, controls pass cleanly
γ_h	0.06	1.21	5/10 (null)	borderline

Table 3: Per-band drift-control summary on d_S. *Drift* R^2 is the cohort-median coefficient of determination of the linear within-baseline drift fit (Control 1). *rsPost / rsPre MAD* is the cohort-median ratio of the within-rsPost split-half null width to the within-rsPre split-half null width (Control 2); ratios near unity indicate that rsPost is internally stationary at the same scale as rsPre. *xb sym* is the symmetric cross-baseline pass count (Control 3): the number of patients (out of ten) with $d_S(\text{taskTest.late}, \text{rsPost.early}) < d_S(\text{rsPre.late}, \text{rsPost.early})$, both sides on the same half-segment noise budget. The *post-controls indication* updates 2 with the additional information from the three controls.

Combining the three controls, the trace indications of β and low-γ survive without caveat; α carries a single caveat (a non-negligible but residual-overshooting drift component); and δ is read as a soft trace. 3 summarizes the controls per band, and Fig. 18 shows the symmetric cross-baseline scatter on d_S directly.

The substrate is band-dependent. The |ImCoh| edge-weight distribution narrows in the high-γ band relative to the lower-frequency bands, with consequences for both the dynamic range of the distances and their stability. The borderline status of high-γ in § 4.4 should be read with this in mind: the band carries less coupling weight to begin with, so the trace scalar is intrinsically noisier than in the lower bands. Cross-band comparisons should not be read as cross-band rankings of effect strength.

Distance-class disagreements are interpretively meaningful. Two example cells make the role of the three-distance triplet concrete. Fig. 19 shows in its top row the Pat14 low-γ (taskLearn↔rsPost) comparison, with low $d_P = 0.07$ and high $d_S = 0.66$: the scatter of upper-triangular entries is monotone but compressed

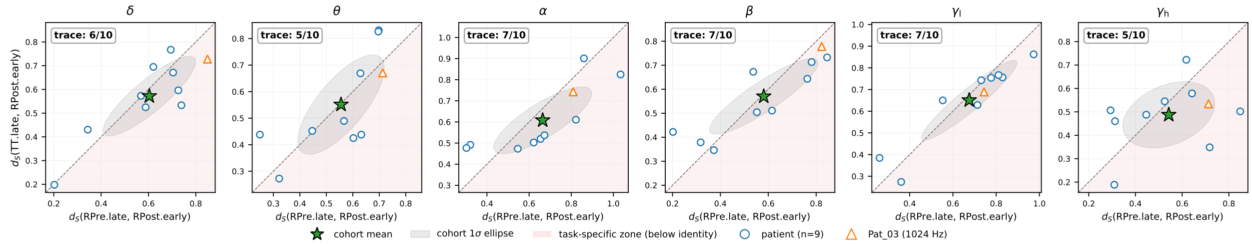


Figure 18: Symmetric cross-baseline scatter on the rank distance d_S, per band. Each point is one patient ($n = 10$; Pat03 marked as orange triangle) in the plane of $d_S(\text{rsPre.late}, \text{rsPost.early})$ on the horizontal axis versus $d_S(\text{taskTest.late}, \text{rsPost.early})$ on the vertical axis; both quantities use the same half-segment budget so the comparison is noise-symmetric. Points falling below the dashed identity lie in the task-specific zone, where the late-task to early-post-rest distance is shorter than the maximum-time-gap baseline-baseline distance with no task between the halves. The green star marks the cohort mean and the gray ellipse the cohort 1σ covariance ellipse. The trace count $n/10$ is annotated per panel. The α, β, and low-γ bands clear the $\geq 7/10$ task-specificity threshold; δ lands at $6/10$, matching its T_d count from § 4.4; θ and high-γ fall back to the binomial $5/10$ null, consistent with their drift-only and borderline indications.

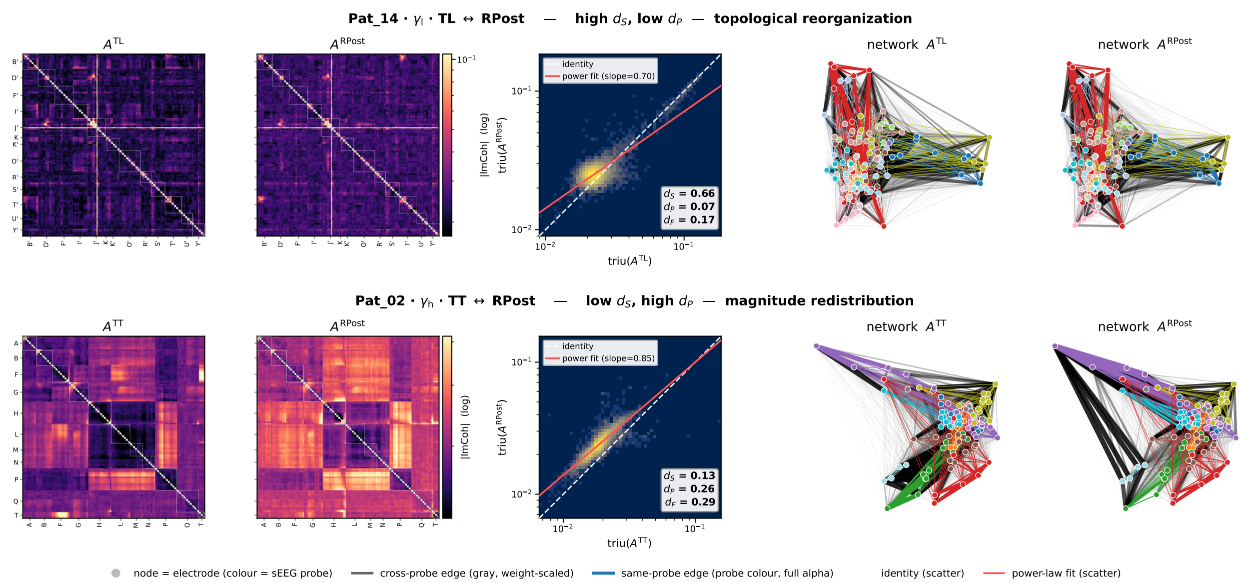


Figure 19: Two phase-pair comparisons illustrating how the three distances capture different aspects of $W(\mathcal{B})$ reorganization. Top row: Pat14, low-γ band, (taskLearn↔rsPost). Bottom row: Pat02, high-γ band, (taskTest↔rsPost). Columns 1–2 show the two adjacency matrices on a shared logarithmic colorscale; column 3 shows a log–log density-shaded scatter of the upper-triangular entries with a power-law fit overlaid (red) and the distance triplet inset; columns 4–5 show the two networks at a shared spring-layout embedding computed on the average matrix, with nodes colored by sEEG probe and same-probe edges drawn in the probe color.

in log–log, indicating topological reorganization without amplitude redistribution. The bottom row shows the Pat02 high-γ (taskTest↔rsPost) comparison, with low $d_S = 0.13$ and high $d_P = 0.26$: the scatter is rank-preserving but non-linear, indicating magnitude redistribution without topological reorganization. Disagreements between the three distances are therefore informative about which aspect of the matrix has changed, and motivate carrying the full distance triplet into the cohort-level analyses rather than collapsing onto a single scalar.

The audit motivates the LRG step. The audit establishes that the substrate carries a directionally consistent, band-dependent task-shaped pattern that is not reducible to monotone session drift at the cohort level. It does not establish corrected-significance persistence, and it does not localize the reorganization beyond the edge level. The LRG analysis presented in the next section reads the same rest–task–rest geometry through the diffusion propagator and the communication ultrametric, asking whether the trace acquires multiscale structure beyond what the substrate already shows.

Section summary. The substrate audit identifies β and low-γ as clean trace bands surviving all three drift controls, α as a trace band with a single drift caveat ($R^2 = 0.20$), δ as a soft trace, θ as drift-only, and high-γ as borderline. No (band, distance) cell survives Benjamini–Hochberg correction at $m = 18$ on the cohort one-sided Wilcoxon, but the four-phase geometric ordering (within-task tightest, within-rest widest) and the rank–amplitude correlation along the trace direction jointly rule out monotone session drift as the cohort-level explanation. The substrate verdict gates the LRG investigation of § 5: every band carried into the diffusion layer is read against the substrate-level indication established here.

5 Multiscale Laplacian phase-distance investigation

The substrate audit reads the displacement at the level of individual edge pairs in $W(\mathcal{B})$, where each entry is treated as an independent observation. This is conservative but blind by construction: the matrix is fully connected and weight-heterogeneous, and most of what makes the network a network — the structure of preferential information-flow paths between contacts, the loops the diffusion process traverses, the way edge-weight differences (rather than topological sparsification) shape communication — lives at the level of pairs of pairs and beyond, not at the level of single edges. The diffusion machinery of § 3 is the natural tool to read this higher-order structure: the propagator $K(\tau) = e^{-\tau L}$ integrates over all paths between two contacts weighted by the edge-weight heterogeneity of $W(\mathcal{B})$, and the communication distance $\mathcal{D}(\tau)$ it induces measures pairwise dissimilarity at the level of full information-flow geometry rather than at the level of direct edge weights. Reading the displacement through $\mathcal{D}(\tau)$ therefore asks a strictly stronger question than the substrate audit: does the cognitive episode reorganize not just which contact pairs are coupled but how information flows between them across the entire network? The investigation of this section is the falsification test for the corresponding equivalence at the diffusion layer, with the controls separating a genuine memory imprint in the communication geometry from shared-baseline correlation, within-session drift, and volume-conduction residue. The investigation proceeds at four geometric levels of the LRG output, each asking the same question of a different facet of $\mathcal{D}(\tau)$.

The per-pair correlation on $\mathcal{D}(\tau)$ under independent half-baselines (§ 5.3) is the primary controlled test: it asks whether task-induced and rest-induced distance shifts pull pairwise distances in the same direction relative to the baseline, and reports the only LRG probe with the full control battery (split-baseline, drift-floor, cross-probe) applied uniformly at every band. The dendrogram tree distance via the Kendall–Colijn family (§ 5.2) asks whether the imprint also restructures the hierarchical encoding of $\mathcal{D}(\tau)$; the partition-cut and spectral-subspace views (§ 5.4) ask whether it registers across resolution and spectral cutoffs; the anatomical localization (§ 5.5) asks where in the cortex the imprint lives.

The four levels are not independent statistical tests of the same null—they are derived from the same $\mathcal{D}(\tau)$—but they ask geometrically distinct questions and their convergence is a structural consistency check that the imprint is a coherent reorganization of the communication geometry rather than an artifact of any one probe.

Throughout this section the diffusion resolution is fixed at $\tau' = 1/\lambda_{\max}$, the finest scale resolved by the propagator. The choice is forced by the substrate: as developed in § 5.1, the |ImCoh| Laplacian spectrum is gap-free and the partition stability index Ψ does not select a privileged scale, so any analysis that requires picking such a scale is not well-grounded here. The methodological adaptation is documented in § 5.1 and the subsection that introduces it should be read as setup; the biological claim is carried by § 5.3 and § 5.5, with § 5.2 and § 5.4 reading the same imprint at additional geometric levels, and § 5.6 closing the section with a per-patient module-level illustration of how the imprint distributes across single subjects.

The headline result is band-resolved. The per-pair imprint of the cognitive episode survives every control at α, β, and low-γ ($q = 0.027$). The β imprint registers at every geometric level the analysis tests: per-pair correlation, tree topology, tree heights, partition cut, and spectral subspace. The low-γ imprint registers at the per-pair level and the heights of the dendrogram, with anatomical localization to left fusiform cortex ($p_{\text{hyper}} = 5.2 \times 10^{-6}$). The α imprint lives almost exclusively in the per-pair geometry: it does not restructure the dendrogram tree distance, but it shifts the per-pair communication distances coherently, controlled at $p = 0.007$. The δ and θ bands do not carry a controlled imprint; high-γ is borderline. The cohort claim is therefore a non-stationarity statement at three frequency bands: in α, β, and low-γ, the rsPost is not statistically equivalent to the rsPre and the four-phase paradigm is non-ergodic at the rest level in those bands.

5.1 LRG on a continuous spectrum: from partition selection to communication geometry

The standard LRG workflow recapped in § 3 proceeds in two stages: the propagator $\hat{\rho}(\tau)$ defines a one-parameter family of network states indexed by the resolution τ, and the partition stability index $\Psi(n;\tau)$ of Eq. (6) selects a privileged dendrogram cut at which substantive interpretation can take place. This procedure is well posed when the Laplacian spectrum has a clear separation between intra-module and inter-module modes, which produces gaps in the linkage merge-height profile and a sharp peak in Ψ at the modular scale. On the fully connected weight-heterogeneous $|\text{ImCoh}|$ substrate of § 2 this condition is not met, and the partition selection step does not apply across the cohort. The remainder of this subsection states the failure mode quantitatively and identifies the LRG objects that remain informative once partition selection is set aside.

Figure 20 shows the failure mode in a representative cell. The merge-height profile t_n (right panel, blue) decays smoothly and concavely from the root height $t_0 \approx 1$ down to the leaf-pair level $t_{N-1} \approx 0.08$, with no inflection. The associated $\Psi(n)$ profile (right panel, red) consists of two large spikes at the boundaries of the merge sequence and small fluctuations in between, with no dominant interior maximum. The Ψ-best cut on this cell sits at the leaf-pair extremum ($n = N - 3$; left panel, dashed red line), separating a single tight pair from the rest of the network — not a partition into modules in any useful sense. The mechanism

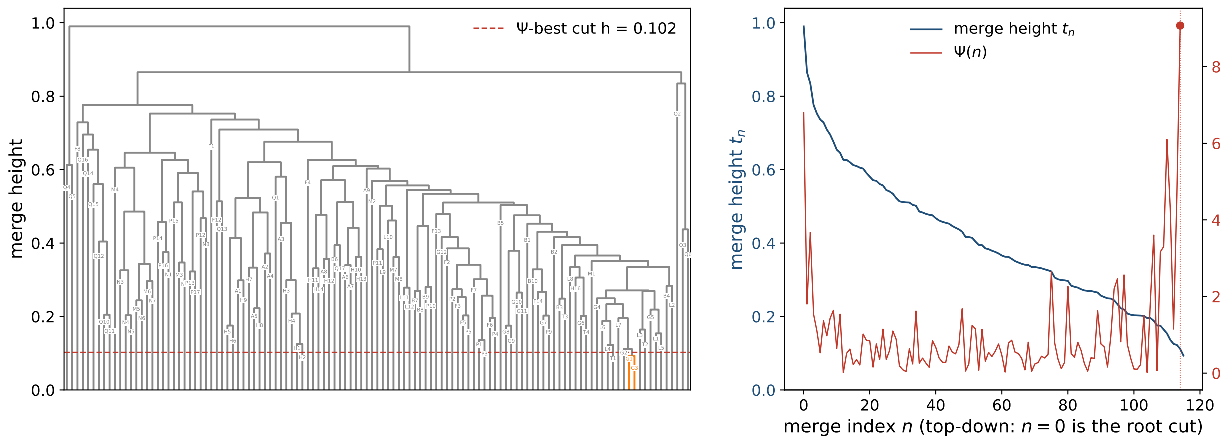


Figure 20: Ψ profile on the $|\text{ImCoh}|$ substrate. *Left:* dendrogram for Pat02 β rsPre with the Ψ-best cut $h = 0.102$ marked as a dashed red horizontal line, slicing off a single tight pair (orange) at the bottom of the tree. *Right:* the merge-height profile t_n (blue, log scale) decays smoothly from $t_0 \approx 1$ at the root cut to $t_{N-3} \approx 0.1$ at the leaf-pair cut without inflection, while the $\Psi(n)$ profile (red) peaks at $n = 0$ and $n = N - 3$ with no dominant interior maximum.

is a substrate property, not a pathology of the dendrogram. $\Psi(n) = \mathcal{N}[\log_{10} t_n - \log_{10} t_{n+1}]$ is a log-gap detector: it was designed in [10] to locate the order-of-magnitude jumps in merge height that mark real scale boundaries in topologically structured networks. In a network with well-separated communities, the propagator $\hat{\rho}_{ij}(\tau)$ between nodes on opposite sides of a community boundary decays exponentially with τ because no direct edge short-circuits the diffusion; the communication distance $\mathcal{D}_{ij}$ blows up across the boundary and stays small within a community, and successive merge heights inherit that amplification, with gaps spanning orders of magnitude. On the fully connected weight-heterogeneous $|\text{ImCoh}|$ substrate every contact pair carries a nonzero baseline coupling, so there are no edge-absence boundaries for the propagator to vanish across; $\mathcal{D}$ stays in a bounded finite range and t_n decays approximately linearly in n rather than log-linearly. Applied to a linearly spaced merge sequence, Ψ does not find scale boundaries because there are none of the topological kind — it instead amplifies the smallest-magnitude tail of the sequence (the leaf-pair merge), because $\Delta \log_{10} t$ is mechanically largest wherever t is smallest, even when Δt is tiny in absolute terms. The boundary-pinning of $\arg\max_n \Psi$ is therefore a generic property of Unweighted Pair Group Method with Arithmetic Mean (UPGMA) on a fully connected weight-heterogeneous graph, not a defect of any individual cell.

This is a cohort-level phenomenon and not a property of the example cell. Fig. 21 reports the distribution of the Ψ argmax position across all 180 (patient, band, phase) cohort cells (rsPre, taskTest, rsPost; taskLearn omitted as the trace investigation does not condition on it). Of the 180 cells, 68 have argmax exactly at the leaf-pair extremum ($n = N - 3$), 21 at the root extremum ($n = 0$), and 91 at strictly interior positions; of the 91 interior cells, only 2 have argmax more than ten merges away from the nearer boundary, with the remaining 89 sitting within ten merges of an extremum. The cohort piles at both ends with a near-empty middle. Bottom-pinning dominates top-pinning by roughly 3 : 1 (68 versus 21), the directional signature of Ψ acting as a log-gap detector that amplifies the smallest-magnitude tail of the merge sequence. The pinning strengthens monotonically with frequency: strict argmax-at-an-extremum counts run from 8/30 in δ to 21/30 in high-γ, with θ (19/30) and high-γ the most boundary-favoring bands and δ the most interior-favoring. Bottom dominance holds for every patient except Pat15, the only top-dominant patient in the cohort (8 top versus 5 bottom across his $(band, phase)$ cells). On this substrate the natural LRG object is

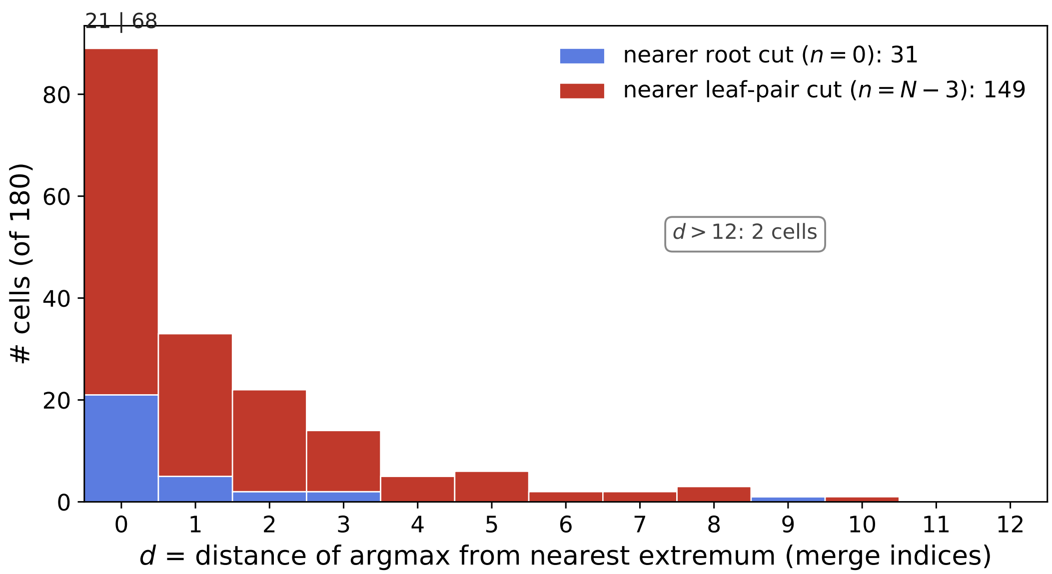


Figure 21: Cohort distribution of the Ψ argmax position folded to the distance $d = \min(n,\, N-3-n)$ from the nearer of the two Ψ extrema, across all 180 (patient, band, phase) cohort cells (rsPre, taskTest, rsPost; taskLearn omitted from this audit). Bars are stacked by which extremum the cell is nearer (blue: root cut at $n = 0$; red: leaf-pair cut at $n = N - 3$). 89 cells (49%) sit at $d = 0$; 144 (80%) within $d \leq 2$; 169 (94%) within $d \leq 5$; only 2 cells (1%) beyond $d = 12$. Bottom-side dominance is $\sim 5:1$ cohort-wide (149 leaf-pair-side versus 31 root-side) and $\sim 3:1$ at the strict $d = 0$ extremum (68 versus 21).

therefore not the single Ψ-selected partition of $\hat{\rho}(\tau)$ but the multiscale structure encoded by $\hat{\rho}(\tau)$ and its derived dendrogram — read as a continuous hierarchy rather than collapsed onto one privileged cut. $\mathcal{D}_{ij}(\tau)$ is large when communication between contacts i and j is hindered by the random walk on $W(\mathcal{B})$, small when it is facilitated, and the resulting matrix encodes a continuous geometry of preferential information-flow paths whose structure is carried by edge-weight heterogeneity rather than by any decomposition into discrete modules. The dendrogram remains useful, but as a hierarchical encoding of $\mathcal{D}(\tau)$ rather than as a partition selector.

The three subsections that follow read the trace at the three geometric levels this reframing identifies—the dendrogram and $\mathcal{D}(\tau)$ as a tree-and-matrix encoding (§ 5.2), $\mathcal{D}(\tau)$ as a continuous per-pair geometry under controls (§ 5.3), and the partition view as a multiscale profile (§ 5.4)—all at fixed $\tau = 1/\lambda_{\max}$ and none requiring a privileged dendrogram cut.

What follows. § 5.2 reads the imprint via the Kendall–Colijn tree distance on the dendrogram; § 5.3 reads the controlled per-pair test on $\mathcal{D}(\tau)$ directly.

5.2 Geometric decomposition: tree distance and matrix distance

If the cognitive episode imprints a memory on the communication geometry of $\mathcal{D}(\tau)$, the imprint should register in the dendrogram $\mathcal{T}(\tau)$ that hierarchically encodes $\mathcal{D}(\tau)$ and in the matrix-distance pattern between contact pairs. A natural probe family that side-steps the Ψ issue of § 5.1 reads the dendrogram $\mathcal{T}(\tau)$ without picking a privileged cut: the Kendall–Colijn (KC) tree distance [6] compares two dendrograms in a leaf-pair vector space whose entries can be tuned to weight tree topology versus ultrametric heights. The probe asks whether the same leaf-pairs share their most-recent common ancestor at the same relative depth and at the same height across phases. A complementary read of $\mathcal{D}(\tau)$ directly, via matrix distances on its upper-triangular entries, is deferred to § 5.3 where the controlled per-pair test on the same object is developed.

The KC family interpolates between two limits via a parameter $\lambda \in [0, 1]$. At $\lambda = 0$ the metric reduces to the L^2-distance between the two trees in the leaf-pair common-ancestor depth basis (pure topology), and at $\lambda = 1$ to the L^2-distance in the leaf-pair common-ancestor height basis (pure heights); intermediate λ values blend the two contributions after per-tree normalization. We use the two extremes only, so that the decomposition between topology and heights is clean. For each (patient, band) cell, the phase-triangle scalar at parameter λ is

$$T_{\mathrm{KC}}(\lambda) = d_{\mathrm{KC}}\big(\lambda;\, \mathcal{T}^{\mathrm{taskTest}},\, \mathcal{T}^{\mathrm{rsPost}}\big) - d_{\mathrm{KC}}\big(\lambda;\, \mathcal{T}^{\mathrm{rsPre}},\, \mathcal{T}^{\mathrm{taskTest}}\big). \tag{11}$$

$T_{\mathrm{KC}}(\lambda) < 0$ means the rsPost tree sits closer to the taskTest tree than the rsPre tree does, at the corresponding KC parameter; this is the imprint direction.

Figure 22 reads the cohort outcome on the $(\widetilde{T}_{\mathrm{KC}}(\lambda{=}0), \widetilde{T}_{\mathrm{KC}}(\lambda{=}1))$ plane, where the tilde denotes the cohort median over the ten patients and the sign convention is flipped throughout the figure so that the imprint direction is positive on both axes. Statistical significance is assessed by a one-sided Wilcoxon signed-rank test at each axis with null $H_0 : T_{\mathrm{KC}}(\lambda) \geq 0$ (no imprint or anti-imprint direction) against alternative $H_1 : T_{\mathrm{KC}}(\lambda) < 0$ (imprint direction); each band is a single point at its cohort-median position, with markers filled when $p \leq 0.05$ on at least one axis and hollow otherwise. The Wilcoxon signed-rank test ranks the magnitudes of T_{KC} in addition to their signs, so a smaller patient count at one axis can produce a smaller p-value than a larger patient count at the other axis when the per-patient $|T_{\mathrm{KC}}|$ values are individually larger; this is the case for β at $\lambda = 0$ ((7/10) patients but with larger per-patient magnitudes) versus β at $\lambda = 1$ ((8/10) patients with smaller magnitudes).

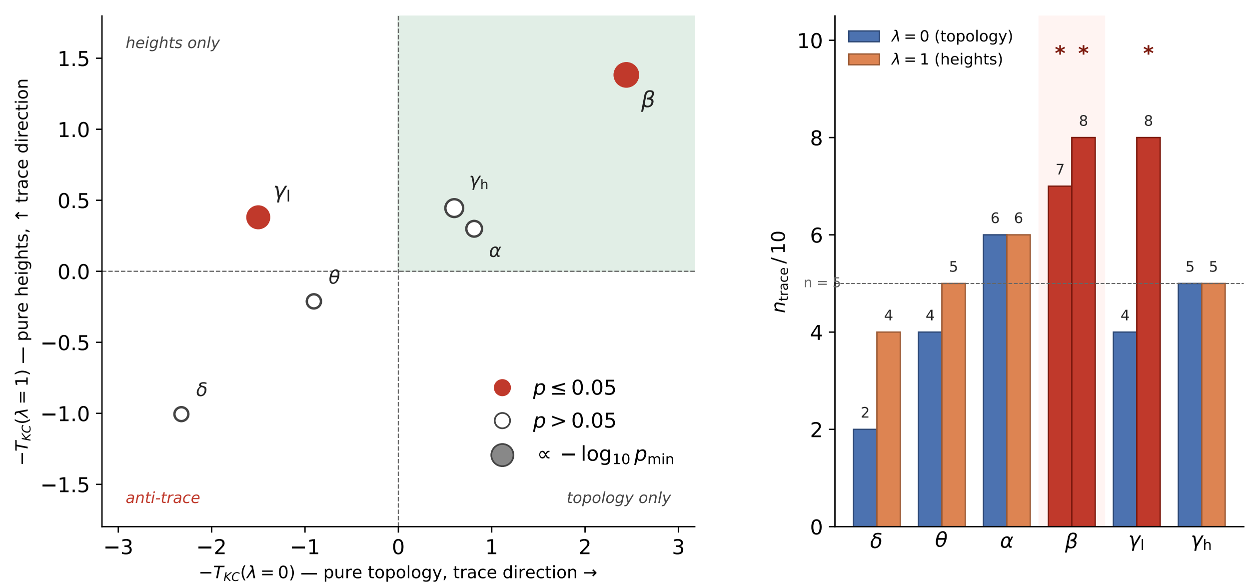


Figure 22: Geometric decomposition of the KC tree-distance imprint, per band. *Panel a:* cohort-median imprint direction at KC $\lambda = 0$ (pure topology, horizontal axis) versus $\lambda = 1$ (pure heights, vertical axis); positive values on both axes are the imprint direction. β sits in the upper-right quadrant: topology $p = 0.019$ (7/10 patients), heights $p = 0.042$ (8/10). low-γ sits in the upper-left quadrant: heights $p = 0.042$ (8/10), topology in the anti-imprint half-plane (4/10 patients). α sits slightly positive on both axes but neither axis reaches significance (6/10 patients on each). δ and θ sit in the lower-left, both axes anti-imprint at the cohort-median level. *Panel b:* per-band patient counts $n_{\mathrm{imprint}}/10$ on each axis. All p-values are one-sided Wilcoxon signed-rank, $H_1 : T_{\mathrm{KC}}(\lambda) < 0$.

The band-level reading of Fig. 22 is a structural decomposition. The β trace shifts both tree topology and tree heights coherently: 7/10 patients in the trace direction at $\lambda = 0$ ($p = 0.019$) and 8/10 at $\lambda = 1$

($p = 0.042$). The cleaner-by-magnitude $\lambda = 0$ cell carries cohort mean $|T_{\mathrm{KC}}|$ of 5.61 over its 7 imprint-direction patients against 1.02 over the 3 dissenters, a 5.5× ratio; at $\lambda = 1$ the corresponding ratio is only 1.6× (2.87 versus 1.83).

Both β cells survive a stronger control, the within-baseline-null controlled test introduced for the per-pair correlation in § 5.3 and applied here to the KC probe family in parallel: for each patient, rsPre is split into two independent halves and a baseline null distribution of $T_{\mathrm{KC}}(\lambda)$ is constructed by computing the same triangle scalar with the half-baselines used in place of the genuine task and rest legs; the test then asks whether the observed $T_{\mathrm{KC}}(\lambda)$ sits more in the trace direction than this drift-floor null does. The paired Wilcoxon over the ten patients at β gives $p = 0.001$ at both $\lambda = 0$ and $\lambda = 1$ under H_0: observed and null exchangeable, against H_1: observed in the trace direction; all ten patients fall below their own null, and Pat03 dropout (motivated by the 1024 Hz sampling-rate asymmetry, § 4.2) preserves the test at $p = 0.002$ on both axes, confirming that the signal does not depend on Pat03.

Multiple-comparison correction is applied across the KC probe family at $m = 12$ (band, λ) cells: six bands $\{\delta, \theta, \alpha, \beta, \gamma_l, \gamma_h\}$ crossed with two KC parameter values $\{\lambda = 0, \lambda = 1\}$. The Benjamini-Hochberg (BH) procedure controls the expected False Discovery Rate (FDR) among the rejected nulls at level q: a finding with corrected $p_{\mathrm{BH}} \leq q$ belongs to a set of rejections in which the expected proportion of false positives is at most q; we use $q = 0.05$ throughout.

Both β cells survive at $p_{\mathrm{BH}} = 0.006$. The cross-probe restriction (excluding same-probe leaf pairs from the KC computation) preserves the directional signal at 7/10 and 8/10 and the test direction at uncorrected $p < 0.05$ ($p = 0.024$ at $\lambda = 0$, $p = 0.042$ at $\lambda = 1$), with within-probe $p_{\mathrm{BH}} = 0.146$ and 0.168 respectively. Unlike under MSC, where same-probe pairs are dominated by volume-conduction artifact (§ 2.2), under |ImCoh| the same-probe contribution to the β signal is genuine short-range lagged coupling: at $\lambda = 0$ the same-probe pairs contribute a coherent cohort-mean imprint of $+0.205$ per patient (cohort standard deviation 0.195) in the trace direction, so the cross-probe restriction discards real signal rather than removing bias. The loss of corrected significance under restriction therefore reflects reduced power, not a same-probe-driven effect.

The cohort KC $\lambda = 0$ scalar of Fig. 22 integrates over all $N(N-1)/2$ leaf-pairs into one number per patient. Decomposing the same scalar into per-leaf, per-pair contributions surfaces which leaves and which partner sets carry the β topology trace at the dendrogram level. For each leaf i and each phase Φ, let $m_\Phi(i, j)$ denote the number of edges from the root to the most-recent common ancestor of (i, j) in $\mathcal{T}^\Phi$. A leaf-pair (i, j) is trace-positive when $|m_{taskTest} - m_{rsPost}| \leq 1$ and $|m_{rsPre} - m_{taskTest}| > 1$, with a subtree-size guard restricting the taskTest and rsPost MRCA subtrees to at most half of the contact set. The per-leaf score f_i is the fraction of i's partners satisfying this predicate; the focal leaf i^* is $\arg\max_i f_i$.

Figure 23 shows the per-leaf decomposition at three β trace patients and at one high-γ reference cell. At Pat07 β, focal leaf $i^* = 62$ carries $f^* = 0.32$ over $|P| = 37$ trace-positive partners: the red bundle traces a localized mid-tree subtree at near-identical depth in taskTest and rsPost, while the same partners sit on an opened-up path at very different depth in rsPre. Pat05 β (focal 66, $f^* = 0.26$, $|P| = 31$) and Pat10 β (focal 37, $f^* = 0.34$, $|P| = 38$) reproduce the same structural signature on different cortical territories, confirming that the β topology trace is reproducible across patients in the cohort and not a single-patient artifact. Pat14 high-γ at the same predicate carries $f^* = 0.01$ at $|P| = 1$: a single trace-positive pair across the entire $N(N-1)/2$ pair set, rendered as one thin red path to a lone partner. The contrast at fixed predicate and fixed cohort layer is direct, with 37/31/38 versus 1 trace-positive partners at the focal leaf.

The complementary direction $T^-(i, j) = (|m_{\mathrm{rsPre}} - m_{\mathrm{rsPost}}| \leq 1) \wedge (|m_{\mathrm{taskTest}} - m_{\mathrm{rsPre}}| > 1)$ reads the anti-trace direction (leaf-pairs whose rsPre and rsPost MRCA depths agree against an isolated taskTest); the cohort-scalar $T_{\mathrm{KC}}(\lambda = 0)$ of Fig. 22 takes the net of the two directions. The cohort-aggregate ratio T^+/T^- at the median over the ten patients is 2.89 at β, 0.90 at θ, 0.94 at δ, and 1.30 at α; β is the only band whose cohort ratio combines a large value with substantial T^+ and T^- counts across patients. θ reads as drift-only at the dendrogram level not because the per-leaf predicate fails to register pairs but because the two directions cancel within the same cohort: per-leaf θ figures would render coherent bundles in either direction at the per-patient level, with the cohort net vanishing. The per-leaf figures of Fig. 23 are therefore visually faithful to the cohort scalar at β where $T^+ \gg T^-$. The α and low-γ bands are silent and anti-trace at KC $\lambda = 0$ respectively, and do not produce coherent leaf-pair trace bundles at this geometric level; the α trace lives in the per-pair communication geometry developed in § 5.3, and the low-γ trace at the dendrogram lives in the heights axis discussed earlier in this subsection.

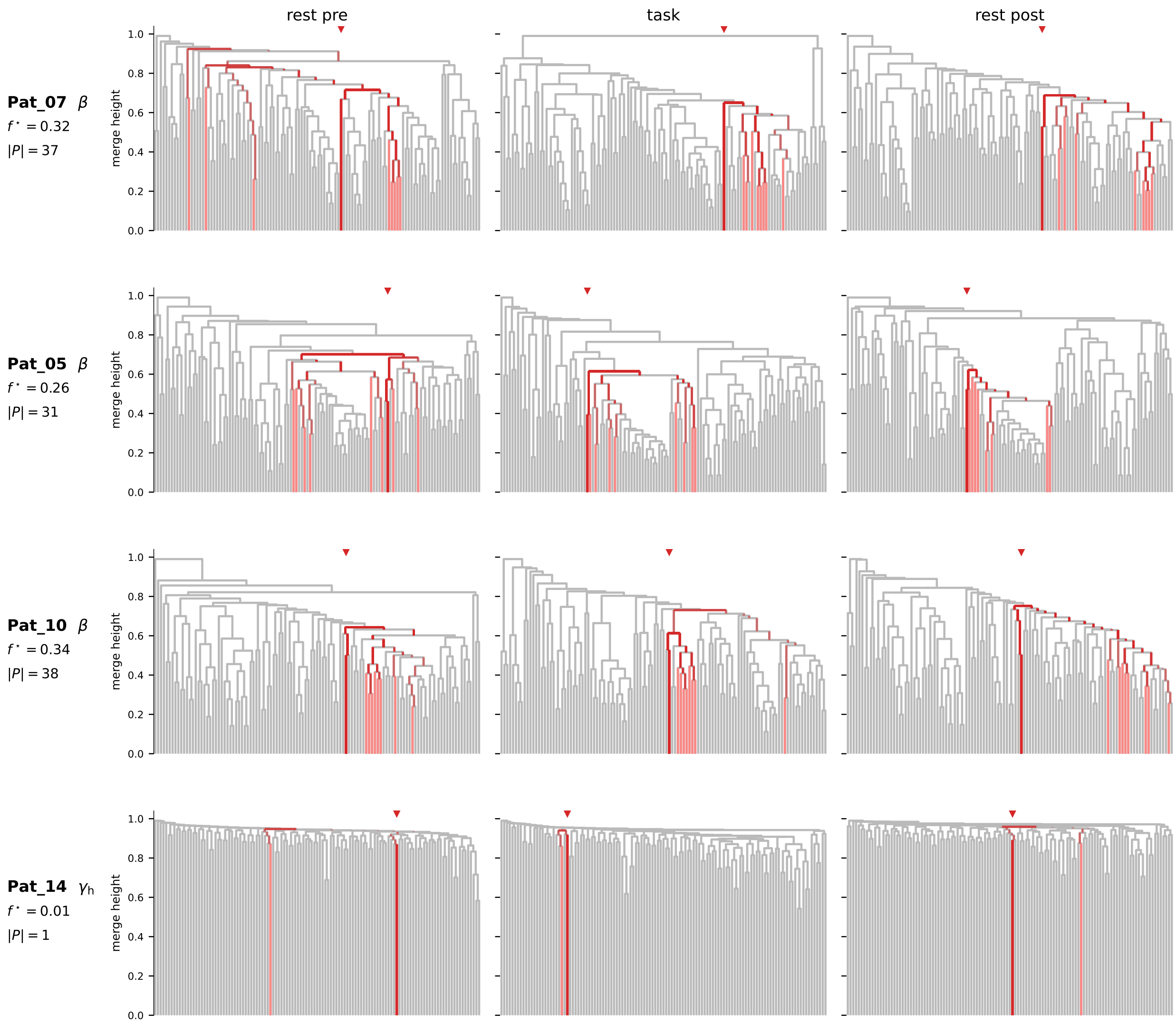


Figure 23: Per-leaf decomposition of the KC $\lambda = 0$ topology trace. Rows 1–3 show three β trace patients (Pat07, Pat05, Pat10); each row shows the dendrogram of $\mathcal{D}(\tau)$ at rsPre, taskTest, and rsPost, with the focal leaf i^* and its trace-positive partner bundle highlighted in red. The bundle traces a localized mid-tree subtree at near-identical MRCA depth in taskTest and rsPost and an opened-up path in rsPre. Per-row labels report i^*, f^*, and the trace-positive partner count $|P|$. Row 4 shows Pat14 high-γ, where the same predicate at the same partner cutoff returns one trace-positive pair across the entire $N(N-1)/2$ pair set ($f^* = 0.01$, $|P| = 1$), rendered as one thin red path to a lone partner near the top of the tree. The trace-positive predicate is $|m_{\text{taskTest}} - m_{\text{rsPost}}| \leq 1$ and $|m_{\text{rsPre}} - m_{\text{taskTest}}| > 1$ with a subtree-size guard at half of the contact set.

The low-γ band shows a partial decomposition: the heights axis is in the trace direction ($8/10$, $p = 0.042$) but the topology axis is in the anti-trace half-plane ($4/10$, $\widetilde{T}_{\text{KC}}(\lambda = 0) = -1.50$ on the unflipped scale where positive is anti-trace). The low-γ trace at the dendrogram level is therefore a shift in the ultrametric heights without a corresponding rearrangement of which leaf-pairs share their most-recent common ancestor. Under the within-baseline-null control the heights probe sits at uncorrected $p < 0.05$ at the within-probe $p_{\text{BH}} = 0.055$ (just above the $q \leq 0.05$ threshold). Under the cross-probe restriction the directional count moves to $9/10$ at $p = 0.024$, but the shift is driven by a single patient (Pat07) whose T_{KC} sign-flips by less than 0.2 at the noise floor of this cell, while the patients carrying the cohort signal in either direction (Pat05, Pat02 on the trace side; Pat15 on the anti-trace side) are unaffected by the restriction. The cohort is bimodal at the heights axis with one patient on the zero-crossing rather than the restriction acting as a denoising step, so we carry the heights cell as a directional companion finding rather than a corrected cohort claim.

The α band is silent on tree distance: both axes sit at the cohort-median trace side of zero in the figure but neither approaches significance (topology $p = 0.461$, heights $p = 0.500$, 6/10 patients on each axis against the 5/10 chance line). The α substrate trace of 8/10 patients in § 4.4 therefore does not crystallize into a tree-distance shift at the LRG layer, and the cohort signal is not a methodological artifact of the controlled KC test: under a band where the substrate audit gave a clear directional verdict, the controlled KC test sits at the chance line on both axes, so the β and low-γ signals reported above are not produced by the test machinery on null cohorts. The δ and θ bands sit in the anti-trace half-plane on both axes, with topology counts of 2/10 (δ) and 4/10 (θ); this is consistent with δ not surviving the LRG diffusion (the substrate δ trace was edge-only and the propagator washes it out, § 5.1) and with θ being drift-only across both substrate and LRG layers.

The KC probe family identifies β as the only band whose trace registers in both tree topology ($\lambda = 0$, within-probe BH-$q = 0.006$) and tree heights ($\lambda = 1$, $q = 0.006$) under the within-baseline-null control, with all ten patients below their split-baseline null on both axes. Low-γ registers in heights only, with directional support that does not pass within-probe correction ($q = 0.055$) and is carried as a directional companion. α is silent on the dendrogram tree distance, and δ and θ sit in the anti-trace half-plane on both axes. The geometric decomposition that emerges—β carrying both topology and heights, low-γ heights-only, α tree-silent—is the structural contribution of § 5.2; the per-pair test on $\mathcal{D}(\tau)$ of § 5.3 reads the same object directly without the dendrogram intermediate, and the band-resolved synthesis comparing the two reads is given at the end of § 5.3.

5.3 Controlled cohort trace via the per-pair correlation on $\mathcal{D}(\tau)$

The KC probe of § 5.2 reads the trace at the dendrogram level and identifies β as the multifacet band, with low-γ carried as a heights-axis directional companion. α, low-γ topology, and the dendrogram-level reading of the trace at the per-pair geometry of $\mathcal{D}(\tau)$ are not addressed by the KC family. We now apply the per-pair correlation test on $\mathcal{D}(\tau)$ directly under the full split-baseline, drift-floor, and cross-probe control battery; this is the load-bearing controlled cohort claim of § 5.

For each (patient, band) cell we split rsPre into two halves of equal duration, build the LRG on each half separately, and form the per-pair shifts $\Delta_{\text{task}}(i, j) = \mathcal{D}_{ij}^{\text{taskTest}}(\tau) - \mathcal{D}_{ij}^{\text{rsPre}_A}(\tau)$ and $\Delta_{\text{rest}}(i, j) = \mathcal{D}_{ij}^{\text{rsPost}}(\tau) - \mathcal{D}_{ij}^{\text{rsPre}_B}(\tau)$ on independent half-baselines. The construction eliminates by design the shared-baseline contribution that would correlate the two shift maps trivially when both reference the same rsPre build. The per-pair correlation

$$\rho_{\text{split}}(p, b) = \rho_S\big(\Delta_{\text{task}},\, \Delta_{\text{rest}}\big) \tag{12}$$

asks whether task-induced and rest-induced shifts agree in sign and rank across all $N(N-1)/2$ contact pairs; a positive ρ_{split} is the trace direction. The corresponding scalar in the T-convention of § 5 is $T_{\text{CTM}} = -\rho_{\text{split}}$.

Two further controls operate on the same per-(patient, band) outputs. The drift-floor null $\rho_{\text{drift}}(p, b) = \rho_S\big(\mathcal{D}^{\text{rsPre}_B} - \mathcal{D}^{\text{rsPre}_A},\, \mathcal{D}^{\text{rsPost}_B} - \mathcal{D}^{\text{rsPost}_A}\big)$ is the rest-only counterpart of ρ_{split}, built from rsPre and rsPost halves alone with no task data; on the same halved-data noise budget as ρ_{split} it captures the within-session drift component of the per-pair correlation. The cross-probe restriction $\rho_{\text{xprobe}}(p, b)$ is the same ρ_{split} evaluated on cross-probe pairs only, filtering out same-probe pairs whose ultrametric distances carry anatomy-dominated residual coupling unrelated to the cognitive episode. The trace direction at the cohort level requires that ρ_{split} sit above ρ_{drift} under a one-sided paired Wilcoxon and that ρ_{xprobe} preserve the cohort-median sign and patient count.

Figure 24 reads the cohort outcome on the controlled ρ_{split}. At α, cohort-median $\rho_{\text{split}} = +0.115$, eight of ten patients positive and the same eight above their drift floor, paired Wilcoxon $p = 0.007$; the cross-probe restriction preserves both the cohort-median sign and the patient count ($\rho_{\text{xprobe}} = +0.105$, eight of ten positive). At β, $\rho_{\text{split}} = +0.222$, $p = 0.014$, the same 8/10 patient agreement on both conditions. At low-γ, $\rho_{\text{split}} = +0.140$, $p = 0.010$: nine of ten patients above their drift floor and seven of ten with ρ_{split} positive in absolute terms, with two patients (Pat07 and Pat13) slightly negative on ρ_{split} but more negative on ρ_{drift} and therefore passing the controlled criterion while failing the absolute one. Under within-probe BH

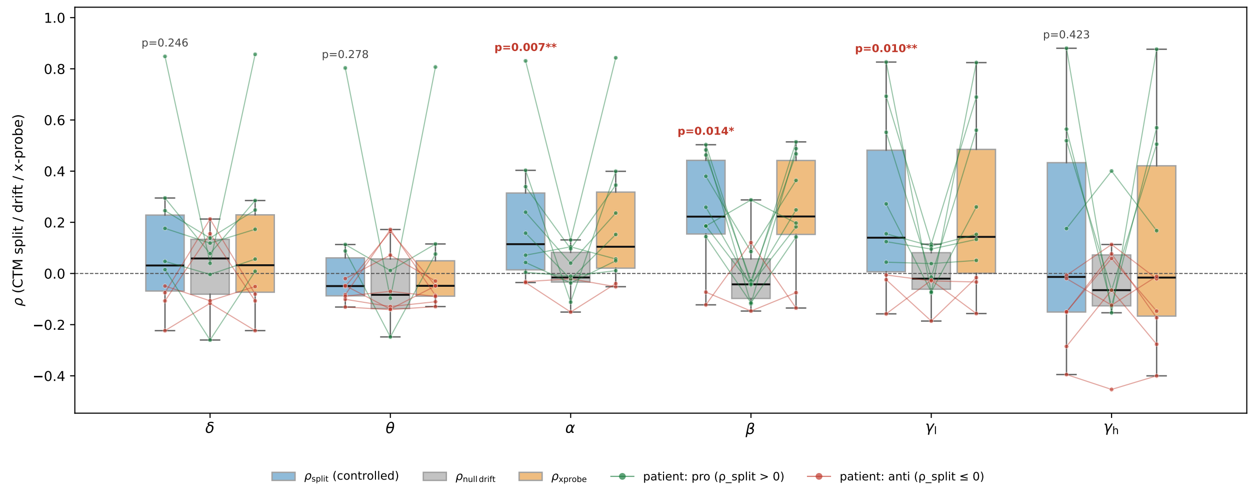


Figure 24: Controlled cohort ρ_{split} at the LRG ultrametric layer, per band. Each panel shows three boxplots: ρ_{split} (blue), ρ_{drift} (gray), and ρ_{xprobe} (orange). Per-patient lines connect the three values for each patient, colored green when $\rho_{\text{split}} > 0$ (trace direction) and red otherwise. One-sided paired Wilcoxon p-values for $\rho_{\text{split}} > \rho_{\text{drift}}$ are annotated above the blue box, with significance asterisks at $p < 0.05$ (*) and $p < 0.01$ (**).

at $m = 6$ bands, all three trace bands sit at $q = 0.027$. The δ cohort-median is marginally positive ($+0.031$, $6/10$, $p = 0.246$), θ is negative at the cohort median, and high-γ is essentially zero; none reaches the trace direction.

The split-versus-drift gap is larger than ρ_{split} alone. The drift-floor cohort medians at the three trace bands are negative ($\widetilde{\rho}_{\text{drift}} = -0.016$ at α, -0.043 at β, -0.020 at low-γ), so at β the patients move from a drift floor below zero to a split-baseline median at $+0.222$, a controlled gap of $+0.265$: the Wilcoxon fires not because ρ_{split} is mildly positive against zero, but because patients move upward from drift into the trace direction on the same halved-data noise budget that produced the drift floor. This mechanism is most visible at low-γ, where Pat07 and Pat13 are negative on ρ_{split} but still above their own drift floors: the controlled cohort claim at low-γ rests on the upward shift from drift, not on the positivity of ρ_{split} in absolute terms.

The cross-probe restriction reads how the cohort signal distributes between near-anatomy and across-implant contact pairs. Same-probe pairs share recording-shaft anatomy and short-range lagged coupling; cross-probe pairs are by construction not exposed to either contribution. The same-probe-to-cross-probe cohort-median ratio is $1.9\times$ at α ($\widetilde{\rho}_{\text{same}} = +0.198$ versus $+0.105$ cross-probe), $1.25\times$ at β ($+0.278$ versus $+0.222$), and $1.21\times$ at low-γ ($+0.173$ versus $+0.143$). The signal at α is therefore more concentrated in near-anatomy pairs than at β and low-γ, but in every controlled band the cross-probe restriction preserves both the cohort-median sign and at least seven of ten patients in the trace direction: the cohort claim does not depend on same-probe pairs at any of the three controlled bands.

The per-pair object underlying ρ_{split} is directly visualizable. Fig. 25 reads $\Delta_{\text{task}}(i, j)$ against $\Delta_{\text{rest}}(i, j)$ as a scatter cloud across all $N(N-1)/2$ contact pairs at three (patient, band) cells, each chosen as the patient sitting closest to the cohort-median ρ_{split} at the corresponding band: Pat06 at β ($\rho_{\text{split}} = +0.259$), Pat05 at α ($\rho_{\text{split}} = +0.071$), and Pat08 at θ ($\rho_{\text{split}} = -0.051$). All three panels span the full $[-0.3, +0.3]$ range on both axes; the contrast between trace and null bands is therefore about the diagonal tilt of the cloud, not about its overall spread. The two trace-band panels show a visible pull along $y = x$: per-pair shifts at β and α tend to agree in sign and rank between task and rest. The θ panel shows the same dynamic range with no diagonal pull, the cloud spreading symmetrically across the four quadrants. Cross-probe pairs (dark blue) outnumber same-probe pairs (light gray) by roughly twelvefold across all three panels and dominate the visible cloud; the diagonal pull at α and β survives cross-probe restriction ($\rho_{\text{xprobe}} = +0.057$ and $+0.249$ respectively) because cross-probe pairs are the bulk of the cloud carrying the cohort signal, not a small subset of it.

Two robustness checks complete the controlled framing. Pat03 dropout, motivated by the 1024 Hz sampling-rate asymmetry of that patient (§ 4.2), preserves the directional signal at all three controlled

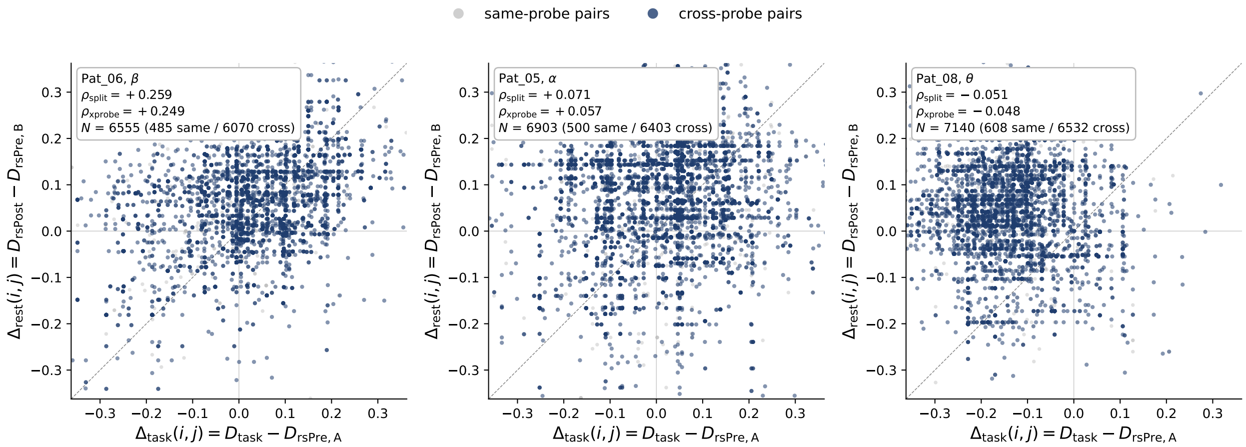


Figure 25: Per-pair scatter of $\Delta_{\text{task}}(i, j)$ versus $\Delta_{\text{rest}}(i, j)$ at three exemplar (patient, band) cells, one panel per cell. Each point is one of the $N(N-1)/2$ contact pairs; same-probe pairs in light gray, cross-probe pairs in dark blue. The dashed diagonal marks $y = x$ (perfect agreement between task-induced and rest-induced shifts). *Left:* Pat06 at β, the patient closest to the cohort-median $\rho_{\text{split}} = +0.222$. *Center:* Pat05 at α, closest to the cohort-median $\rho_{\text{split}} = +0.115$. *Right:* Pat08 at θ, closest to the cohort-median $\rho_{\text{split}} = -0.049$. All three panels span the full $[-0.3, +0.3]$ range on both axes; the trace-band panels show a diagonal pull along $y = x$, and the θ panel shows comparable cloud spread without diagonal structure. The horizontal and vertical banding visible in all three panels reflects pairs sharing a common contact, which carry correlated communication distances by construction on $\mathcal{D}(\tau)$ — this is a structural feature of the per-pair object and not a plotting artifact.

bands: the paired Wilcoxon p on the remaining nine patients is 0.014 at α, 0.027 at β, and 0.014 at low-γ; α and low-γ remain BH-surviving and β remains directional and uncorrected-significant. The pro-cohort restriction ($n = 8$, excluding Pat07 and Pat15 as cross-probe-aggregate anti-aligned at the substrate level) also preserves the signal: $p = 0.027$ at α, $p = 0.039$ at β, $p = 0.027$ at low-γ. The dropout criterion here is the cross-probe-aggregate one, not a Continuous Trace Matrix (CTM)-specific one: under the per-pair ρ_{split} measurement, both Pat07 ($\rho_{\text{split}} = +0.186$ at β) and Pat15 ($+0.144$ at β) are themselves CTM-trace patients at β, so the pro-cohort restriction is a cross-substrate-aggregate sensitivity check rather than a removal of CTM anti-trace patients.

The matrix distances on the upper-triangular entries of $\mathcal{D}(\tau)$ — rank d_S, Pearson d_P, Frobenius d_F — read the per-pair object directly without the dendrogram intermediate. The β cohort sits at 8/10 trace direction on each of the three distances, with seven patients in the trace direction simultaneously on all three; the smallest cohort p is 0.042 on d_P and d_F. Low-γ sits at 8/10 on d_F ($p = 0.053$) but only 5/10 on d_S: the diffusion preserves the magnitude pattern of the band's communication distances while leaving the rank ordering of edges essentially unchanged across phases, the matrix-level mirror of the heights-only KC reading at low-γ. α sits at 7/10 on d_S ($p = 0.423$) without reaching uncorrected significance on any of the three distances, consistent with the per-pair correlation being the load-bearing α reading at the LRG layer.

The honest joint-multiple-comparison statement carries two scopes. Under within-probe BH-FDR at the six bands of CTM ($m = 6$), the trace bands α, β, and low-γ all survive at $q = 0.027$. Under joint BH across the forty-eight (probe, band) cells of the five LRG probe families assembled in this section — KC at $\lambda \in \{0, 1\}$, CTM, the three matrix distances on $\mathcal{D}(\tau)$, VI(k), and the Grassmann probe at one summary scalar per band — no single cell survives at $q \leq 0.05$; the smallest joint q is 0.164, reached by the three CTM trace bands. The within-probe correction is the load-bearing controlled cohort claim; the joint correction is the conservative honesty statement under aggregation across probe families that ask geometrically distinct questions.

The picture that emerges from § 5.2 and § 5.3 together is band-resolved at the LRG layer. The β trace appears at every facet the analysis tests: tree topology (within-probe $q = 0.006$), tree heights ($q = 0.006$), the matrix d_P and d_F distances ($p = 0.042$ on each), and the per-pair correlation (controlled $p = 0.014$, $q = 0.027$). The low-γ trace lives in the heights of the dendrogram (within-probe $q = 0.055$, directional companion), in the matrix Frobenius distance ($p = 0.053$), and in the per-pair correlation (controlled $p = 0.010$, $q = 0.027$) — but not in tree topology and not in the rank distance. The α trace lives in the per-pair correlation alone (controlled $p = 0.007$, $q = 0.027$), consistent with the substrate α trace being an edge-rank phenomenon that crystallizes into the diffusion communication geometry without restructuring the linkage hierarchy or the matrix-level summaries. The cohort claim of § 5 at the LRG layer is therefore not band-uniform: each of the three trace bands carries a different mechanistic signature, and the per-band signature is itself the contribution of § 5 to the trace investigation begun at the substrate level in § 4.

5.4 Multiscale companion views: variation of information across cut levels and Grassmann distance across spectral dimensions

The controlled per-pair test of § 5.3 identifies α, β, and low-γ as the three bands at which the cognitive episode imprints a memory on the communication geometry of $\mathcal{D}(\tau)$. That test reads the per-pair distances directly without reference to any partition or spectral cutoff. The two natural multiscale extensions read the same imprint at intermediate cluster counts and at intermediate eigenmode counts: the variation of information $\mathrm{VI}(\pi^{\Phi}(k), \pi^{\Phi'}(k))$ across cuts of the dendrogram reads cluster-assignment label-space mismatch at every tree depth, and the Grassmann chordal distance $d_{\mathrm{G}}(U_k^{\Phi}, U_k^{\Phi'})$ across spectral cutoffs reads the cumulative principal-angle rotation between the leading-eigenmode subspaces of L. Neither probe selects a privileged k, and neither carries the split-baseline control battery of § 5.3; the role of § 5.4 is to read whether the three-band identification extends as a coherent multiscale signature across spans of k rather than concentrating at any one resolution.

We use two probes on the same diffusion output. The first is the variation of information between dendrogram partitions across phase pairs. For a (patient, band) cell let $\pi^{\Phi}(k)$ denote the partition into k clusters obtained by cutting the dendrogram of phase Φ at the corresponding linkage height; the variation of information between two partitions on the same leaf set is $\mathrm{VI}(\pi, \pi') = H(\pi \mid \pi') + H(\pi' \mid \pi)$, where H is the conditional entropy of the cluster-assignment random variables. The triangle scalar at cut level k is

$$T_{\mathrm{VI}}(k; p, b) = \mathrm{VI}\big(\pi^{\mathrm{taskTest}}(k),\, \pi^{\mathrm{rsPost}}(k)\big) - \mathrm{VI}\big(\pi^{\mathrm{rsPre}}(k),\, \pi^{\mathrm{taskTest}}(k)\big), \tag{13}$$

with $T_{\mathrm{VI}}(k) < 0$ the trace direction. The second probe is the Grassmann chordal distance between leading-eigenmode subspaces of L. Let U_k^{Φ} denote the $N \times k$ matrix whose columns are the k slowest non-trivial diffusion modes of L^{Φ} — the eigenvectors associated with the k smallest non-zero eigenvalues of the symmetric Laplacian. The chordal distance between two such subspaces is

$$d_{\mathrm{G}}(U, U') = \sqrt{\sum_{i=1}^{k} \sin^2 \theta_i}, \tag{14}$$

where θ_i are the principal angles between the subspaces; the form is gauge-invariant under sign flips and basis rotations within degenerate eigenspaces, and reads the cumulative principal-angle rotation between the subspaces. The triangle scalar at spectral cutoff k is

$$T_{\mathrm{G}}(k; p, b) = d_{\mathrm{G}}\big(U_k^{\mathrm{taskTest}},\, U_k^{\mathrm{rsPost}}\big) - d_{\mathrm{G}}\big(U_k^{\mathrm{rsPre}},\, U_k^{\mathrm{taskTest}}\big). \tag{15}$$

$\mathrm{VI}(k)$ reads label-space mismatch between cluster assignments at one cut depth in the tree, and $d_{\mathrm{G}}(k)$ reads the cumulative angle between eigenmode subspaces at one spectral cutoff; both are evaluated on the continuous grid $k \in \{2, \ldots, 79\}$. At fine cut levels the partition consists mostly of singletons and VI becomes dominated by the Hamming-style mismatch between near-identity partitions; we mask cells where the singleton fraction in either partition exceeds 0.5, which on this cohort ($N = 113$–122 contacts per patient) removes $k \gtrsim 80$ and is stable to threshold variations between 0.4 and 0.6. The Grassmann view additionally resolves the per-mode contribution at each k via the cohort-median signed principal-angle shift in radians $\Delta\theta_i(k) = \theta_i^{(\mathrm{rsPre}\to\mathrm{taskTest})} - \theta_i^{(\mathrm{taskTest}\to\mathrm{rsPost})}$, at each principal-angle index $i = 1, \ldots, k$. Fig. 26 reports the cohort $\mathrm{VI}(k)$ outcome; Fig. 27 reports the Grassmann mode-resolved view.

Figure 26 reads the cohort outcome at the partition-level view. β carries the longest contiguous pro-trace span: a sixteen-cell block at $n_{\mathrm{trace}} \geq 7/10$ between $k = 17$ and $k = 32$, with a one-cell dip at $k = 33$ and resumption at $k = 34$–35, peaking at $n_{\mathrm{trace}} = 9/10$ at $k = 22$, and sparser pro-trace cells continuing out to

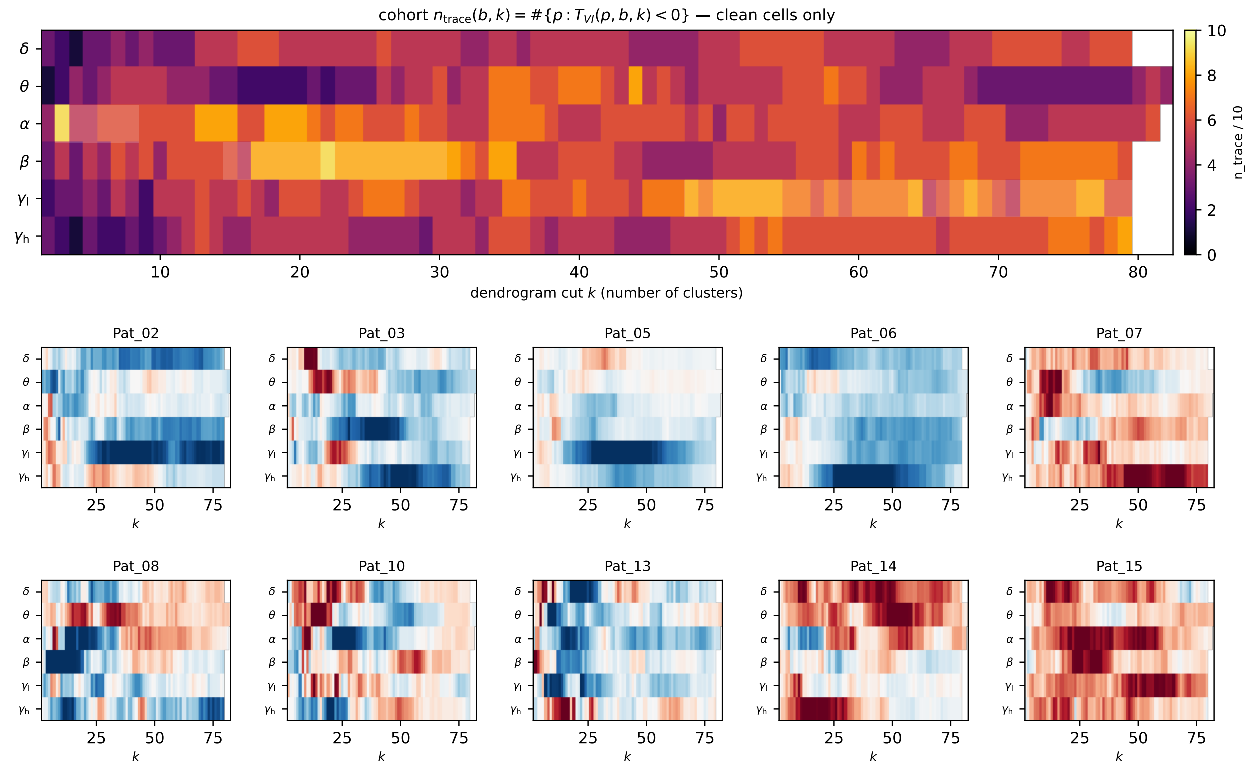


Figure 26: Multiscale partition trace via variation of information at every dendrogram cut level. *Top panel:* cohort heatmap of $n_{\mathrm{trace}}/10$, the count of patients with $T_{\mathrm{VI}}(k) < 0$, at each (band, k) cell; brighter cells indicate stronger cohort agreement in the trace direction. White cells at $k \gtrsim 80$ are masked due to singleton-dominated partitions (singleton fraction ≥ 0.5). *Bottom panels:* per-patient $T_{\mathrm{VI}}(k)$ heatmap (RdBu divergent; blue = trace direction, red = anti-trace direction) for each of the ten patients, with bands on the vertical axis and k on the horizontal axis.

$k = 78$. α sits at the fine end of the cut continuum, peaking at $n_{\mathrm{trace}} = 9/10$ at $k = 3$ and pro-trace through $k = 37$, with isolated null cells breaking strict contiguity. Low-γ sits at intermediate-to-coarse cuts: a twelve-cell block at $k = 45$–56 reaching $n_{\mathrm{trace}} = 8/10$ at $k = 48$, with continuation through $k = 77$. δ is anti-trace at fine k and null at coarser cuts, high-γ shows a sparse pro-trace span only at the coarse edge of the unmasked range, and θ is essentially null with two short pro-trace clusters at $k = 34$–36 and $k = 39$–41 that do not extend to a sustained signal. The three pro-trace bands occupy distinct ranges of the cut continuum — α at fine cuts, β at intermediate cuts, low-γ at intermediate-to-coarse cuts — so the partition-level signature of the cognitive episode is band-resolved at every dendrogram depth at which a signal surfaces. Fig. 27 reads the cohort outcome at the spectral subspace view across the continuous range $k \in \{2, \ldots, 79\}$. The principal-angle heatmaps reveal a structural feature that the chordal scalar $d_{\mathrm{G}}^2 = \sum_i \sin^2 \theta_i$ collapses out: the cohort-median $\Delta\theta_i(k)$ is concentrated along the diagonal $i \approx k$ at every spectral cutoff, that is, the trace contribution at each k comes from the most-misaligned shared direction (the largest principal angle in the included subspace), not from any fixed eigenmode index. The sign of the diagonal band is band-specific. The β panel carries the most coherent positive diagonal of the cohort: above a fine-k cold zone at $k \lesssim 12$ where $n_{\mathrm{trace}} \leq 5/10$, the top-mode $\Delta\theta_k(k)$ is in the trace direction at cohort-median $7/10$ for every $k \geq 12$, with 81% of unmasked cells reaching $n_{\mathrm{trace}} \geq 7/10$ and the chordal-scalar cohort agreement sustained at $n_{\mathrm{trace}} \approx 7$–$8/10$ through $k = 79$. The low-γ panel shows the broadest positive diagonal band, extending 20–40 modes back from $i = k$, with $n_{\mathrm{trace}} \approx 7$–$8/10$ across $k \approx 10$–79; the one-sided cohort Wilcoxon on the chordal scalar reaches $p = 0.014$ at $k = 13$ (8/10) and $k = 14$ (9/10), the strongest single cells in the cohort. The α panel shows a positive diagonal band emerging at $k \approx 10$ and sustained across the full remaining range, with cohort agreement $n_{\mathrm{trace}} \approx 7$–$8/10$ throughout. Across the continuous k-grid, roughly 35 Grassmann

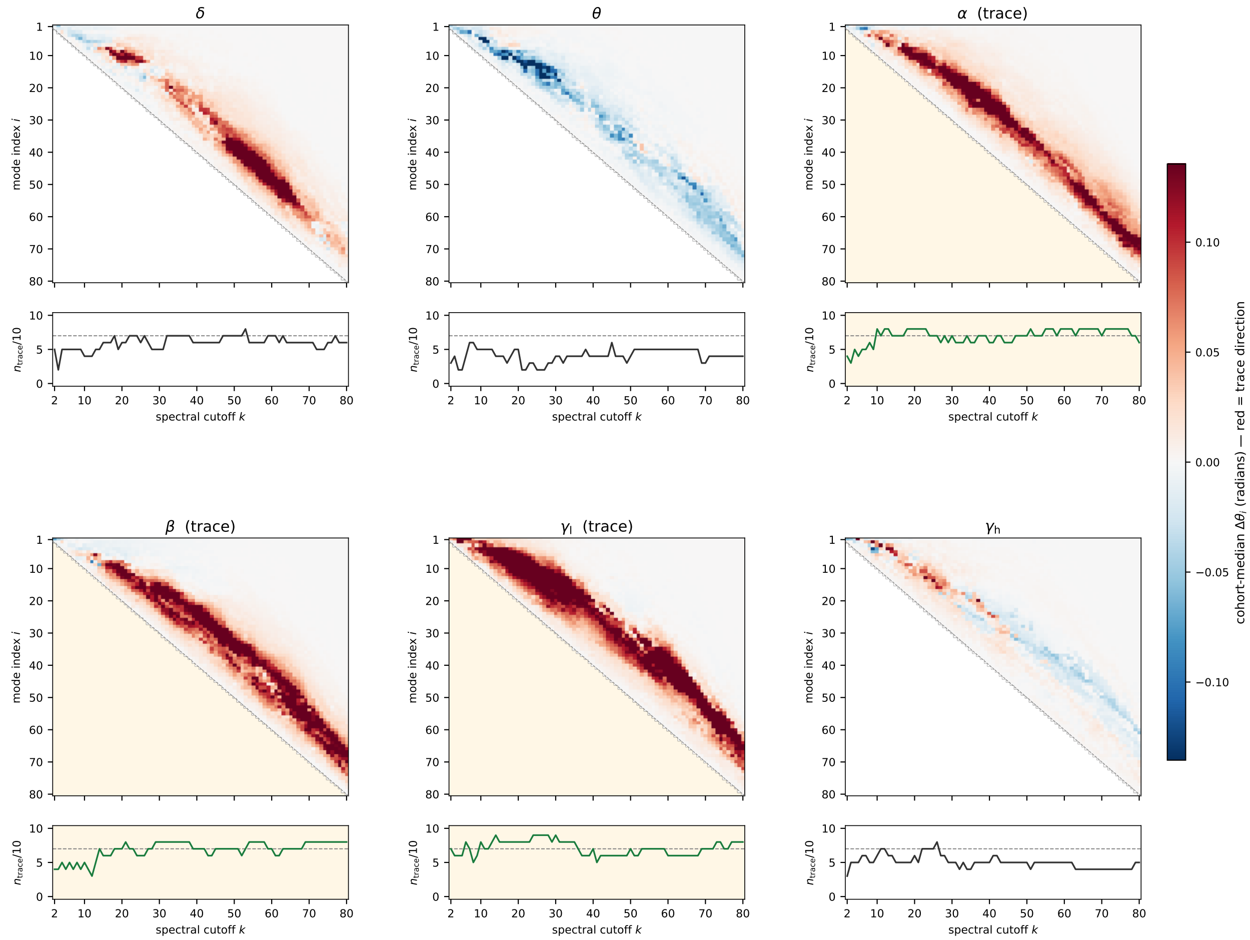


Figure 27: Multiscale spectral trace via Grassmann principal angles on the leading k Laplacian eigenmodes, $k \in \{2, \ldots, 79\}$. Six panels, one per band; each panel contains a heatmap of the cohort-median signed principal-angle difference $\Delta\theta_i(k) = \theta_i^{\text{rsPre}\to\text{taskTest}} - \theta_i^{\text{taskTest}\to\text{rsPost}}$ (in radians) across spectral cutoffs k on the horizontal axis and principal-angle index i on the vertical axis, with a strip below showing cohort agreement $n_{\text{trace}}(k)/10$ on the chordal scalar (horizontal dashed line at $7/10$). Principal angles are sorted in ascending order, so θ_1 is the most-aligned shared direction and θ_k the most-misaligned. Red indicates the trace direction at that mode index (the post-task subspace sits closer to the task-phase subspace than the pre-task subspace does, at the i-th principal angle); blue indicates the anti-trace direction; white indicates no asymmetry. Cells where $i > k$ are out-of-range and rendered blank. The three controlled trace bands of § 5.3 (α, β, low-γ) are highlighted with light-shaded panel backgrounds.

(band, k) cells reach uncorrected one-sided $p < 0.05$ on the chordal scalar, dominated by β at $k \geq 40$ and low-γ at $k \approx 10$–28; these counts are not reported as controlled cohort claims (no split-baseline null is applied at this layer) but they fix the order of magnitude of the spectral imprint. The θ panel shows a sustained negative diagonal band across $k \gtrsim 12$ at $n_{\text{trace}} \approx 4/10$, consistent with systematic anti-alignment rather than absence of signal. The δ and high-γ panels show no sustained diagonal: δ hovers at $n_{\text{trace}} \approx 5$–$6/10$ at coarse k without contiguity, and high-γ shifts from positive at low k to negative at high k along the diagonal, an unstable spectral signature. The band-level convergence with the controlled CTM bands of § 5.3 is therefore carried by the trio with positive diagonal bands (α, β, low-γ), with β the strongest of the three at the spectral subspace level.

The three trace bands of § 5.3 are the same three bands that carry pro-trace cohort agreement at $\mathrm{VI}(k)$ and at $d_{\mathrm{G}}(k)$: α at the fine end of the cut continuum, β at intermediate cuts and across the spectral subspace above the fine-k cold zone, low-γ at intermediate-to-coarse cuts and across the broadest principal-angle diagonal. The single probe in this section that disagrees with this three-band identification is the KC $\lambda = 0$ global tree distance of § 5.2, which calls only β trace; α and low-γ preserve their global pairwise tree distance while reorganizing cluster assignments at specific cut depths and rotating eigenmode subspaces at specific spectral cutoffs. β is the only band that crystallizes simultaneously into the global tree distance, the per-cut partitions, and the eigenmode subspace. The one caveat that does not already follow from the descriptive role announced in the opening of this subsection is the meaning of k itself: cutting the dendrogram at the same k on networks of different size samples the tree at different relative depths, and the leading k Laplacian eigenmodes capture different fractions of the diffusive response on networks of different size and edge-weight density, so the pro-trace contiguity reported above is a cohort-aggregated patient-count claim, not a per-patient claim about the same partition resolution or spectral cutoff across the cohort.

5.5 Anatomical localization

The controlled cohort claim of § 5.3 reads the trace as a single per-pair scalar ρ_{split} aggregated across all $N(N-1)/2$ ultrametric distance pairs simultaneously. That aggregation hides where in the cortex the imprint lives: a band whose memory imprint admits a single-region anatomical address would localize at the cohort level, while a band whose imprint distributes its contributions across many regions would not, even at the same controlled ρ_{split} magnitude. We resolve this question on the same diffusion substrate as § 5.3, at the same $\tau = 1/\lambda_{\max}$, through a per-leaf calibrated catalog that flags individual contacts by their participation in ρ_{split} and tests anatomical concentration via hypergeometric enrichment against the cohort cortical contact pool. By construction, the localization layer and the controlled cohort claim are mutually consistent rather than independent; the question is band-resolved.

For each (patient, band) cell, the per-leaf score is the contact-restricted mean of the per-pair Spearman correlation that defines ρ_{split}, demeaned by the patient/band leaf-mean to remove cohort-trace baseline. The within-patient null is the same split-half ρ_{drift} construction of § 5.3 applied per-leaf, demeaned identically. We flag a contact as a trace-leaf at band b when its score exceeds the within-patient p_{95} of the null distribution, one-sided upper tail, calibrated separately at every (patient, band) cell. After dropping white-matter (`Wm`) and parcellation-failure (`Unk`) labels, the cohort cortical contact pool is $N_{\text{total}} = 866$, with cortical baseline trace rates of 3.46% at α ($K = 30$ trace-leaves), 5.31% at β ($K = 46$), and 8.66% at γ_l ($K = 75$). For each anatomical region r with at least five cohort contacts and at least two contributing patients, we test over-representation through the one-sided hypergeometric tail $p_{\text{hyper}}(r) = \mathrm{P}\big(X \geq n_{\text{trace}}(r) \,\big|\, N_{\text{total}}, K, n_{\text{contacts}}(r)\big)$, where $n_{\text{trace}}(r)$ is the observed trace-leaf count in region r. The multiple-comparison correction is Bonferroni across the $m = 48$ (band, region) cells passing the strict-eligibility filter, with corrected threshold $\alpha/m \approx 1.04 \times 10^{-3}$. Two systematic sensitivity regimes are applied throughout: dropping Pat03 (motivated by the 1024 Hz sampling-rate asymmetry of that patient, § 4.2) and a pro-cohort restriction excluding the cross-probe-aggregate anti-aligned patients Pat07 and Pat15 (identified at the substrate layer in § 4.4). The single anatomical cell that survives Bonferroni correction is γ_l at left fusiform cortex (ctx-lh-fusiform), with 13/38 cortical fusiform contacts flagged as trace-leaves at the cohort level (34.2% trace rate against the 8.66% cohort baseline; enrichment 3.95×, $p_{\text{hyper}} = 5.2 \times 10^{-6}$). The cell strengthens slightly without Pat03, to 11/30 at 4.03× ($p_{\text{hyper}} = 4.4 \times 10^{-5}$), and remains comfortably below α/m under both sensitivity regimes. The cell rests on a disclosed double dependence on Pat02: he contributes 18 of the cohort's 38 cortical fusiform contacts (47.4% of the cohort fusiform implant) and 9 of the 13 trace-leaves (69% of the peak), with a per-patient fusiform trace rate of 50.0% — twice the rate of the next two contributors (Pat03 and Pat13, each 2/8 at 25.0%). The dependence is therefore not a single-patient artifact in disguise: Pat02 is both the cohort's anatomically densest fusiform implant and its highest-rate fusiform tracer, and either departure (anatomical density without high rate, or high rate at typical density) would weaken the cell. Without Pat02 the cell drops to 4/20 over the residual two-patient signal at enrichment 3.32× and $p_{\text{hyper}} = 0.027$, failing Bonferroni at $m = 48$ and surviving only at uncorrected $p < 0.05$. The yellow halos in Fig. 28 flag Pat02's contribution explicitly. The γ_l ctx-lh-fusiform cell is therefore Bonferroni-significant at the cohort level under multi-region correction, with the anatomical density and the per-patient trace rate of Pat02 disclosed as jointly load-bearing.

The β anatomy is uncorrected-only at every cell. At full cohort, the three uncorrected candidates are Hippocampus (5/27, 3.49×, $p_{\text{hyper}} = 0.011$), left fusiform (5/38, 2.48×, $p_{\text{hyper}} = 0.045$), and left superior temporal (6/53, 2.13×, $p_{\text{hyper}} = 0.055$), none of which reaches the Bonferroni threshold. Under Pat03

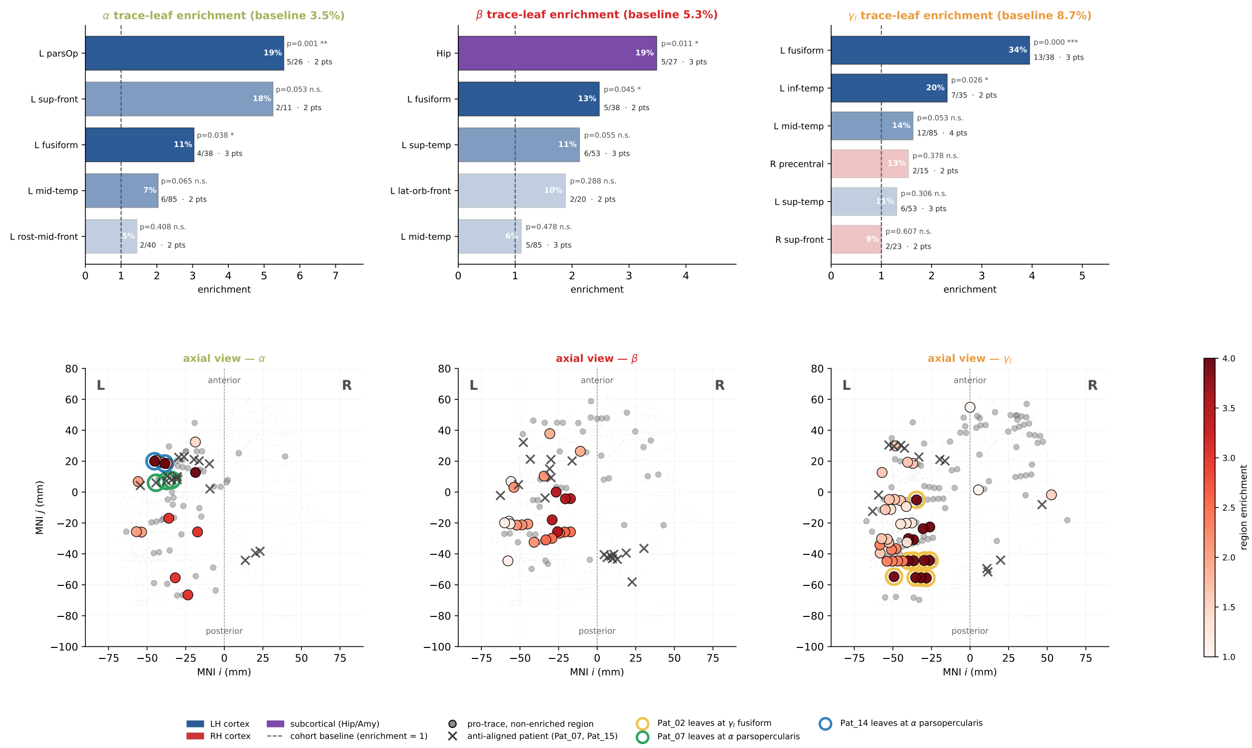


Figure 28: Anatomical localization of CTM trace-leaves. *Top row:* per-region enrichment for the three trace bands α (left), β (middle), and γ_l (right), restricted to regions with at least five cohort contacts and at least two contributing patients. Bars are sorted by trace rate against the per-band cortical baseline (annotated under each panel title); the per-cell trace fraction $n_{\text{trace}}/n_{\text{contacts}}$ and the number of contributing patients are annotated beside each bar. One-sided hypergeometric p-values are flagged with $*p < 0.05$, $**\ p < 0.01$, $***p < 0.001$; the Bonferroni threshold across $m = 48$ cells is $\alpha/m \approx 1.04 \times 10^{-3}$, and only the γ_l ctx-lh-fusiform cell crosses it. *Bottom row:* axial MNI projections of the cortical contact pool per band; colored markers are trace-leaves shaded by region enrichment (red intensity), gray markers are pro-cohort non-trace contacts, X-markers identify trace-leaves of the anti-aligned patients Pat07 and Pat15. Yellow halos in the γ_l panel flag the nine ctx-lh-fusiform trace-leaves contributed by Pat02, which carry $9/13$ of the γ_l fusiform peak. Green halos in the α panel flag the three ctx-lh-parsopercularis trace-leaves contributed by Pat07 and blue halos flag the two contributed by Pat14; together these constitute the entire two-patient base of the only uncorrected-suggestive α anatomy cell, with Pat07 anti-aligned at the cross-probe-aggregate level and Pat14 pro-aligned at the § 5.3 controlled criterion.

dropout, Hippocampus and left fusiform collapse to non-significance ($p_{\text{hyper}} = 0.062$ and 0.470 respectively), because Pat03 carries 2 of the 5 Hippocampus trace-leaves and 3 of the 5 fusiform trace-leaves at β; we therefore drop both regions from the β anatomical headline. The third candidate, left superior temporal, has none of its six trace-leaves contributed by Pat03, so its numerator is unchanged under that dropout and its p-value shifts only modestly to $p_{\text{hyper}} = 0.035$; even so, it remains uncorrected only and well above α/m. The pro-cohort restriction ($n = 8$, dropping Pat07 and Pat15) reproduces the full-cohort β pattern at essentially unchanged enrichment values, with no region recovering Bonferroni significance. The two anti-aligned patients contribute almost no cortical β trace-leaves to begin with: each has nine total β trace-leaves, of which most are white-matter or unparcellated and excluded by construction, with their residual cortical trace-leaves regionally scattered (one per region in Pat15 across three right-hemisphere regions; a single ctx-lh-parsopercularis contact in Pat07). The two anti-aligned patients are therefore not tracing in a competing cortical pattern; they are not tracing cortically at all at the calibrated threshold (Fig. 28, X-markers in the bottom row). The β anatomical reading is that the band's memory imprint, despite being the cohort-strongest at every functional layer of § 5 (tree topology, tree heights, per-pair correlation, partition-cut, spectral subspace), distributes its cortical contributions broadly enough that no single region carries it at multi-region correction.

The α anatomy admits one uncorrected-suggestive cell that we flag with explicit downgrade: ctx-lh-parsopercularis at 5/26 contacts (19.2% trace rate against the 3.46% cohort baseline, 5.55× enrichment, $p_{\text{hyper}} = 1.4 \times 10^{-3}$), narrowly missing the Bonferroni threshold of 1.04×10^{-3}. The cell rests on a two-patient base of mixed CTM alignment. Pat07 contributes three of the five trace-leaves: he is the cross-probe-aggregate anti-aligned patient identified at § 4.4, with $\rho_{\text{split}} = -0.035$ at α (negative, opposite to the cohort imprint direction) but passing the drift criterion only because his own drift floor $\rho_{\text{drift}} = -0.151$ is more negative still. Pat14 contributes the remaining two trace-leaves and is unambiguously pro-aligned at α: $\rho_{\text{split}} = +0.240$ — the third-strongest pro-trace value in the α cohort — and well above his own drift floor ($\rho_{\text{drift}} = +0.040$). The cell therefore aggregates one anti-aligned and one pro-aligned trace into a hypergeometric overage that has no clean interpretation as an instance of the controlled cohort signal at α. The fragility under either dropout confirms this: dropping Pat07 collapses the cell to 2/20 on a single-patient base and $p_{\text{hyper}} = 0.121$; dropping Pat14 drops it to 3/16 on a single-patient base and $p_{\text{hyper}} = 0.019$, still single-patient. Dropping Pat02, who contributes zero α parsopercularis trace-leaves despite implanting six parsopercularis contacts, mechanically sharpens the rate ratio to 7.35× and pushes the cell across Bonferroni at $p_{\text{hyper}} = 3.4 \times 10^{-4}$ — the denominator collapses faster than the numerator, which does not constitute robustness. The green and blue halos in Fig. 28 flag the Pat07 and Pat14 contributions respectively. We record the cell as a directional indication only.

Two side observations close the section. First, Pat07's single cortical β trace-leaf sits at left parsopercularis, the same region that anchors his α contribution: a patient-internal cross-band consistency at parsopercularis that is consistent with the framing of § 4.4 of Pat07 and Pat15 as scientifically meaningful anti-tracers shaped by implant geometry rather than as noise. The observation is not a cohort claim. Second, the band-resolved anatomical reading is itself the substantive finding: γ_l admits a Bonferroni-survived peak at left fusiform; β populates twenty cohort-eligible regions without any single peak clearing 3.5× baseline; α populates twelve cohort-eligible regions with three peaks at 2–5× baseline that all rest on two-patient bases. The negative claim at β and α is therefore "no single region carries the cohort imprint", not "the imprint is brain-wide" — the data test single-region concentration, not distributional spread, and a substantial minority of cohort-eligible cortex (twelve to twenty regions of forty-eight) carries the trace at every controlled band. The localization asymmetry between γ_l and β/α is consistent with γ_l reflecting more local cortical processing than β, and is the band-resolved contribution of § 5.5 to the trace investigation: the cognitive episode's memory imprint at the diffusion layer of § 5.3 admits a single-region anatomical address only at γ_l, with β and α imprinting through patterns distributed across cohort-eligible cortex rather than through any one region.

5.6 Cross-phase module taxonomy on the dendrogram

The per-pair tests of § 5.2 and § 5.3 read the trace at the level of individual leaf pairs in the communication distance $\mathcal{D}(\tau)$. At that level, the trace is a coherent shift of pair-distance correlations between phases, statistically detectable cohort-wide for α, β, and low-γ under controls. We now ask a structurally distinct question: when the per-pair memory signal is coarsened onto modules of the dendrogram, can it be classified into discrete behavioral patterns visible in the connectivity matrix of an individual patient? The answer is the four-class taxonomy of cross-phase modules — *trace*, *reset*, *rearrange*, *anchor* — together with a residual *diffuse* category, applied at the per-patient level on each (patient, band) cell. This subsection is a structural illustration; the cohort claims are upstream at § 5.3 and unaffected by what follows.

The taxonomy operates on subtrees of the dendrogram of $\mathcal{D}(\tau)$ at $\tau = 1/\lambda_{\max}$ at KC $\lambda = 0$. For each phase, the candidate subtree pool is every internal node of the linkage with leaf-set size in $[3, \lfloor n/2 \rfloor]$, where n is the patient's contact count. For each source subtree of size s at one phase, the closest counterpart at the destination phase is found by greedy arg max of the Jaccard similarity of leaf sets, restricted to a destination size band $[s/1.5, s \times 1.5]$; destinations are not removed across successive matches, so the same destination subtree can in principle match multiple sources. A leaf-set match is classified by three-way Jaccard thresholds: *coherent* if Jaccard ≥ 0.6, *fragmented* if Jaccard < 0.5, and *ambiguous* (rejected from the candidate pool) if in the buffer $[0.5, 0.6)$. Two further operational rules complete the matching procedure. A containment guard rejects a candidate of size s when the smallest disrupted-phase subtree fully containing all of its leaves has size below $2s$, which prevents a small leaf set from being declared fragmented when the leaves are in fact co-located inside a tight slightly-larger subtree. A selection-time deduplication sorts the surviving candidates by (−size, −score) and accepts each only if its union leaf set overlaps the cumulative accepted leaf set by less than 30%, capped at twelve modules per class per cell. The four module-level classes are then defined by the matching status of a candidate's source phase against the matching status of its destination matches; the residual diffuse class is leaf-level, collecting leaves not assigned to any module of the four. The five classes are summarized in 4.

The anchor class uses a variant of the same procedure that tests three phases simultaneously rather than pairwise: for each rspre source subtree L_{pre}, the destination search is over the joint pool $(L_{\text{tt}}, L_{\text{post}})$ within the same 1.5× size band, and the score is the geometric mean of the three pairwise Jaccards, with each individual Jaccard required to be at least 0.6. The selection-time deduplication and module count cap are identical to the other classes. The *strict-intersection leaf set* of each module-level class is the load-bearing definition

class	match rsPre–taskTest	match taskTest–rsPost	match rsPre–rsPost	meaning
trace	fragmented	coherent	—	task-seeded coherence surviving into re
reset	—	fragmented	coherent	rest topology disrupted by task and re
rearrange	fragmented	—	—	emergent post-task coherence with tecedent
anchor	coherent	coherent	coherent	phase-invariant backbone preserved phases
diffuse	—	—	—	leaves not in any of the above intersection sets

Table 4: The five classes of cross-phase module behavior on the dendrogram of $\mathcal{D}(\tau)$ at KC $\lambda = 0$. The *coherent* / *fragmented* entries describe the Jaccard match status of the candidate's leaf set against the destination phase, with the matching rule and Jaccard thresholds (coherent ≥ 0.6, fragmented < 0.5) defined in the body. Trace, reset, rearrange, and anchor are module-level classes; diffuse is leaf-level. Dashes denote phase pairs whose match status is not part of the class's defining gates.

for visual rendering and for diffuse-residual computation. For trace, the strict set is $L_{\text{tt}} \cap L_{\text{post}}$; for reset, $L_{\text{pre}} \cap L_{\text{post}}$; for anchor, $L_{\text{pre}} \cap L_{\text{tt}} \cap L_{\text{post}}$. The rspre role for trace, the taskt role for reset, and the absence of an explicit rspre–rspost match for trace and rearrange are gating conditions on the matching procedure (the candidate has to pass the corresponding fragmentation or coherence test to qualify), not contributors to the strict leaf set. The rearrange strict set is L_{post} minus the union of the strict trace, reset, and anchor leaves at the same cell, kept only if the residual is at least three leaves; rearrange therefore collects rspost coherence with no antecedent at either earlier phase. The diffuse leaf set is whatever leaves of the contact set remain unclassified after the previous four passes. Throughout the figures of this subsection, the dendrogram leaf bars and the connectivity matrix block reordering are keyed to the strict-intersection leaf set, while the network panels use the union convention — $L_{\text{tt}} \cup L_{\text{post}}$ for trace, $L_{\text{pre}} \cup L_{\text{post}}$ for reset, $L_{\text{pre}} \cup L_{\text{tt}} \cup L_{\text{post}}$ for anchor, L_{post} (after residual subtraction) for rearrange — so that fringe leaves that participate in one matched-subtree pair but not in the strict intersection remain visible as colored network nodes. The strict / union convention asymmetry is documented in each figure caption.

The decoupling between this module-level taxonomy and the per-pair cohort claim of § 5.3 requires emphasis. The cohort trace at α, β, and low-γ is a per-pair statistical statement about the upper-triangular entries of $\mathcal{D}(\tau)$, established at the cohort level under controls regardless of what modules are extractable from any single patient's dendrogram. Module-level findings in this subsection are at the per-patient level only; modules are not naturally well-defined in a fully connected weighted network, and we are reading them off the tree topology induced by the linkage construction on $\mathcal{D}(\tau)$. A module identified at one band for one patient does not corroborate or invalidate the cohort per-pair claim; it is a different reading of the same underlying communication geometry, at a coarser scale and a different unit of analysis. The contribution of § 5.6 is structural: the per-pair memory signal admits a module-level reading at the per-patient level, and that reading takes one of five behavioral forms.

We illustrate each class in turn at one representative (patient, band) cell, drawn from the cohort without privileged anchoring at β or any other band. Fig. 29 shows trace at Pat07 β, where six clades of multiscale sizes (4, 4, 5, 5, 6, 24 leaves; 32 trace leaves of 116 total) fragment in rspre, crystallize coherently in taskt, and persist into rspost; the connectivity matrix renders the trace leaves as densely intra-connected blocks that emerge in taskt and saturate in rspost. Fig. 30 shows reset at Pat10 low-γ, one of the cohort's two top-ranked reset profiles (five reset clades, tied with Pat13 by clade count): the reset clades are coherently grouped in rspre, fragment in taskt, and reassemble in rspost, with eighteen reset leaves of one hundred thirteen total. Fig. 31 shows rearrange at Pat03 high-γ, one of the cohort's three top-ranked rearrange cells (twelve rearrange clades, tied with Pat02 and Pat06; sixty-five rearrange leaves of one hundred twenty-two total), giving the cleanest single-clade dominant visual through its twenty-leaf rearrange clade. Pat03 is the cohort's 1024 Hz sampling-rate outlier (§ 4.2); the dendrogram of $\mathcal{D}(\tau)$ is unaffected by the sampling-rate asymmetry, and the cell is included for the cleanest pedagogical visual. Fig. 32 shows anchor at Pat02 β, where nine clades (forty-two anchor leaves of one hundred seventeen total) are coherently preserved across all three phases. Fig. 33 shows the diffuse category at Pat15 β, where fifty-two leaves of one hundred eighteen do not belong to any of the four strict-intersection module classes; the diffuse class is rendered as a single visualization unit because diffuse leaves do not satisfy the leaf-set matching contract that defines a module. Across the cohort, rearrange is the most abundant of the four module-level classes by a factor of three to five (560 rearrange clades against 144 anchor, 107 trace, 83 reset across the sixty (patient, band) cells), reflecting the abundance of post-task coherence that has no leaf-set antecedent at either earlier phase. With the five classes anchored visually, we examine one (patient, band) cell that simultaneously expresses all five classes on a single dendrogram, allowing direct comparison of where on the tree each class lives. Pat13 at β populates all five classes with two trace modules, four reset, three rearrange, seven anchor, and thirty-six diffuse leaves of one hundred nineteen. Fig. 34 shows the full taxonomy in one panel. The taxonomy of Fig. 34 resolves a per-patient structural inventory that the per-pair cohort tests of upstream subsections aggregate without separating: at Pat13 β, the inventory is dominated by anchor and diffuse leaves, with smaller but visible trace and reset populations. We do not generalize the relative balance of these classes to the cohort; substantial inter-patient variability is documented in the per-class summary tables and is itself a structural finding rather than a power problem. Pat07 β is the cohort's unique cell with $\min(n_{\text{trace}}, n_{\text{anchor}}) = 6$, expressing the joint-richest trace-and-anchor profile in the cohort; Pat15 β carries a single trace module, a single anchor module, and fifty-two diffuse leaves of one hundred eighteen, identifying him as a patient whose dendrogram structure is dominated by the diffuse residual rather than by any of the four classifiable cross-phase patterns. The taxonomy of this subsection is structural and descriptive: it does not introduce a new statistical test, does not attempt cohort-level claims at the module-count level, and does not refine or replace the per-pair controlled cohort claim of § 5.3. The contribution of § 5.6 is to demonstrate that the per-pair memory signal admits a module-level reading at the per-patient level on the dendrogram of $\mathcal{D}(\tau)$, and that this reading takes one of five distinct behavioral forms each visible in single patients in the connectivity matrix at the corresponding phase. The anatomical mapping of trace-class versus anchor-class modules to Desikan-Killiany regions is a separate open question, deferred to future work.

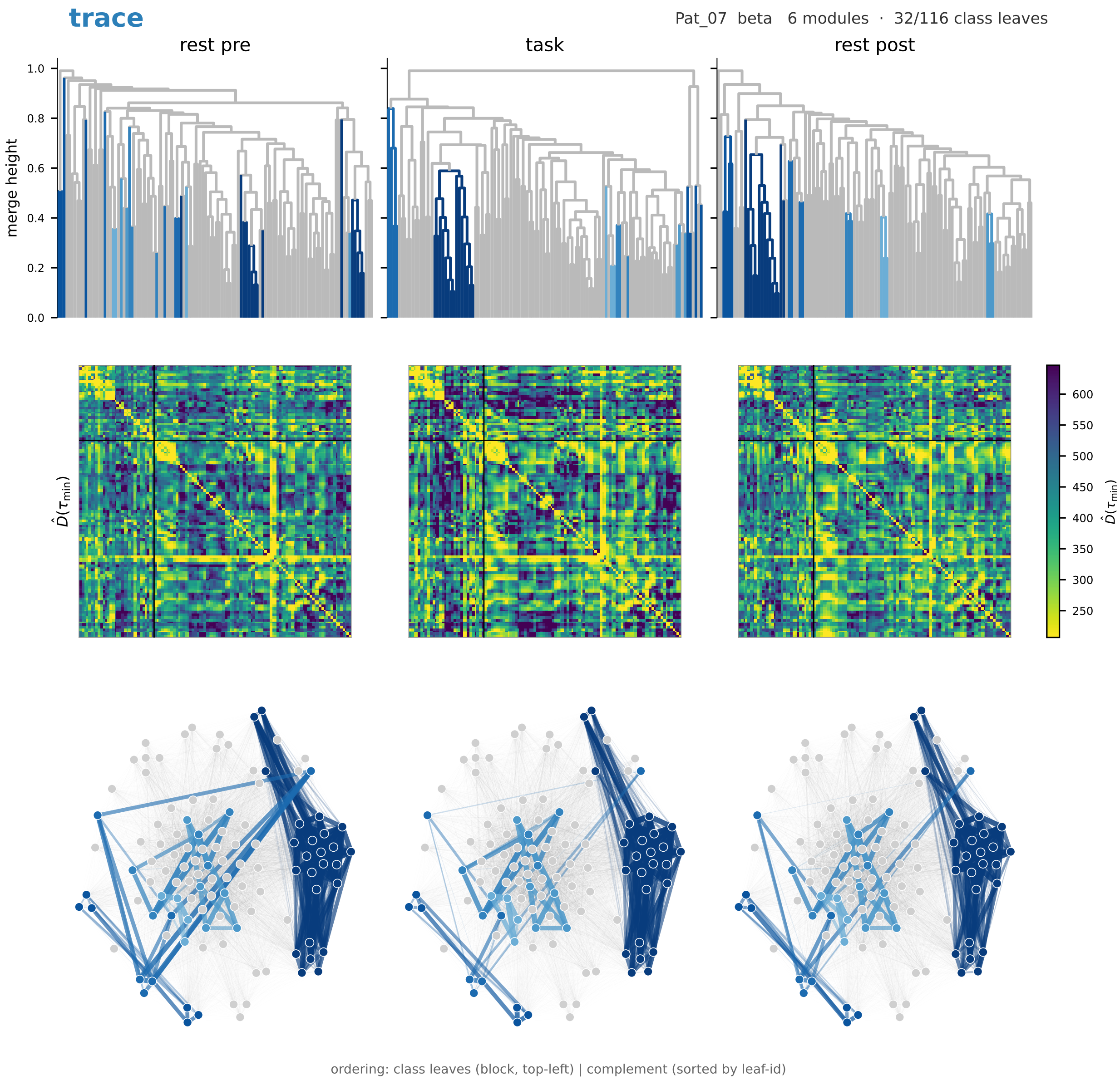


Figure 29: Trace class shown at Pat07 β. Three columns by two rows. Top row: the dendrogram of $\mathcal{D}(\tau)$ at rsPre, taskTest, rsPost, with the trace clade highlighted. Bottom row: the connectivity matrix of $\mathcal{D}(\tau)$ at the same three phases, restricted to the leaf indices of the trace clade and its complementary leaves, with rows and columns ordered by clade membership. The trace leaves are fragmented in rsPre, form a coherent clade in taskTest, and persist as a coherent clade in rsPost; the connectivity matrix shows the corresponding densely intra-connected block emerging at taskTest and saturating at rsPost. Network membership reflects the union $L_{\rm tt} \cup L_{\rm post}$; matrix-block membership reflects the strict intersection $L_{\rm tt} \cap L_{\rm post}$.

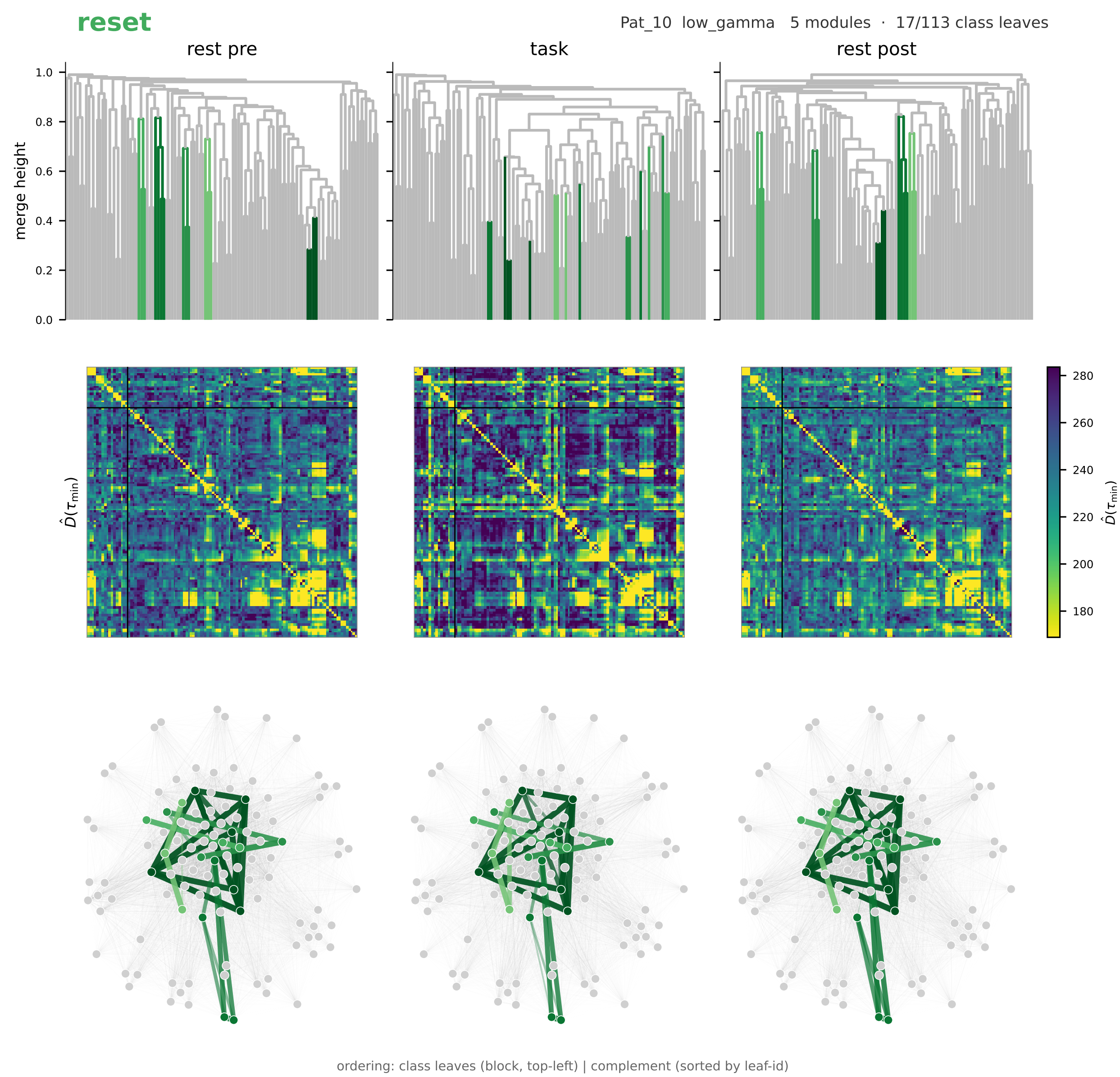


Figure 30: Reset class shown at Pat10 low-γ. Same layout as Fig. 29. The reset clades are coherent in rsPre, fragment in taskTest, and reassemble in rsPost; the connectivity matrix shows the corresponding densely intra-connected block at rsPre and rsPost with a scattered intra-connection pattern at taskTest. Network membership reflects the union $L_{\mathrm{pre}} \cup L_{\mathrm{post}}$; matrix-block membership reflects the strict intersection $L_{\mathrm{pre}} \cap L_{\mathrm{post}}$.

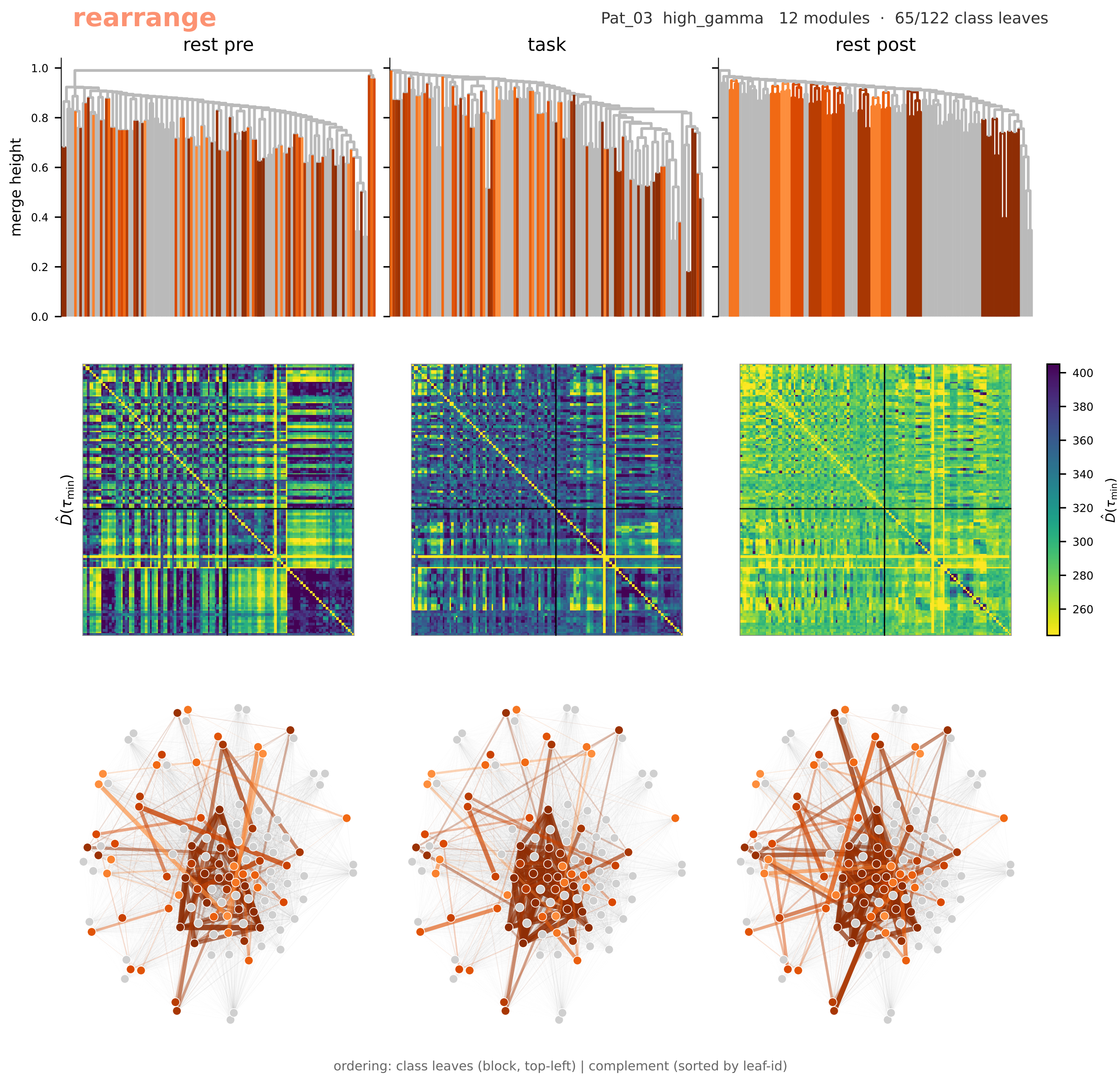


Figure 31: Rearrange class shown at Pat03 high-γ. Same layout. The rearrange clade is fragmented in rsPre and taskTest and emerges as coherent in rsPost; the connectivity matrix shows scattered intra-connections at the two earlier phases and a densely intra-connected block emerging only at rsPost. Network membership reflects L_{post} after subtraction of strict trace, reset, and anchor leaves; matrix-block membership reflects the same residual.

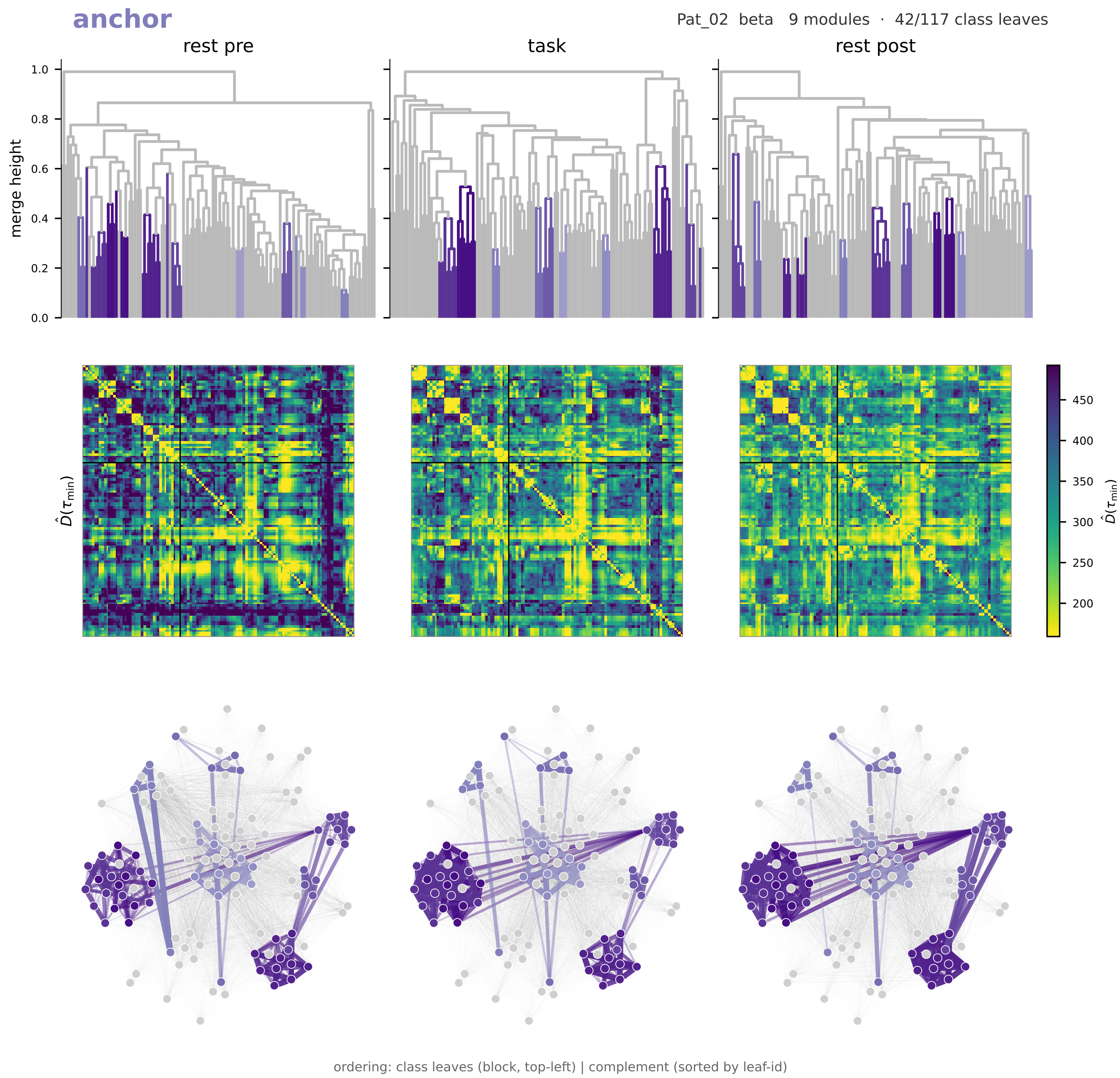


Figure 32: Anchor class shown at Pat02 β. Same layout. The anchor clade is coherently grouped at all three phases; the connectivity matrix shows a phase-invariant densely intra-connected block. Network membership reflects the union $L_{\text{pre}} \cup L_{\text{tt}} \cup L_{\text{post}}$; matrix-block membership reflects the strict intersection $L_{\text{pre}} \cap L_{\text{tt}} \cap L_{\text{post}}$.

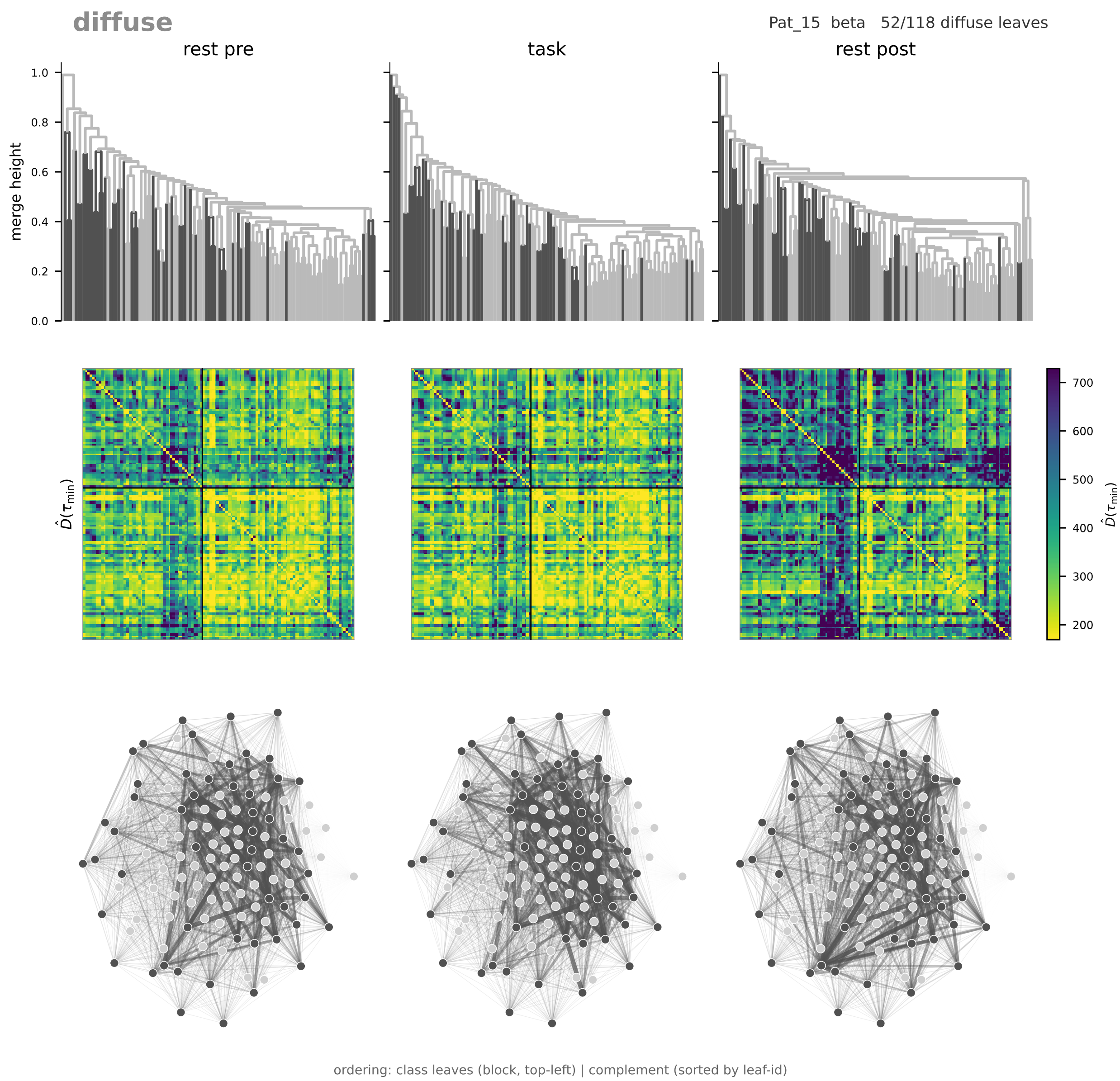


Figure 33: Diffuse leaves shown at Pat15 β. Same layout. Diffuse leaves do not belong to any of the four strict-intersection module classes; the connectivity matrix shows scattered intra-connections at all three phases without a coherent block structure. Diffuse is a leaf-level rather than module-level descriptor and is rendered as a single visualization unit.

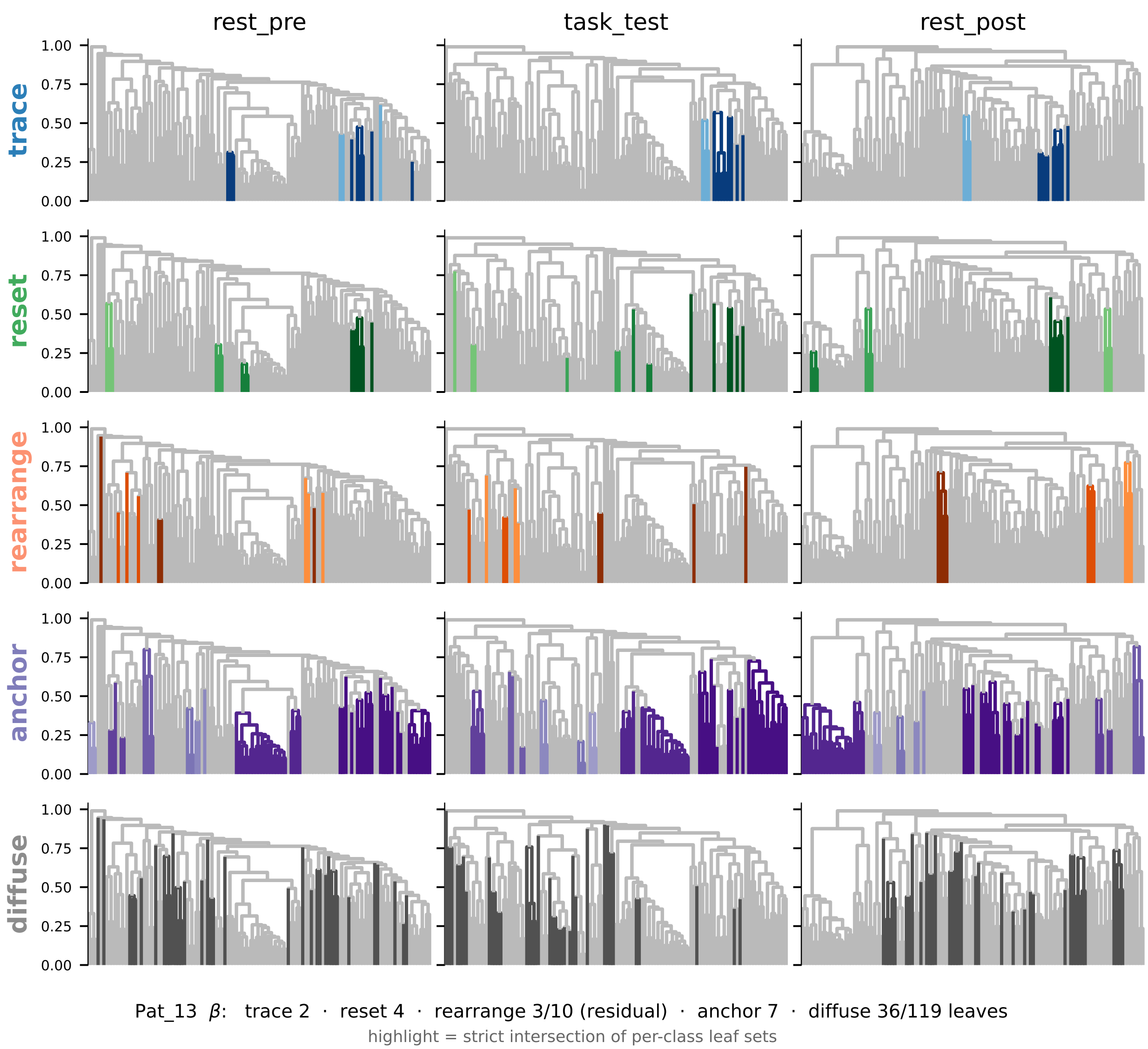


Figure 34: Full cross-phase module taxonomy at Pat13 β. Five rows (one per class: trace, reset, rearrange, anchor, diffuse) by three columns (phases: rsPre, taskTest, rsPost). Each cell is the dendrogram of $\mathcal{D}(\tau)$ at the corresponding phase, with the leaves of the row's strict-intersection class set highlighted; within a row, modules are sorted by strict-intersection size and shaded from light to dark, and the same strict-intersection leaf set receives the same shade across all three phase columns so a reader can trace a single module visually from left to right. In columns where the module's matched subtree at that phase is fragmented or absent (the rspre column for the trace and rearrange rows; the taskt column for the reset and rearrange rows), the shaded leaves still appear at their actual dendrogram positions but no longer form a connected colored subtree. Patient totals at β: two trace modules, four reset, three rearrange, seven anchor, thirty-six diffuse leaves of one hundred nineteen. The figure shows that a single (patient, band) cell can simultaneously express all five taxonomic classes, with anchor and diffuse populations dominating the leaf budget and the trace and reset populations smaller in count but occupying structurally distinct positions on the dendrogram.

6 Synthesis: band-resolved trace, diffusion geometry, and outlook

The substrate audit of § 4 and the multiscale Laplacian investigation of § 5 together test a single biological claim: the **cognitive episode imprints a structural memory on the post-task resting state**, so that rsPost is not statistically equivalent to rsPre and the four-phase paradigm is non-ergodic at rest. The central falsification of the equivalence is the controlled per-pair correlation on $\mathcal{D}(\tau)$ at $\tau' = 1/\lambda_{\max}$, which surfaces the imprint at α, β, and low-γ at within-probe BH-$q = 0.027$ under the split-baseline, drift-floor, and cross-probe controls (§ 5.3). The KC tree-distance decomposition (§ 5.2), the multiscale VI(k) and Grassmann views (§ 5.4), the anatomical localization (§ 5.5), and the per-patient module taxonomy (§ 5.6) are evidence layers attached to the same memory claim at distinct geometric levels of the same diffusion output, not independent statistical tests of distinct nulls. This section assembles the band-resolved reading the article will carry, frames why the LRG layer enriches what the raw |ImCoh| substrate alone could not unfold, and lists the controls and directions still owed.

6.1 Band-resolved synthesis

The β band is the cohort's multifacet imprint. It registers at every geometric level the analysis tests: the per-pair correlation on $\mathcal{D}(\tau)$ ($\rho_{\text{split}} = +0.222$, controlled $p = 0.014$, $q = 0.027$), the tree topology and tree heights of the dendrogram (within-probe BH-$q = 0.006$ on both KC axes, 10/10 patients below their split-baseline null), the matrix Pearson and Frobenius distances on $\mathcal{D}(\tau)$ ($p = 0.042$ on each), the partition-cut view across the dendrogram continuum (sustained 7–9/10 cohort agreement on VI(k) for $k \in [17, 32]$), and the leading-eigenmode subspace (positive principal-angle diagonal on the Grassmann probe sustained from $k \approx 12$ out to $k = 79$, above a fine-k cold zone). The β trace passes the Pat03-dropout robustness check at every layer and survives the cross-probe restriction. Anatomically, no single region carries it under multi-region correction: the strongest full-cohort candidate (Hippocampus, 3.49×) does not survive Pat03 dropout, and the only Pat03 -robust candidate (left superior temporal, 2.13×) remains uncorrected. The data test single-region concentration rather than distributional spread, so the appropriate negative claim is that the β imprint does not localize to a single cortical region under the controls applied here, not that it is brain-wide.

The low-γ band carries the imprint at the per-pair level (controlled $p = 0.010$, $q = 0.027$; 9/10 patients above their drift floor), at the heights axis of the dendrogram (KC $\lambda = 1$, 8/10, within-probe $q = 0.055$, carried as a directional companion), at the matrix Frobenius distance ($p = 0.053$), at intermediate-to-coarse partition cuts (VI(k) span $k \in [45, 56]$), and at the broadest principal-angle diagonal of the spectral subspace probe. It does not register at tree topology (KC $\lambda = 0$ anti-trace) and does not register at the rank distance d_S. The geometric signature is therefore distinct from β: the cognitive episode preserves which leaves co-merge in the linkage hierarchy but shifts how deep the merges sit, and rearranges magnitudes of communication distances without rearranging their rank ordering. The single-region anatomical address that distinguishes low-γ from β sits at left fusiform cortex, the only (band, region) cell to clear Bonferroni correction across the $m = 48$ cohort-eligible cells ($p_{\text{hyper}} = 5.2 \times 10^{-6}$), with the disclosed double dependence on Pat02 as both the cohort's anatomically densest fusiform implant and its highest-rate fusiform tracer.

The α band is the geometrically narrowest of the three controlled bands. It registers at the per-pair correlation alone (controlled $p = 0.007$, $q = 0.027$) and is silent on the global tree distance at both KC axes (6/10 patients on each, neither approaching significance). It surfaces in the multiscale companions at fine partition cuts (VI(k) peak 9/10 at $k = 3$) and in the spectral subspace diagonal above $k \approx 10$, and admits one suggestive but downgraded anatomical cell at left parsopercularis whose two-patient base aggregates one anti-aligned and one pro-aligned CTM contribution. The α imprint therefore lives in how information flows between contact pairs in the diffusion geometry, without restructuring the linkage hierarchy or concentrating in a single cortical territory.

The δ substrate trace was edge-only and does not survive the LRG diffusion: the propagator washes out a signal that lived at the level of single edges. The θ band is drift-only at substrate and drift-only at the diffusion layer; high-γ is borderline at both layers, with the reduced edge-weight dynamic range of the band as a documented substrate caveat. The Pat07 and Pat15 cross-probe-aggregate anti-alignment is preserved by every LRG probe and is read as a scientifically meaningful structural finding rather than as cohort noise, consistent with the implant-geometry interpretation introduced at § 4.4.

6.2 Why diffusion geometry enriches the raw |ImCoh| reading

The interpretive heart of the layered substrate-and-LRG reading is the geometric distinction between asking which contact pairs are coupled and asking how information flows between them. On a fully connected weight-heterogeneous network every pair already carries a nonzero baseline |ImCoh| weight, so structural reorganization is not about which edges exist—all edges exist—but about which preferential pathways the edge-weight heterogeneity carves through the network. The substrate audit reads the upper-triangular entries of $W(\mathcal{B})$ as independent observations and identifies four bands with directionally consistent task-shaped patterns at the cohort level. That reading is conservative and necessary, but blind to the structure of communication itself: edge weights establish hierarchies of preferential information transit, the diffusion propagator $K(\tau) = e^{-\tau L}$ integrates over all walks weighted by those hierarchies, and the communication distance $\mathcal{D}(\tau)$ reads pairwise dissimilarity in the geometry of preferential pathways rather than in direct coupling alone. The α band is the cleanest illustration of the enrichment: its substrate trace is the strongest of the cohort at the directional level (8/10 patients on d_S) but does not register at the dendrogram tree distance under controls; only the per-pair geometry of $\mathcal{D}(\tau)$ crystallizes the cohort signal into a controlled claim. The α memory imprint is invisible to a direct edge-rank reading and to a cluster-assignment reading; it lives in the diffusion geometry that reads how information flows. The Ψ-irrelevance result of § 5.1 is the methodological cornerstone of this enrichment. The standard log-gap detector $\Psi(n;\tau)$ was designed to find topological scale boundaries on networks where the propagator vanishes across edge-absence gaps; on the dense weight-heterogeneous |ImCoh| substrate no such gaps exist, the merge-height profile decays linearly rather than log-linearly, and Ψ pins to dendrogram boundaries as a generic property of UPGMA on this substrate (89/180 cells at strict argmax-at-an-extremum). The reframing that follows—reading $\mathcal{D}(\tau)$ directly rather than through a privileged partition cut, and using the dendrogram as a hierarchical encoding rather than a partition selector—is not a workaround but the principled adaptation of LRG to dense weighted graphs, and it is the layer at which the band-resolved memory imprint surfaces.

6.3 Outlook

The cohort claim of § 5 is established under the within-baseline null and drift-floor controls applied to the per-pair correlation; two cohort-wide drift insulations remain owed before the persistence language can be elevated from conditional to established. The first is a cohort-wide drift-triangle null at the LRG layer, mirroring Control 1 of § 4.6, currently run only at Pat06 where $R^2 \leq 0.05$ in nearly every (band, distance) cell. The second is a symmetric cross-baseline check at the diffusion layer, mirroring Control 3 of § 4.6, comparing $\mathcal{D}(\tau)$ computed on (taskTest-late, rsPost-early) against (rsPre-late, rsPost-early) on matched half-segment noise budgets. Until these complete, the cohort claim of § 5.3 stands as a controlled per-pair signal under the documented null architecture and not as a fully drift-insulated persistence statement; the manuscript will frame it accordingly. Three open directions inherit naturally. First, the per-patient cross-phase module taxonomy of § 5.6 admits an anatomical mapping to Desikan–Killiany regions that resolves trace-class versus anchor-class versus reset-class spatial distributions at the per-patient level, deferred from the structural illustration role taken there. Second, the cross-probe-aggregate anti-alignment of Pat07 and Pat15 merits an implant-geometry analysis that disentangles scientifically meaningful inter-subject heterogeneity from coverage-driven artifact; the parsopercularis cross-band consistency at Pat07 (β and α cortical contributions both anchored at the same region) is a starting handle. Third, the diffusion resolution is fixed at $\tau' = 1/\lambda_{\max}$ throughout § 5 for the reasons developed in § 5.1, but a controlled τ-sweep that reads $\mathcal{D}(\tau)$ at coarser scales tests whether the band-resolved imprint sharpens or dissolves as the propagator integrates over longer pathways, and is the natural extension of the per-pair reading to a fully multiscale diffusion-geometry claim.

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